PHINMA UNIVERSITY OF ILOILO
College of Information Technology Education

CAPSTONE PROJECT PROPOSOL

AUTOMATED RESOURCE MANAGEMENT SYSTEM

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Research Subject

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CHAPTER 1

INTRODUCTION

1.1 Background of the Study

The advent of advanced automation technologies has the potential. The need for more efficient tools to manage inventory and advancement of automation technologies pushes the emergence of Automated Resource Management Systems in various forms over several decades, evolving from 1960s manufacturing inventory control into the sophisticated, AI-driven platforms used today. Inventory management is critical to running any business, especially in auto supply businesses,where maintaining an optimal stock level directly impacts profitability and customer satisfaction. Inventory management involves tracking, ordering, storing, and selling a company's stock.In the context of auto supply businesses, effective inventory management ensures that essential parts and products are available when needed while avoiding the high costs associated with excess inventory or stockouts. Moreover, the recommendation to adopt a "just-in-time" (JIT) inventory system is insightful, for it can help reduce costs by minimizing the amount of inventory held at any given time, lowering storage costs, and reducing the risk of overstocking. This approach ensures that businesses only order inventory as needed, improving cash flow and preventing excess inventory from becoming obsolete or taking up valuable space.

Because of this people are using web-based applications a lot. These applications are used by institutions and organizations. They make it easy for people to talk to each other, store information and manage data. For example schools using these applications. In many organizations, they still use traditional ways when it comes to managing equipment and resources. They do not use web-based applications for this. Managing equipment and resources is still done by hand. Due to the manual process, problems often arise such as improper recording of information, data loss, slow processing of transactions, and manual data entry leads to version control issues, while disconnected systems create information silos that hinder strategy. decision-making.This is the case for organizations that use mobile and web-based applications.Such problems can affect the organization and effectiveness of an institution in managing their resources.

Digital technologies and automation systems and data analysis capabilities have brought about significant changes to the global manufacturing sector.The existing. manufacturing systems that depend on human control and centralized management systems cannot achieve current requirements for flexible production and effective operational processes and delivery of high-quality products and quick market entry.Autonomous manufacturing has become the primary operational model of the Industry 4.0 framework because it enables production systems to execute self-monitoring and self-analysis and self-optimization with minimal human involvement.The Manufacturing Execution System (MES) has evolved into its present form which functions as the fundamental technology that permits autonomous manufacturing systems to operate without human control.Conventional MES solutions primarily focus on production tracking, scheduling, and compliance reporting.Intelligent MES platforms use artificial intelligence machine learning and real-time data processing to support shop floor decision-making which adapts to changing conditions on the shop floor.Intelligent MES systems create dynamic scheduling systems through operational data analysis which enables them to adapt production schedules and detect anomalies and forecast equipment failures and enhance quality control processes.The intelligent MES system gains better performance through its implementation of cloud-native analytics.Cloud-native systems provide their users with two benefits through elastic computing capacity and permanent system up-time and advanced analytical features which traditional on-site systems cannot deliver. Manufacturers use cloud platforms to process real-time data that comes from different types of machines and enterprise systems.Cloud-native analytics support advanced use cases which include predictive maintenance and cross-plant performance benchmarking and end-to-end supply chain visibility needed for autonomous operations.Smart manufacturing networks use intelligent MES solutions which connect production systems through their continuous operational data exchange with business processes.The system enables manufacturers to achieve real-time operational monitoring through automated decision-making processes which enhance their ability to optimize manufacturing operations and handle unexpected events from inside their facilities and outside their business environment.Organizations need to solve multiple problems which include data protection measures and system operational dependability and employee role preparation and ability to connect different systems to complete the project successfully.The research investigates how intelligent manufacturing execution systems and cloud-native analytics technologies enable manufacturers to achieve self-operating production processes.The research studies technological foundations and architectural designs together with their operational advantages while it identifies major difficulties and prospective future paths.The research investigates how modern manufacturing enterprises can develop self-optimizing production systems which utilize data to establish resilient operational capabilities.The integration of predictive analytics into energy supply chain management is increasingly recognized as a crucial tool for mitigating risks and ensuring operational efficiency.

Energy supply chains face numerous challenges, including supply disruptions, fluctuating demand, price volatility, and environmental concerns, which can impact both short-term operations and long-term sustainability. Predictive analytics, leveraging data-driven insights, machine learning algorithms, and statistical models, offers a proactive approach to addressing these challenges by forecasting potential risks and enabling timely interventions.This framework focuses on the application of predictive analytics to identify, assess, and mitigate risks in energy supply chains. Key components of the framework include data collection, analysis of historical trends, real-time monitoring, and forecasting of potential disruptions. By analyzing large datasets from various sources such as market trends, weather patterns, geopolitical factors, and production data, predictive analytics can forecast risks related to energy production, transportation, and consumption, thereby providing valuable insights into potential bottlenecks, price fluctuations, and demand-supply imbalances.One of the primary advantages of predictive analytics in energy supply chains is its ability to improve decision-making and resource allocation.It enhances risk management by allowing organizations to anticipate and prepare for disruptions before they occur, reducing operational downtime and ensuring a more resilient supply chain.Additionally, predictive models help optimize inventory management, demand forecasting, and supplier relationships, contributing to cost reduction and improved overall efficiency.Despite its potential, the adoption of predictive analytics in energy supply chains faces challenges, such as data quality, technological infrastructure, and the need for skilled professionals to interpret and act on predictive insights.This paper explores these barriers and outlines strategies to overcome them, ensuring the effective implementation of predictive analytics.Ultimately, the framework presented aims to foster a more agile, resilient, and efficient energy supply chain, capable of adapting to emerging risks and contributing to the long-term sustainability of the energy sector.

The dynamic landscape of global supply chains necessitates innovative solutions to tackle challenges in demand forecasting and inventory optimization. Traditional methods, often constrained by limited adaptability and scalability, struggle to manage the complexities of modern supply chains. Artificial Intelligence (AI) has emerged as a transformative force, enabling predictive supply chain management through advanced data analytics, machine learning algorithms, and real-time decision-making capabilities. By harnessing AI-driven tools, businesses can accurately forecast demand patterns, reduce stockouts, and minimize excess inventory, thereby improving operational efficiency and customer satisfaction. AI-powered systems leverage historical data, market trends, and external factors such as economic shifts and weather conditions to provide precise predictions. These tools enhance responsiveness by identifying potential disruptions and enabling proactive measures, ensuring supply chain resilience. Furthermore, AI facilitates seamless integration across supply chain nodes, fostering collaboration and enabling data-driven insights that were previously unattainable.From predictive analytics for demand forecasting to intelligent automation in inventory management, AI-driven tools are revolutionizing the traditional supply chain model. Case studies reveal substantial reductions in holding costs, improved lead times, and enhanced supply chain visibility. However, challenges such as data quality, system integration, and ethical considerations in AI deployment remain critical areas for exploration.The insights presented underscore the pivotal role of AI in driving efficiency and innovation in an increasingly complex and competitive global economy. All markets today are dynamic.Change is in the air everywhere, and change affects strategy. A winning strategy today may not prevail tomorrow. It might not even be relevant tomorrow.There was a time, not too many decades ago, when the world held still long enough for strategies to be put into place and refined with patience and discipline. The annual strategic plan guided the firm.That simply is no longer the case.New products, product modifications,subcategories, technologies, applications, market niches, segments, media, channels, and on and on are emerging faster than ever in nearly all industries—from snacks to fast food to automobiles to financial services to software.Multiple forces feed these changes, including digital technologies, the rise of China and India, trends in healthy living, energy crises, political instability, and more.The result is markets that are not only dynamic but risky, complex, and cluttered.Such convoluted markets make strategy creation and implementation far more challenging.Strategy has to win not only in today's marketplace but also in tomorrow's, when the customer, the competitor set, and the market context may all be different. In environments shaped by this new. In reality, some firms are driving change. Others are adapting to it. Still others are fading in the face of change.How do you develop successful strategies in dynamic markets? How do you stay ahead of competition? How do you stay relevant to the customer? Implementing an Enterprise Resource Planning system (ERP) in organizations is a strategic move to enhance efficiency and productivity.It discusses how ERP systems streamline processes, centralize data, and facilitate informed decision-making.ERPs eliminate redundancy and improve communication by integrating various functions like finance, HR, and supply chain management. The resulting optimization of resource allocation and reduced manual effort leads to increased productivity.Furthermore, ERP systems enable real-time tracking of key performance indicators, fostering proactive responses to market changes. In summary, ERP deployment empowers organizations to achieve heightened operational efficiency and productivity.

To address these challenges, an Automated Resource Management System can be used that will provide a more systematic way of monitoring, recording, and managing equipment and resources. It offers a lifeline, providing surer footing, a more navigable path, streamline inventory tracking, dramatically improve accuracy, replaces manual spreadsheets with real-time visibility across multiple projects, unified workflows, and provide the firepower that allows companies to respond swiftly to market changes based on deeper insights into consumer behavior and trends. The aim of this study is to develop an automated system that will help facilitate and improve resource management.

1.2 Statement of the Problem

Despite the availability of various technologies that can be used in information management, many organizations will face problems related to manual resource management systems.

This study is going to find out if the Automated Resource Management System really works in making resource management and how the organization runs better. The Automated Resource Management System is what we are looking at. We want to know some things about the Automated Resource Management System.

1. Does the Automated Resource Management System make a difference in how resources are given out and used?

2. Does the Automated Resource Management System make inventory monitoring, control and management better?

3. Does the Automated Resource Management System make tracking and reporting of resources accurate?

4. Does the Automated Resource Management System help reduce delays, mistakes and waste of resources in what the organization does?

5. How better does the Automated Resource Management System make the allocation and use of resources that are available?

6.How well does the Automated Resource Management System manage inventory and the resources of the organization?

7. What kind of effect does the Automated Resource Management System have on how accurate and reliable tracking and reporting of resources are?

8. How much does the Automated Resource Management System help reduce mistakes, delays and waste of resources, in operations?

9. What kind of impact does the Automated Resource Management System have on how productive the organization's how well it performs overall?

Challenges with managing inventory inaccurate inventory tracking, having an excess of unsold merchandise, and insufficient stock to fulfill client needs are the three main issues with inventory management.

- Inadequate or outdated inventory management techniques
- Demand fluctuations caused by customers' changing desires and needs
- Difficulty locating certain commodities in a warehouse
- Depending on the company's field and employer, there are four main kinds of inventory that they may oversee: raw materials, work in progress, finished goods, maintenance, repair, and operations (MRO/Repair) goods

1.3 Objectives of the Study

1.3.1 General Objective

This study aims to design and develop an Automated Resource Management System that can help in more efficient, orderly, and systematic management of equipment and resources within an organization and to develop an effective system that will streamline the client's business operations and increase efficiency.

1.3.2 Specific Objectives

To achieve the general objective, this study aims to:

1.To design a system that integrates sales tracking and inventory management with proper categorization and maintaining the data quality during the transaction.

2.To develop a user-friendly web-based system that incorporates a Point of Sale (POS) functionality for efficient inventory management, facilitating faster transactions.

3.To incorporate predictive analytics to compensate for the customer's demand for stocks and prevent overstocking of particular product supplies.

4.To test and evaluate the functionality and accessibility of the proposed system.

1.4 Scope and Delimitation

The objective of this project was to create an Automated Resource Management System that could be used to monitor and control resources/equipment at the institution. The system will allow users to login/register, track/monitor resources, store data in a database, and access an administrative dashboard for system administration. The project will only produce a prototype of the Automated Resource Management System for academic purposes; it will not be used in actual production or integrated with other external systems at any other organization.The data collected for the analysis may or may not be the right time for the analysis. The analyzed data may or may provide the accurate results for decision making and the scope of this study focuses on the development and implementation of an automated system, a smart inventory management system designed for mini grocery stores. The system covers core inventory functions such as product registration, stock level monitoring, low-stock alerts, expiration tracking, and the generation of basic inventory reports. It is intended for daily use by store owners, cashiers, and staff with basic computer literacy.

1.5 Significant of the Study

1This study is going to be helpful to the following people:

Users - The Automated Resource Management System will make it easy for users to access, monitor and manage the resources they have. It will give them a way to check if resources are available, track how they are used and see important information right away. This means users will not have to spend much time and effort on manual tasks and they will have a better experience overall. The Automated Resource Management System is really going to make things easier for users.

Administrators - The system will give administrators a way to manage the resources of their organization. They will be able to keep track of everything in one place, which will make their job easier. The Automated Resource Management System will help administrators record, update and track resource information. It will also reduce the amount of work they have to do, make mistakes and help them make good decisions because they will have all the information they need.

Organizations - This study will help organizations manage their resources better. The Automated Resource Management System will reduce errors, make sure resources are not wasted and keep records of how resources are used. This will make the organization more productive, use resources better and work efficiently. The Automated Resource Management System is going to improve how organizations work.

BSIT Students - This study can be really helpful to Bachelor of Science in Information Technology students who want to learn about system development, database management and process automation. The Automated Resource Management System can teach them about designing, developing and implementing automated systems. It can also guide them when they work on their projects and research studies. BSIT Students can learn a lot from the Automated Resource Management System.

Future Researchers - The results of this study can be used as a starting point for research on resource management systems, information systems and automation technologies. Future researchers can use this study to learn about features, improve existing methods or develop more advanced systems that meet the needs of organizations. The Automated Resource Management System can be a reference for Future Researchers.

1.6 Definition of Terms

Application Development - We create programs that help companies work better by planning, organizing, testing and keeping an eye on Application Development. The main job of Application Development is to make sure companies can work well by giving employees the tools they need to do their jobs in Application Development. An Automated System is a machine that can work on its own. It usually needs people to supervise the Automated System. The Automated System needs instructions and special tools to work efficiently and accurately in the Automated System. This is important for the Automated System to work properly.

Resource Management - is about taking care of the things a company needs to work like buildings and equipment in Resource Management. This includes making sure everything is ready and in condition and getting new things when needed for Resource Management. The people in charge of Resource Management have to control and keep track of everything a company owns from getting supplies to selling things in Resource Management. This helps things run smoothly and makes sure everything is in the place at the right time for Resource Management.

Resource Management - means keeping track of everything a company has, checking if it is working well and making sure it is safe and secure in Resource Management. The people in charge of Resource Management need to know what is going on. This helps prevent things from getting lost or wasted in Resource Management.

Database - is a collection of files on a computer that stores information about a company. It is like a library where people can find and update information in the Database. The Database can be used for Application Development or Resource Management.

Databases - are very useful for storing information. An Administrator is someone who takes care of a computer system and makes sure it is safe. The Administrator is in charge of making decisions, setting rules and managing resources.

Monitoring System is a tool that helps companies keep track of how they're managing their resources. It gives them real-time information, which helps them make decisions about Resource Management. We also keep a list of all the equipment a company has including what it is, how much of it we have, what condition it is in and if it is available for Resource Management. This helps us keep track of things and make decisions about what to buy and what to get rid of in Resource Management.

Resource Management - is also about keeping track of all the things a company has and using them well in Resource Management. This is very important for companies.

Report Generation - is about making reports that show how well a company is managing its resources in Report Generation. This includes information about equipment, inventory and how things are being used in Report Generation. These reports help leaders make decisions about what to do using Report Generation.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Foreign Studies

According to a study by Smith and Anderson (2021), the use of automated systems in information management helps to improve the efficiency and accuracy of processes within an organization. Their study showed that digital systems can speed up data processing and reduce errors caused by manual operations.

On the other hand, Johnson et al. (2020) emphasized that the use of intelligent systems and recommendation technologies helps in improving the user experience and user trust in a system.Highly dependent on its inventory management. Meeting consumer demand is a critical business function, and inventory is the glue that holds the purchasing, supply, and production processes together. The advantages of having inventory, such as improved customer service and protection from supply risks, are often outweighed by the significant costs of having too much inventory. Not only that, but new research has demonstrated that ineffective inventory management has far-reaching consequences that impact the performance and market value of both firms in the long run.

A study by Nakamura and Lee (2018) focused on the use of Just-In-Time (JIT) inventory systems in Japan. The research revealed that JIT helps companies minimize inventory holding costs, improve production efficiency, and strengthen supplier coordination. By maintaining only the necessary level of stock at the right time, businesses were able to reduce waste and optimize their supply chain processes. The study concluded that JIT is an effective inventory strategy widely adopted in global industries to improve operational efficiency.

The importance of inventory management and control in improving operational efficiency for companies in different industries is explored in this study. Efficient inventory management is crucial for companies that want to stay competitive and make money in today's market because of how globalization and technology are changing the dynamics of the industry. A company's success or failure is highly dependent on its inventory management. Meeting consumer demand is a critical business function, and inventory is the glue that holds the purchasing, supply, and production processes together.The advantages of having inventory, such as improved customer service and protection from supply risks, are often outweighed by the significant costs of having too much inventory. Not only that, but new research has demonstrated that ineffective inventory management has far-reaching consequences that impact the performance and market value of both firms (Aatif Latif Shaikh, (April 2024).

Also, several African business units and firms have failed, and ineffective inventory management contributes to these failures (Orobia et al., 2020). Holding too much inventory resulted in more capital being tied up, which caused deterioration, obsolescence, and damage. Conversely, a lack of inventory is linked to sales interruption brought on by stockouts, poor customer service, and underused machinery and equipment (Dong & Su, 2010; Gul et al., 2013; Karim et al., 2017; Orobia et al., 2020). In connection with this, their investigations examined whether inventory management influences the link between managerial skill and profitability.

Meanwhile, Malaysian SMEs struggle with irregular inventory, inaccurate estimates, slow customer response times, and improper accounting recording processes that lead to poor performance (Mohamad et al., 2016). Similarly, Abdulrasheed et al. (2013) noted that poor performance from ineffective inventory management solutions causes firms to struggle with irregular delivery, decreasing consumer productive capacity, and excessive production costs. Since there are not enough materials to satisfy the unexpected rise in consumer demand, there needs to be more profits to cover costs and returns on equity; there are material wastes and thefts brought on by surplus stock, and most businesses have communication chains that are not active. Thus, inventory management has become essential for managers in charge of production. Therefore, the focus of their research is on how inventory management affects the profitability of SMEs.

Effective inventory management in warehouses is presently a top priority for Indian firms. Operation managers must address concerns, including reducing inventory carrying costs and product availability (Panigrahi et al., 2021). The results of their research are used to establish how effective inventory management strategies affect steel manufacturing businesses' operational performances. Their study focuses on how inventory automation and inventory distribution turnover approaches may give large steel manufacturing enterprises a competitive edge and boost operational performance. The same problem is experienced by most Nepalese public manufacturing enterprises, resulting in massive losses due to inefficient inventory management (Risal & Acharya, 2021).

Eliminating future hurdles would satisfy public manufacturing enterprises. Inventory management is a broad topic that has received little attention in the Nepalese context (Risal & Acharya, 2021; Sunday & Joseph, 2017). Thus, their studies analyzed public enterprises' inventory management and profitability.In Saudi Arabia, inventory errors significantly impact a company's financial reports. Most errors result from poor inventory data management, but they can also be brought on by staff needing to count the goods physically. A proper inventory control system must be implemented to minimize costs while maximizing profit. However, many issues still arise, such as a lack of consistency due to no standard operating procedures, a fixed management system, and material discrepancies (Althaqafi, 2020). Therefore, the study investigated how inventory management affects financial performance.

This study defines inventory management, operational efficiency, and financial performance with the problems identified by past studies. Operational efficiency refers to arranging all production units within a company to guarantee that they operate harmoniously to accomplish critical business objectives (Panigrahi et al., 2021). Additionally, Yu et al. (2018) clarify that operational efficiency refers to the firm's capacity to enhance results by establishing regular routines and standard operating procedures across functional domains. This study defines operational efficiency as evaluating how well a company uses its resources to provide the greatest results. A separate study conducted by Kwon and Lee (2019) and Panigrahi et al. (2021) concluded that different factors have an impact on operational efficiencies, such as Research and Development (R&D) and Inventory Automation Practices (IAP). The studies concluded that these factors (R&D and IAP) significantly positively affect operational efficiencies and influence a firm's operational efficiency and overall performance.

Effective inventory management enables businesses to achieve higher levels of customer service, lower costs, and improved operational efficiency (Chase et al., 2021). Realizing the expenses of effective inventory management is still essential for Micro, Small, and Medium-Sized Enterprises (MSMEs) in the Philippines to maximize profit. After reviewing the related literature, there are few studies and insufficient data on the impact of inventory management on operational efficiency and financial performance (Elsayed, 2015; Kljenak et al., 2013; Shockley & Turner, 2015). Inventory management is crucial and frequently consumes a significant portion of an organization's operational budget; thus, it has to be closely regulated. It has been acknowledged that keeping accurate inventory records and using efficient inventory control strategies are crucial. It is also found that state firms adopted technology less readily than other industries in inventory management, which may impact how well they manage their inventories (Gatari et al., 2022). Similarly, Khan and Siddiqui (2019) assert that effectiveinventory management does not treat every product similarly; instead, it utilizes control and analytical techniques by the relative economic relevance of each good. In addition, proper inventory planning and control is one of the most critical aspects of a company that requires skill and ability to increase overall performance. Inventory planning and management benefits include improved contract compliance, lower procurement costs, better worker engagement, and lower handling costs (Namusonge et al., 2015).

In addition, Islam et al. (2019) examined the elements contributing to poor inventory management in SMEs. They discovered that the lack of a cohesive firm information system and qualified human resources impede effective inventory management. An integrated information system is essential for providing management with real-time information. Furthermore, it enhances departmental cooperation. To enable the creation of an integrated information system, qualified human resources are needed. Employers must continue to train and retain their workers. By improving quality, saving time, and reducing prices, efficient inventory management may distinguish the business from competitors. From the consumers' perspective, due to daily price increases, society would face economic backwardness due to high inflation, resulting in product shortages and consumer hoarding (Anagaw, 2023). Businesses must lower the number of inventories in the supply chain to rationalize storage costs and maintain them (Hugos, 2018). A robust inventory management system's advantages include efficient stock storage, simple warehouse product access, elimination out-of-date inventory, and a general reduction in stock expenditures (Mulandi & Ismail, 2019).Lastly, inventory management is essential for a firm to succeed, and numerous strategies have been devised to maximize stock levels. Some of the inventory management techniques used by businesses are Just-in-time (JIT), First in, First out (FIFO), ABC analysis, Economic Order Quantity (EOQ), and Vendor-Managed Inventory (VMI). For retailers, Just-In-Time (JIT) is a technique that involves receiving inventory just before selling it rather than keeping it on hand for weeks or months until the business needs it (Cvetkovic, 2022). JIT stresses a lean manufacturing system and reduces inventory waste and handling costs (Patterson et al., 2010).

The utilization of the First-In-First-Out (FIFO) strategy is prevalent in inventory management as it guarantees the consumption or sale of the oldest inventory before newer ones. Adopting this method allows companies to mitigate the potential risks associated with inventory becoming obsolete or spoiled while enhancing the accuracy of their financial reporting. FIFO facilitates preserving product freshness, reducing carrying costs, and providing more precise inventory valuation in financial statements (Smith & Johnson, 2019). In contrast, ABC analysis made effective inventory control possible by categorizing inventory based on significance and assigning priority levels to goods (Kumar & Garg, 2017). The Economic Order Quantity (EOQ) approach determines the right number of orders to place to reduce overall inventory expenses (Wang et al., 2017). Vendor Managed Inventory (VMI) is another technique where the supplier manages inventory levels for the buyer, allowing for a more efficient supply chain (Prajogo & Olhager, 2012).Understanding the benefits and limitations of these inventory management techniques is essential for businesses to choose the most appropriate technique for their specific needs, as it significantly impacts a firm's profitability and ability to meet customer demand (Smith & Johnson, 2019).

Thus, having a thorough grasp of operational efficiency and the variables that affect it may aid firms in identifying areas for development and putting into practice efficient tactics to increase performance.In addition, operational efficiency is a critical factor in the success of businesses across industries. Businesses that emphasized operational efficiency often had better levels of productivity, lower costs, and higher customer satisfaction (Karia et al., 2018). Businesses must be completely aware of their processes, resources, and goals to achieve and sustain operational efficiency. This entails choosing and tracking key performance indicators (KPIs) such as cycle time, throughput, and inventory turnover, as well as routine monitoring and analyzing data to spot areas for improvement. Moreover, successful teamwork, ongoing improvement, and good communication are essential for establishing and maintaining operational efficiency (Sohal, 2016). Overall, firms may gain a considerable competitive edge by realizing the value of operational efficiency and putting out effective plans to achieve it.

To maintain track of its raw materials, stock, work-in-progress, and finished items, every business must have efficient inventory management (Rasool, Hussain, Ullah & Shehzadi, 2024). Moreover, Mamuda and Adamu (2023) contended that inventory management is essential to enhancing the effectiveness and efficiency with which businesses handle their stock. To satisfy consumer needs, Tadayonrad and Ndiaye (2023) emphasize the critical role that correct inventory information plays. They also stress the clear link between improved customer satisfaction and efficient inventory management. Accurate inventory control is essential for companies as it affects not just financial statements but also customer contentment and overall productivity. Inventory management errors may have crippling financial effects and damage a company's reputation by causing stockouts and overstocking, among other problems (Groenewald & Kilag, 2024). Alam, Thakur, and Islam (2024) state that the biggest obstacle to managing inventory at an optimal level is the difficulty in forecasting demand and aiming customers' expectations of product availability in the market. One challenge in inventory management is keeping the supply and demand for inventory in balance. In an ideal world, a company wouldn't lose sales due to stockouts and would have adequate inventory to match client demand. However, because carrying inventory is expensive, the company does not want to keep an excessive amount of goods on hand. The methods for controlling, storing, and keeping track of inventory products are the three primary components of inventory management, according to Zoho Corporation (2024). Managing storage space effectively is a daunting challenge. Utilizing inventory management platforms to plan and construct storage areas gives you more control over when new goods are delivered, which is a crucial factor in determining available space. While ordering to delivery is just one part of the extensive cycle that is inventory management, storage is one of the most important. Regretfully, many of the biggest inefficiencies are also located here (Vyas, 2023). According to Jenkins (2022), new challenges for warehouse design and storage arise when considering biodegradable packaging or doing away with packaging entirely to decrease waste. It can also include purchasing new machinery or a reduced shelf life for some things. One issue with the supply chain may be the loss of inventory as a result of theft, damage, or spoiling. Problem areas must be identified, monitored, and measured.

According to Anshur, Ahmed, and Dhodi (2018), improper inventory tracking increases the likelihood of inventory issues, which also have an impact on wastefulness and additional costs overall. It is laborious,redundant, and prone to mistakes to use manual inventory tracking techniques across several applications and spreadsheets (Vyas, 2023). Accurate monitoring is crucial in the cutthroat industry of today. The use of manual processes continues to be a barrier; many companies still enter data by hand using spreadsheets or antiquated software, which can result in errors (Jenkins, 2022). Periodic and Perpetual systems are two types of inventory control systems. Various methods are used in inventory control to keep an eye on the movement of stocks in a warehouse. Existing stock at a warehouse is managed via inventory control (Estrellas, 2024). Inventory and warehouse managers utilize a variety of inventory management control approaches, including inventory forecasting, inventory reorder points, information technology, and inventory turnover, to monitor and enhance organizational performance (Anshur et al., 2018). There are several approaches to inventory management, and each has advantages and disadvantages of its own. Managers should be sure to take into account the kind of product they offer, the size of the company, the total budget, and the degree of precision required to operate a successful supply chain when selecting the best strategies (Tarver & Aditham, 2023). It is a function that is very vital and significant to the performance of any kind of organization (Hayes, 2024; Porzuczek, 2022). Munyaka and Yadavalli (2022) stated that deterministic inventory theory bases inventory operations on a known (certain) demand. A deterministic model yields the same result due to certainty about the factors, conditions, and parameters involved, which are stated explicitly at the beginning. One of the parameters is demand. Antic, Djordjevic, and Lesec (2022) state that deterministic demand is expressed as a monthly sales forecast for each product. The deterministic demand model seeks to minimize the total costs associated with production time, setup time, and overtime, as well as inventory-related costs like ordering costs, carrying costs, and stock-out costs (overstocks and shortages).According to David (2019), deterministic theory is the inventory theory that most suppliers or retailers use. In essence, this is maintaining inventory levels and placing further orders as needed. When shipment times and client orders can be predicted, this hypothesis performs well. You need to have enough inventory, for instance, if it takes a week for things to ship to a store. If not, you will have to decline orders from consumers when the item runs out of stock.

According to Dan C. (2014), E-commerce can provide substantial benefits even to small enterprises through improved efficiencies and raised revenue. It enables a new way of working to emerge as businesses face the future and embrace the new economy. E-commerce enables small business entrepreneurs to gain access to better quality information, thus, empowers them to be informed about their decision in their business. Most importantly, E-commerce can give a competitive advantage. It can help strengthen market position and open new business opportunities with its potential to improve profits. Likewise, the statement provided by KPMG International (2017) describes the capabilities of E-commerce when it comes to accessibility. Actually, even businesses without physical location could operate because of this technology. According to Sharma G. (2015), E-commerce is considered an excellent alternative for individuals and companies to reach new customers. Service quality delivery through the Internet is an essential strategy to success, more than 93 important than price and web presence. The E-commerce Website has been identified as having a significant impact on business activities in solving the geographical problem. A number of performance problems have been observed for E-commerce Websites, and much work has gone into characterizing the performance of web servers and Internet applications.

According to Pandey D. and Agarwal V. (2014), global presence has been yet another versatile and important advantage of E-commerce, which enables accessibility of a very wide range of products and services virtually, anywhere in the world, even in the remote areas. The feature of global presence has a boosting effect in the entire gamut of international trade, which in turn pose a subsequent impact in the enhancement of an even greater level of global business presence of products and services offered. Hence, this relationship is vice-versa. Technology driven E-commerce can provide automated continuous business transactions round-the-clock. Round-the-clock busi-ness operation and the feature of global market presence have enabled mankind to exceed the physical limitation of Intraday market operations and extend it into a 24/7 process. However, Inventory optimization and management have a significantly large impact on the retail industry. Every retailer aims to optimize their return on investment, and in order to do so, they must make sure that they are informed of the changes and that their inventory investments are kept at the best, or optimal, level, which is a difficult objective to achieve. Retailers encounter significant difficulties managing and storing their inventory. In a related study by (Stanelyte, 2021), it was determined how the demand of the consumer drives the entire phenomenon of inventory in the retail industry. Since products in the retail sector tend to sell quickly, it is crucial to predict client demand precisely in order to develop the ideal model for inventory replenishment. In addition, in-depth data analytics systems that may be used to predict future client needs based on purchase history data are only one of the tools available to retailers to help them satisfy customer demand. Other factors, including the price that customers pay for products, the expense of maintaining inventory, and the management of product loss, can still lead to poor demand management performance even when inventory is managed properly. Retailers can utilize a variety of inventory management techniques to assist their stores satisfy customer demand. The local market environment and the retailer's product line determine the best inventory levels for a certain store. (Chawla, et.al., 2019).

That is why to efficiently track inventory, each online business needs a different approach to inventory management (Lopienski, 2021). Furthermore, because it has complete access to real-time data, implementing an inventory management system is a proven approach to distribute inventory efficiently as the supply chain expands (Lopienski, 2021). Inventory management affects how well a company runs its operations, treats its customers, and boosts revenues.In order to meet demand, inventory optimization seeks to cost-effectively order the appropriate amounts of the right products. This is accomplished by incorporating supply and demand volatility into stock rule and inventory parameter adjustments. Inventory optimization seeks to achieve optimal availability while lowering inventory costs and reducing the risk of expensive surplus or obsolete stock. It does this by balancing service level (product availability) goals with inventory investment restrictions (Drakeley, 2023). Integrating digital technologies to reduce errors, delays,and administrative burdens, to enhance cost-efficiency and responsiveness. Likewise, Abubakar et al. (2019) emphasized that the lack of integrated inventory systems in educational institutions leads to higher operational risks and limits the institutions' capacity for strategic asset planning. In addition, the integration of data mining techniques into inventory management has been shown to enhance forecasting accuracy, enabling institutions to prevent shortages and overstocking. This approach also improves monitoring efficiency, speeds up reporting, and supports more effective resource allocation (Tungcul & Kummer, 2021). Effective tracking of assets prevents resource loss, redundancy, and inefficiency, enabling organizations to maintain optimal resource allocation.

Traditionally, inventory management practices, such as spreadsheets, are labor-intensive, error-prone, and inefficient. These methods often lead to data inaccuracy, oversight, and internal shrinkage, significantly impacting business operations (Madamidola et al., 2024). In response to these trends, some auto supply businesses are adapting their inventory management practices to better align with their specialization needs. By adopting new inventory management techniques, these businesses can gain greater control over stock levels and streamline their ordering processes, ultimately enhancing efficiency and operational effectiveness.Understanding the significance of physical location in inventory management is crucial for auto supply businesses. Smith and Johnson (2018) emphasized that the location of retail establishments greatly influences inventory turnover rates, demand forecasting accuracy, and transportation costs. Urban locations often have higher demand but may face challenges related to space constraints and traffic congestion (Rahman et al., 2021). Conversely, rural areas might offer lower operating costs but could struggle with lower customer traffic (Li & Liu, 2012). On the other hand, the longevity of an auto supply business can significantly impact its inventory management practices. Jeong et al. (2019) noted that established businesses often have refined inventory control systems, benefiting from historical sales data for more accurate demand forecasting. Conversely, newer businesses may face challenges such as inventory overstocking or stockouts due to limited historical data (Garcia & Nguyen, 2019). Inventory management is the art and science of maintaining stock levels of a given group of items, incurring the least cost, and being consistent with other relevant targets and objectives set by management (Prempeh, 2015). It is an important part of making all the decisions in handling the inventory in an organization, such as activities to be carried out, policies of inventory management, and procedures in handling the inventory to ensure enough quantity of each item is always kept in the warehouse. (Chan et al., 2017). Effective inventory management practices (EIMP) are essential in the operation of any business. However, implementing such practices faces challenges, notably cost barriers, attitudes, and knowledge of owners/managers (Atnafu et al.; Ahmad & Zabri, 2018). Overcoming these hurdles is vital for sustaining growth and competitiveness in the market. Furthermore, the effectiveness of inventory management practices may directly impact customer satisfaction through ordering decisions, minimizing inventory levels, and eliminating costs (Ahmad et al., 2020; Andersson et al., 2015; Smith & Jones, 2018). While optimizing inventory ordering decisions and minimizing levels are essential for operational efficiency (Orobia et al., 2020), eliminating costs must be balanced to avoid stockouts and maintain customer service levels (Simchi-Levi et al., 2015).

According to (Wang & Hong, 2022), thousands of final products together with extensive types of raw materials and intermediates make large-scale production networks complicated. Traditional inventory models cannot manage the intricate decisions often needed to be made, while traditional simulation methods are too slow for handling the scale effectively. For solving these drawbacks, (Wang & Hong, 2022) proposed a simulation approach which is RNN-inspired: it exploits the computational potential of recurrent neural networks for the purpose of production network structural layouts. This technique proved very fast: by orders, typically by thousands, more rapid compared to conventional simulations and yet will enable an efficient optimization even of the biggest inventory-related problems in the time constraint of real-world applications. Ain, Kiisler., Olli-Pekka, Hilmola. (2020) explained Abstract Research is based on wholesale and distribution operations of real-life case company, and in this setting, the most critical part of company's supply chain is the inventory replenishment to warehouse (Distribution Center) as well as fulfilling and delivering customers' orders. Different Economic Order Quantity (EOQ)-based models have been considered (Reorder Point, Reorder Point with pipeline on order inventory, and "pulse train"). The simulation system evaluates annual total logistics costs. Results show that in an environment, where local warehouse inventory levels are rather high and replenishment order quantity is rather small, it is important to have frequent shipments divided in suitable intervals. In the simulation model, this could be done e.g. with the use of "pulse train" function or incorporating a pipeline on order inventory in order to decide. The research findings are valid for a small-scale supply chain servicing small and geographically limited markets with clients assuming high customer service levels (e.g. 24-hours lead time). For bigger markets, the cross-docking based supply chain models are worth considering in simulations.Recently, Li et al. developed an affordable AI-assisted machine supervision system for SMEs manufacturing that significantly reduced costs and provides actionable insights and engage SMEs in sustainable manufacturing.

However, sales are not a universally ideal indicator of performance. It is affected by factors such as inflation and currency exchange rates, whereas employment remains relatively stable in this.regard. Moreover, sales do not always precede growth in other areas. In high-technology start-ups and the launch of new activities within established firms, it is often observed that assets and employment may grow before sales are realized (Delmar et al., 2003). On the other hand, employment has been argued to be a more direct indicator of organizational complexity than sales, making it a preferable metric when the focus is on the managerial implications of growth or performance (Delmar et al., 2003). Studies also show a strong link among sales growth, an expanding customer base, and increased market reach, which collectively drive the need for additional workforce (Storey, 2016). MSMEs with well-structured strategies and a focus on market penetration demonstrate a higher capacity for job creation. Scholars often emphasize employment growth as a vital measure of business performance and its impact on job creation (Siepel and Dejardin, 2020). This measure is particularly relevant to policymakers, who tend to prioritize employment over sales growth in assessing the economic contributions of businesses. A finding supported by Şeker & Saliola (2018), who highlights the remarkable job growth rates among small firms in developing countries. Similarly, Clayton et al. (2013) point out that while small businesses are often regarded as the primary drivers of job growth, older firms also play a substantial role. Although entrepreneurs and managers may not always prioritize employment growth as a direct indicator of success, it remains a critical measure for understanding medium- and long-term business development.

On the other hand, subjective measures focus on qualitative aspects of effectiveness, such as growth, expansion, efficient service delivery, product quality, survival, and competitiveness (Dobbs and Hamilton, 2007). Also, through extensive literature reviews, Ardichvili et al. 2003). However, for this study, it focused on sales and employment growth only.The measurement of MSME business performance can be seen from the increase in sales, which is the increase in the number of purchases made by customers for the goods sold (Agus et al., 2023)). Studies indicate that a range of factors contribute to sales improvements within MSMEs. Access to finance plays a critical role, enabling investments in inventory, expansion, and marketing efforts (Nwosu and Orji, 2016). Strategic marketing skills, particularly in leveraging digital channels, have become vital for reaching wider audiences and promoting products and services (Purwanti et al., 2022). The development of innovative products or services that cater to evolving customer needs can be a significant driver of sales growth (Halila and Rundquist, 2021). Moreover, building strong customer relationships through personalized service and after-sales support leads to enhanced brand loyalty and repeat purchases. One critical aspect of tangible resources is their contribution to operational excellence.

A study by Wongwilai (2022) emphasizes the importance of modern and well-maintained physical assets and infrastructure, such as machinery, equipment, facilities, and inventory. These assets enable MSMEs to streamline production processes, reduce operational costs, enhance product quality, and meet customer demands efficiently. Effective utilization of physical assets leads to improved operational efficiency and productivity, thereby contributing to overall performance.Furthermore, the strategic allocation of tangible resources contributes to

MSMEs’competitive advantage. Moscare-Balanquit (2021) suggests that MSMEs that strategically invest in tangible assets, optimizing operational costs, and managing supply chains efficiently are better positioned to achieve sustainable competitive edges in the market. Effective resource allocation enables MSMEs to differentiate their offerings, enhance customer value propositions, and respond promptly to market demands, thereby strengthening their competitive positions.One key aspect of optimizing physical capital is ensuring efficient resource utilization. Studies like the one conducted by Mugo et al. (2019) on Kenyan manufacturing MSMEs highlight the importance of capacity utilization. This refers to getting the most out of existing equipment and facilities. By operating at optimal levels, MSMEs can maximize output without compromising on quality. Proper maintenance of physical assets is equally crucial. A study by Salawu et al. (2023) emphasizes how preventive maintenance practices minimize breakdowns and production delays, ensuring smooth operation and maximizing uptime. Strategic investment in physical capital can also propel MSME growth. Upgrading equipment with newer technologies can lead to several benefits. Santanaa et al., (2018) highlight how this can increase efficiency, improve product quality, and even allow MSMEs to expand into new markets with capabilities that cater to evolving customer demands. However, informed decision-making is critical. The same study also emphasizes the importance of careful planning and financial resource consideration to avoid overinvestment in assets that may not be fully utilized or become obsolete too quickly.

Research across various regions reinforces the positive impact of physical capital on MSME performance. Sandu et al., (2022) conducted a study in Kenya that found a strong correlation between effective physical capital management and profitability in small businesses. This highlights the financial benefits of optimizing resource utilization and making strategic investments. Similarly, Mugo et al. 's (2019) study provided evidence of the link between capacity utilization and firm performance, demonstrating how getting the most out of existing equipment translates to better business outcomes.The critical role of financial resources in MSME performance extends beyond just having available cash flow. It also provides the essential capital needed to initiate and sustain business activities, ranging from purchasing inventory and equipment to covering operational expenses and salaries. Adequate financial resources ensure smooth cash flow, enabling MSMEs to meet their immediate financial obligations and seize opportunities for growth (Nkwinika and Akinola 2023).Moreover, the ability of MSMEs to access credit and external financing options is instrumental in fueling expansion, innovation, and investment in new technologies or market opportunities (Bose, 2013). A healthy financial profile with access to credit facilities supports ongoing operations and empowers MSMEs to pursue.strategic initiatives that drive long-term competitiveness and market relevance. Enhanced access to credit enables investment in assets, expansion, and adopting new technologies, facilitating business growth (Adjabeng and Osei, 2022). Studies indicate a positive relationship between financial inclusion and MSME profitability, productivity, and job creation (Nwosu and Orji, 2016; Bruhn et al., 2017). Access to savings accounts allows MSMEs to manage cash flow effectively and build financial resilience during downturn.

This positive impact of employee experience on MSME performance is supported by various research studies. Suroso et al. (2017) found a significant correlation between employee experience and SME business performance. Similarly, Ali Hamid Ibrahim’s (2018) research in

Libya highlighted the significant contribution of experience to an MSME’s success. Egberi’s (2019) study in Nigeria further confirmed that employee experience has a positive and measurable effect on an MSME’s overall performance. The knowledge held by employees stands as an indispensable asset within ’he framework of human capital for any organization. This knowledge comprises various dimensions, including educational attainment, the culture of knowledge sharing, disciplinary acumen, and accumulated experiences (Isaga, 2015). Employees endowed with profound knowledge play a pivotal role in advancing product or service innovation and elevating overall customer satisfaction levels. Furthermore, their expertise serves as a catalyst for streamlining production processes, thus contributing to cost reduction and enabling efficient resource allocation within the company (Ibrahim, 2018).

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While physical assets are essential, an MSME’s true competitive advantage often lies in its intangible resources. These are the non-physical assets that drive innovation, build customer loyalty, and enhance brand reputation. Research in this domain reveals insights into how these intangible resources impact MSMEs’ competitiveness and success (Stan et al., 2024). A strong brand reputation, built through effective marketing, quality commitment, and positive customer experience, is a valuable intangible asset. This reputation can be leveraged in global markets, with individual culture playing a significant role (Agarwal and Osiyevskyy, 2021) Organizational culture, particularly in MSMEs, is crucial for driving employee behavior, engagement, and productivity. Corporate governance principles, including fairness, accountability, responsibility, and transparency, can enhance employee commitment, with corporate reputation and organizational trust mediating this relationship (Yeo et al., 2023). The importance of reputation as an intellectual asset for company competitiveness is also highlighted (Krstić et al., 2021).

Building trust among customers is the foundation for growth. Positive brand reputation fosters customer loyalty, leading to repeat business and increased customer lifetime value. In the service sector, brand reputation plays a key role in fostering customer loyalty, with brand trust being a significant factor in this relationship (Gökerik, 2024). Brand credibility further enhances customer loyalty by reducing switching behaviors and increasing word-of-mouth (Tabelessy, 2025). Trust is also a critical factor in relationship marketing, influencing service performance and fostering commitment. Loyal customers become brand advocates, promoting the MSME through positive word-of-mouth. This organic customer acquisition reduces marketing costs while attracting new customers who trust the brand reputation (Gowthaman and Rau, 2010). A strong brand reputation, as evidenced by a positive corporate image, can lead to a significant market-value premium, superior financial performance, and lower cost of capital (Smith et al., 2010). This is due to the ability of strong brands to command higher prices, create a loyal customer base, and maintain brand stability (Hoeffler and Keller, 2003). Furthermore, firms with good reputations are better able to sustain superior profit outcomes over time (Roberts and Dowling, 2002). This is particularly relevant for MSMEs, as they can benefit from higher profit margins and long-term customer loyalty (Tijani et al., 2023).

The digital age presents a unique opportunity for MSMEs to leverage brand storytelling in building and managing their reputation. Social media platforms and online reviews provide powerful tools to connect with customers, share the MSME's brand story, and showcase its values (Ansarin and Ozuem, 2014). By fostering positive online engagement and addressing customer concerns promptly, MSMEs can cultivate a positive brand image and build trust. Thaichon and Quach (2015) highlight the importance of brand reputation in the competitive game, particularly in stable market segments. They further emphasize the role of brand equity in shaping customer expectations, satisfaction, trust, and value, ultimately influencing customer retention. Keller (2009) provides a practical framework for building and managing strong brands, emphasizing the need to understand consumer brand knowledge structures and the use of online, interactive marketing communications. A positive organizational culture, characterized by support and collaboration, is crucial for fostering employee engagement in MSMEs. Engaged employees, who are committed, passionate, and energetic, contribute to reduced turnover, increased productivity, better customer service, and creative problem-solving (Dickson, 2008). This engagement is particularly important in the manufacturing sector, where it can significantly impact employee retention, productivity, and loyalty (Mohanty et al., 2020). Engaged employees are more motivated, take ownership of their work, and are more likely to go the extra mile for the MSME's success. Furthermore, highly engaged employees can lead to a 147% outperformance of competitors, as well as reduced absenteeism and employee turnover (Elliott, 2023).

The integration of digital technologies into business models for sustainability constitutes both a strategic opportunity and increasing necessity. As companies respond to environmental pressures and evolve stakeholder expectations, digital tools offer capabilities that extend beyond mere optimization to encompass business model innovation. This section draws upon two complementary streams of literature: dynamic capabilities theory, which investigates how firms adapt through sensing, seizing, and reconfiguring (Ellström et al. 2022; Qiu et al. 2022; Teece 2007), and sustainable business model innovation (SBMI), which examines how organizations align value creation with environmental and social outcomes (Bocken et al. 2016; Ferlito and Faraci 2022; Geissdoerfer et al. 2017; Plečko and Bradač Hojnik 2024). Dynamic capabilities empower firms to navigate uncertainty by sensing emerging opportunities, seizing them through resource mobilization, and transforming operations to align with environmental priorities (Knoppen and Knight 2022). This capability-based approach is relevant in digital sustainability, where technologies like AI, blockchain, big data analytics, IoT, and additive manufacturing facilitate operational gains and structural transformation (Schöggl et al. 2024). SBMI emphasizes redesigning value propositions, creation mechanisms, and capture to address environmental and social objectives. According to (Evans et al. 2017), SBMI represents a shift from linear resource-intensive models to regenerative stakeholder-oriented configurations with closed-loop systems and shared value creation. Digital technologies enable this process: AI enables predictive personalization and resource allocation, blockchain supports transparency through decentralized verification, and 3D printing enables circular production.

Digital technologies are increasingly positioned as both enablers and accelerators of this sustainability agenda. Their transformative potential lies not only in improving operational efficiencies but also in reshaping entire value chains to align with circular and regenerative economic models (Matarneh et al. 2024). By integrating digital innovations, such as artificial intelligence, blockchain, and the IoT, into business processes, organizations can pursue data-driven sustainability transitions, enhance stakeholder transparency, and unlock new forms of inclusive value creation (Alojail and Khan 2023). As such, the alignment between digital transfor-mation and the UN-SDGs provides a compelling lens through which to evaluate technological investments in business model innovation. Traditionally, initiatives promoting sustainable consumption focused on regulatory compliance, corporate social responsibility (CSR), and consumer behavior modification (Bocken et al. 2016; Geissdoerfer et al. 2017).

Research demonstrates how technologies contribute to SBMI outcomes. Calandra et al. (2023) showed blockchain can support sustainable business model experimentation through traceability across networks. However, these technologies require strategic orchestration, with firms needing integrated technology portfolios aligned with sustainability goals. Integrating dynamic capabilities with SBMI offers a unified perspective: Firms actively design digital sustainability strategies (Rawashdeh et al. 2024; Zhang et al. 2018). Technologies vary in implementation ease and impact: AI and big data enhance efficiency; IoT improves system integration; blockchain and 3D printing support transformation but require greater investment. Understanding this spectrum is crucial for technology sequencing; firms may progress from scalable tools toward disruptive innovations as capabilities mature (Radnejad et al. 2022).The strategic evaluation framework encompasses four dimensions, efficiency, circularity, scalability, and transformative potential—and was developed through a hybrid methodology. These dimensions are informed by the existing conceptual literature on digital sustainability (e.g., Geissdoerfer et al. 2017) and have been refined inductively through the synthesis of empirical findings. Consequently, they serve as an empirically grounded heuristic typology, rather than a purely deductive theoretical construct. While prior studies have proposed frameworks linking digitalization to sustainability—such as (Geissdoerfer et al. 2017) on circular economy implementation, (Bican and Brem 2020) on innovation, or (Lichtenthaler 2021) on “digitainability”—these models focus on individual technologies or specific sectors. The framework developed here adopts a cross-technology, cross-sector perspective integrating the following four dimensions: operational efficiency, circularity enablement, scalability, and transformative potential. It connects these dimensions to dynamic capabilities and SBMI components, offering a comprehensive tool for theory building and managerial applications. This approach enables nuanced comparison of digital technologies in terms of functional roles and strategic relevance across industries.

While important, these approaches face challenges due to supply chain complexity, market inertia, and voluntary action limitations. Digital technologies can facilitate systemic change by enabling firms to reconfigure processes and scale environmental innovations (Moschko and Blažević 2023; Teece 2007). Digital sustainability solution adoption varies across sectors

(Mhlanga and Ndhlovu 2023; Seguí-Mas et al. 2018). Organizations face challenges including infrastructure limitations, regulatory uncertainties, and skill shortages. Technologies like AI and big data provide immediate benefits, while blockchain and 3D printing enable costly transformations. Strategic decision-makers must balance sustainability goals with technological complexity and investment risk.This study examines how firms integrate digital technologies into sustainability strategies across industries. Using comparative case studies, we investigated the following five key technologies in sectors: smart cities, fashion, food retail, consumer goods, and construction. Each sector faces unique environmental pressure and experiments with digital tools. By analyzing these contexts, we aim to identify patterns for managerial decision making. This study had the following three objectives: mapping strategic functions of digital technologies in supporting sustainability goals across efficiency, circularity, and value creation; identifying sector-specific adoption barriers and enablers; and proposing a framework that categorizes digital technologies across efficiency, circularity, scalability, and transformation potential. Our approach builds on dynamic capabilities of Teece (2007) and sustainable business model innovation frameworks of Lüdeke-Freund (2020). These perspectives emphasize firms' need to identify opportunities, capitalize on digital innovations, and reconfigure routines for environmental and competitive outcomes. Digital technologies serve as strategic assets transforming sustainable value creation.

Organizations in all branches of industry feel the pressure to digitize and know that they have to do it quickly before falling behind innovative and digitally focused competitors or brand new market players. (Leipzig, Gamp, Manz, & Schöttle, 2017) An integrated information system is a complete software solution that meets the overall business model of the company, supports and integrates all its organizational parts and business functions, and connects them into business processes within the organization and business processes that connect the company with business partners. An integrated information system is a prerequisite for the success of modern companies that operate based on electronic business models, and which are characterized by digital technologies. The integrated information system supports business processes internally within the organization and business processes externally towards business partners. The most important integrated software solution on which modern companies should base their more effective business is ERP. The application of the ERP system ensures the integrity of information and a unique database available to employees in any organizational part of the company. The integrity, timeliness, and quality of business data is a necessary prerequisite for making adequate and timely decisions and ensuring competitiveness in the modern market.The application of ERP systems requires the company to improve its business processes.This includes a thorough analysis of existing processes and their partial or radical redesign to determine the best way to implement them and achieve significant improvements in process performance measures. These include cost, quality, and speed.

Redesign of business processes, and changes in entire business models are even more necessary in the conditions of developing ERP systems under the influence of modern digital technologies. The era of digital transformation enables small companies to use modern ERP systems, such as the Cloud ERP system and mobile ERP system, although their use has reduced the cost of maintaining servers and other necessary hardware and software infrastructure (Picek, Mijac, & Androcec, 2017). Digitization is part of the great global trend of the fourth industrial revolution (Industrie 4.0), which provides companies with glorious opportunities for business transformation, but carries the risk of their survival if the transformation is not effective. For this reason, organizations in all branches of industry feel the pressure to digitalize before falling behind innovative and digitally oriented competitors or unknown market players (Leipzig et al., 2017). The potential benefits of digitization are multiple, including: increasing sales, improving productivity, fostering innovation in value creation, and creating fresh forms of customer interaction (Matt, Hess, & Benlian, 2015). The term digital transformation encompasses all transformational effects because of the application of new SMACIT) technologies in business. Digital transformation involves the use of modern technology in order to radically improve the performance or achievements of the company.

AI is commonly defined as "…activity devoted to making machines intelligent, and intelligence is that quality that enables an entity to function appropriately and with foresight in its environment" (Nilsson, 2009, 13). Firms are increasingly relying on artificial intelligence (AI), and managers are trying to identify and manage new business models around it (Li, 2010; Wellers et al., 2017). However, in general, large manufacturing firms have failed to create an AI-based growth agenda (Björkdahl, 2020) because of problems integrating AI into the business model (Wuest et al., 2016). Although AI is linked to changes in business models, few studies have investigated whether and how AI leaves its mark. According to Teece (2010, 172), a business model is defined as the "design or architecture of the value creation, delivery, and capture mechanisms" of a firm. Teece (2018) explains that AI is an enabling technology that can be integrated throughout a network of products and systems and can provide a beneficial service for customers in various parts of the value chain. Understandably, the integration of AI innovations may cause what Foss and Saebi (2017, 201) refer to as "designed, novel, non-trivial changes to the key elements of a business model and/or the architecture linking these elements". For example, manufacturing incumbents who have implemented AI innovations have managed to lower maintenance and inspection costs and reduce the number of expensive production stoppages (Björkdahl, 2020). However, such improvements aim for optimization rather than profit-generating opportunities. Further studies are needed in this area (Baines et al, 2009).

A large body of research has highlighted the positive consequences of AI (Björkdahl, 2020). However, it would appear that using AI in transformed business models may not be that straightforward. Even if companies with transformative business models have superior AI solutions, traditional players may still prevail because of a slow market adoption rate, for example. Therefore, incumbents need to transform not only the technology but also the wider network of business relationships. Indeed, manufacturing incumbents typically perform technology development in collaboration with external stakeholders in innovation ecosystems (Adner, 2017; Björkdahl, 2020). During AI development, such collaboration requires the flow of data between stakeholders in the ecosystem. Hence, it is important to study the linkages between AI, business models, and industrial ecosystems. We address this gap in the literature by proposing that the determining impact of AI on business models is best understood from the perspective of industrial ecosystems. Understandably, as value is created in innovation ecosystems (Adner and Kapoor, 2010). Business-model transformation is a matter of inter-firm coordination. Nonetheless, we lack firm-level business-model insights that adopt an ecosystem perspective on AI development/integration in the manufacturing industry (cf. Björkdahl, 2020).

In line with Chen (2014), this study takes a process perspective on change. In this spirit, we first describe a development that is strongly indicative of AI-based business-model innovation. As recommended by Ritter and Letti (2018), we then combine two different streams of literature in order to expand existing knowledge on business-model innovation. We establish a connection between business-ecosystem theory and business-model innovation theory, discussing value-creation, value-delivery, and value-capture mechanisms. We specify our research focus in relation to each business model dimension. This study focuses specifically on manufacturing incumbents. For the most part, we use practical examples to describe the effects of AI because there is scant research connecting AI and business-model innovation in industrial ecosystems. Generally, studies of innovation ecosystem structures are rare, and none of them focuses explicitly on the manufacturing industry (cf. Sant et al., 2020).

The importance of supply chain management, especially in the areas of demand forecasting and inventory management, has necessitated the introduction of digital technologies into the process. Meanwhile, the widespread digitization of demand forecasting and inventory management process, especially by blue-chip companies and manufacturing giants across the world further emphasized the efficacy of internet technologies in aiding business operational efficiency. Machine Learning (ML), Internet of Things (IoT), Artificial Intelligence (AI), Blockchain Technology (BT), and Cloud Computing have been variously adopted across business fields. Their impacts have continually been tested both by qualitative and quantitative studies. Notably, there have been more advantages than the disadvantages of adopting internet technologies in boosting business operation performance. Akinbolajo (2024) asserted that technological optimization of supply chain processes can enhance manufacturing operations in the US. Although there are numerous internet technologies available for adoption in optimizing demand forecasting and

inventory management in businesses, Artificial Intelligence (AI) has become more widely adopted. Numerous definitions and descriptions have been advanced for AI. While there is no consensus on the definitions, all definition and description attempts have a common denominator, that is, the development of a computer-aided robot that performs the task that would ordinarily be carried out by humans. Google (2024) described Artificial Intelligence (AI) as the backbone of innovation in modern computing with its ability to unlock values for businesses and individuals. Google (2024) further defined Artificial Intelligence as “a field of science that is concerned with building computers and machines that can reason, learn, and act in such a way that would normally require human intelligence or that involves data whose scale exceeds what humans can analyze.” Kanade (2022) also defined Artificial Intelligence as “the intelligence of a machine or a computer that enables it to imitate or mimic human capabilities.” While Artificial Intelligence (AI) can perform functions, the objective of this paper is to examine its role in the management of demand forecasting and inventory management in the United States.

Inventory management and demand forecasting are two integral aspects of the supply components in a production process. From the supply of input resources to the production plant, the supply of semi-finished within subsections of the plant, and the finished products to the distribution chains, maintaining a stable and consistent level of inventory is a natural process of sustaining efficiency in production and supply chain. While inventory management manages the supply side of raw materials and finished products, demand forecasting anticipates the level of demand that is to be met by the supply side. Unal et al (2022) describes inventory management as concerning the planning and controlling the inventory of a company to ensure sustainability. Inventory management was defined by Singh and Verma (2018) as a complex process that encompasses the application of management principles in inventory management for the purpose of minimizing investment inventory and balancing the demand and supply side. Although the system of inventory management and demand forecasting plays critical roles in production and distribution of goods, its scope has been broadened and applied in different business areas such as retails, logistics and supply chain management (Beutel-AL, 2012; Varghese et al, 2012; Acosta et al, 2018; Granillo Macias, 2020; Sridhar et al, 2021; Xie, 2021).

Demand is a household name in the field of economics. It is one of the foundational concepts in the free-market system and economics. Demand may be generally defined as the quantity of goods or services that a consumer is willing and able to buy at a prevailing price per unit of time. The concept of demand forecasting speaks to a foreseeable succession of time-variant specific needs in a process, production or distribution. According to Nguyen (2023) and as corroborated by Perera et al (2019) and Abolghasemi (2020), information relating to future demand is critical to the success of business decisions for producers, traders, customers, and policy makers. Gerasymov (2024) described demand forecasting as a process in the supply chain that uses past records alongside modern trends to predict future changes in demand. The author further asserted that an effective supply chain process will resolve problems relating to customer demand.

Lakshmi et al (2023) defined inventory management as “the process of ordering, storing, using, and selling a company's inventory (i.e., raw materials, components, and final good).” Inventory refers to a list of all items that are used in a particular enterprise. In the context of supply chain management, inventory covers all items of raw materials, semi-finished goods, finished products, byproducts, wastes, and other directly or indirectly useful components in business. Many studies have identified the benefits of engaging in inventory management, which includes enhancing operational efficiency, elimination of financial losses and loss of time in processing orders. Inventory management also provides adequate planning of resources and their deployment during business operations. Integration, as a term, has been variously and contextually defined in the literature. Gulledge (2006) describes integration as the process of making separate applications work together, which were not originally designed to do so, by sharing information through some form of interface. Iintegrations in business may be related to aligning different separate operations to interact across levels to produce a common output. Firms have often adopted either vertical or horizontal integration, or both in their operations to enhance business performance, efficiency and profits. In the context of this paper, the infusion of Artificial Intelligence (AI) in the operations of demand forecasting and inventory management is both innovative and necessary for the enhancement of business operations. This integration may be done at the level of individual firms, and it may be done cooperatively by several aligning firms in the same industry.

Artificial Intelligence (AI) is a composite name for several high-level computer systems of machines that performs supersonic functions typical of humans but more efficiently without failing. These systems cover machine learning (ML), deep learning (DL), large language machine (LLM), robotics,natural language processing, predictive analytics, and more. According to Adenekan et al (2024, pp 1610), “AI can analyze large datasets to identify patterns, predict trends, and provide insights that human operators might miss.” The advent of AI has revolutionized business models with lots of socioeconomic benefits.The capabilities of AI position are suitable in aiding the smooth operation of supply chain, which is mostly organized through demand forecasting and management, and inventory management. Despite the ongoing process of adopting AI in specific areas of supply chain operations, its potentials are enormous.

Global supply chains face mounting pressures from economic instability, environmental challenges, and unpredictable disruptions such as pandemics and natural disasters (Ivanov & Dolgui, 2021).The advent of Industry 4.0 has propelled the adoption of digital twin technology, predictive analytics, and sustainable project management as transformative approaches to enhance efficiency and resilience.Digital twin technology offers real-time virtual replicas of supply chain systems, enabling proactive decision-making and rapid response to disruptions (Ivanov & Dolgui, 2021). For example, a digital twin model of a logistics hub can simulate various scenarios to predict bottlenecks and suggest optimization strategies.Predictive analytics complements this technology by processing vast datasets to identify risks, forecast trends, and optimize resource allocation.This integration has proven effective in minimizing operational uncertainties, such as in freight transportation, where predictive models streamline inventory management and delivery schedules (Tavasszy, 2020).These technologies not only bolster economic performance but also support sustainability by reducing inefficiencies and waste.Sustainable project management, underpinned by frameworks like the Triple Bottom Line, ensures that technological advancements align with environmental and social goals.For instance, green logistics initiatives, which integrate circular supply chains, promote reduced emissions and energy use (Liu, et al., 2021). Together, these innovations represent a paradigm shift toward a data-driven, sustainable future, addressing both immediate operational needs and long-term environmental objectives.The seamless integration of these approaches underscores their potential to create resilient supply chains capable of adapting to evolving market demands while prioritizing sustainability and efficiency.Modern global supply chains are increasingly vulnerable to diverse challenges, ranging from geopolitical uncertainties to climate change and technological disruptions (Chopra & Sodhi, 2021). The interconnected nature of supply networks means that a localized disruption, such as a factory closure, can ripple across regions, leading to significant economic and operational consequences.For instance, Traditional supply chain models struggle to adapt to rapid shifts in consumer demand and unexpected shocks. In contrast, innovative technologies such as digital twins and predictive analytics provide the tools to address these issues effectively.Digital twins enable real-time monitoring and scenario analysis, allowing companies to simulate disruptions and preemptively implement mitigation strategies (Ivanov, D., & Dolgui, A. 2021). Similarly, the adoption of circular supply chains, which focus on resource reuse and waste minimization, has gained traction as a sustainable solution to both economic and environmental challenges (Govindan & Bouzon, 2018).The integration of innovative technologies and sustainable practices represents a critical shift towards building supply chains that are not only efficient and resilient but also environmentally conscious, ensuring long-term viability in an unpredictable global landscape.

The integration of digital twin technology and predictive analytics enhances decision-making by providing organizations with comprehensive, data-driven insights. Digital twins simulate real-world systems in real-time, capturing the complexities of physical processes.Predictive analytics complements this by analyzing vast datasets to uncover patterns and forecast future events.Together, these systems empower managers to make informed, proactive decisions that drive efficiency and mitigate risks (Tao et al., 2019). For instance, in warehouse management, an integrated system can use a digital twin to visualize storage capacity and track inventory movement.Predictive analytics can then forecast demand fluctuations based on historical sales data and market trends, enabling dynamic inventory adjustments (Enyejo, et al., 2024).This synergy prevents overstocking or shortages, optimizing both cost and customer satisfaction.Moreover, integrated systems enhance strategic planning by running "what-if" scenarios.Example, digital twins of production lines combined withpredictive models can evaluate the impact of introducing new technologies or adjusting workflows.These insights help managers identify optimal solutions with minimal disruptions. The ability of integrated systems to combine real-time visualization with predictive foresight fosters robust decision-making, positioning organizations to adapt swiftly to market and operational challenges.

The integration of digital twins and predictive analytics into supply chains delivers substantial economic benefits, including significant cost savings and enhanced competitive advantages.By leveraging real-time data and advanced simulations, companies can optimize resource allocation, reduce waste, and streamline operations.For example, predictive analytics enables accurate demand forecasting, minimizing overproduction and associated storage costs, while digital twins provide a virtual environment to test operational adjustments without disrupting physical workflows (Christopher & Holweg, 2011).Cost savings also extend to maintenance activities.Predictive maintenance, driven by data collected through digital twins, identifies potential equipment failures before they occur.This proactive approach reduces downtime and avoids costly emergency repairs, particularly in industries like manufacturing and logistics.Competitive advantages are achieved through improved agility and customer responsiveness (Akindote, et al., 2024).Companies that adopt these technologies can quickly adapt to market changes, manage disruptions effectively, and ensure timely deliveries.For instance, during periods of heightened demand, predictive models allow organizations to adjust inventory levels dynamically, maintaining customer satisfaction while controlling costs (Dubey et al., 2019). Moreover, these technologies enhance decision-making by providing actionable insights, enabling companies to outperform competitors in efficiency and innovation.Organizations that invest in digital twins and predictive analytics not only lower operational expenses but also position themselves as leaders in a highly competitive global marketplace, ensuring long-term profitability and resilience.

Integrating advanced technologies such as digital twins and predictive analytics into supply chain operations faces significant technical and organizational barriers. One of the primary technical challenges is the lack of standardized protocols for data exchange across various systems. This fragmentation inhibits seamless communication between legacy systems and new technologies, limiting the potential of real-time data-driven insights. Luthra and Mangla (2018) suggest adopting interoperable platforms and developing industry-wide standards to address these technical constraints effectively.On the organizational side, the resistance to change often emerges as a key obstacle. Employees accustomed to traditional workflows may resist adopting digital tools due to a lack of familiarity or perceived complexity. Furthermore, organizational inertia, stemming from entrenched hierarchical structures, can delay decision-making processes critical for technology integration. Effective change management strategies, such as comprehensive training programs and leadership-driven initiatives, can mitigate resistance and encourage a culture of innovation.Adopting advanced technologies such as digital twins and predictive analytics significantly enhances supply chain efficiency and resilience by enabling real-time monitoring, precise forecasting, and adaptive strategies.Predictive analytics enhances resilience by using historical and real-time data to anticipate risks such as supplier failures or demand fluctuations.By dynamically adjusting inventory levels and refining distribution strategies, these businesses sustained high levels of customer satisfaction despite unprecedented challenges.Additionally, these technologies improve resource allocation (Ajayi, et al., 2024). Predictive models optimize fuel usage and transportation routes, reducing costs and environmental impact while ensuring timely deliveries (Bashiru, et al., 2024). Digital twins also facilitate equipment maintenance by predicting failures, minimizing downtime and extending machinery lifespan. The integration of these technologies not only strengthens supply chain operations but also fosters a competitive edge, ensuring organizations remain resilient and adaptive in rapidly evolving market conditions.

The adoption of digital twin technology and predictive analytics in supply chains contributes significantly to achieving global sustainability goals by reducing environmental impacts and enhancing social equity.These technologies enable resource optimization, resulting in lower carbon emissions and waste generation.Advancements align directly with the United Nations Sustainable Development Goals (SDGs), particularly SDG 13, which focuses on climate action (Ayoola, et al., 2024). In terms of social impact, these technologies promote fair labor practices and improve working conditions.Predictive analytics facilitates workforce management by ensuring equitable task distribution and identifying areas where automation can reduce physically demanding or hazardous jobs (Akindote, et al., 2024). By enhancing operational efficiency,businesses can allocate more resources to community development projects, aligning with SDG 8, which promotes decent work and economic growth.Moreover, the transparency provided by these technologies fosters accountability in supply chains, enabling organizations to track their environmental and social performance against sustainability benchmarks (Ajayi, et al., 2024).For instance, companies in the apparel industry use these tools to ensure ethical sourcing of materials, thereby addressing labor exploitation concerns.Through the integration of sustainability into technological frameworks, businesses can meet their operational goals while contributing to broader environmental and social objectives (Awotiwon, et al., 2024).This dual focus ensures long-term viability and positions companies as leaders in sustainable innovation.

Predictive analytics facilitates the optimization of resources in supply chain management by leveraging data-driven insights to enhance efficiency and minimize waste.By analyzing large datasets, predictive tools identify trends and inefficiencies, enabling organizations to allocate resources such as inventory, labor, and capital more effectively (Choi et al., 2018).For example, demand forecasting models allow businesses to anticipate seasonal fluctuations, ensuring that inventory levels align with projected sales (Ijiga, et al., 2024). This prevents overstocking, which can lead to excess storage costs, or understocking, which results in missed sales opportunities.In manufacturing, predictive analytics optimizes production schedules by analyzing machine performance and maintenance needs. Real-time monitoring of equipment enables predictive maintenance, reducing downtime and extending the lifespan of machinery. This ensures that production resources are utilized to their fullest potential (Gupta & George, 2016).Furthermore,transportation logistics benefit significantly from predictive analytics. By analyzing traffic patterns, fuel consumption, and delivery timelines, companies can optimize routes to reduce transportation costs and environmental impact. For instance, dynamic routing adjustments based on live traffic data can shorten delivery times while conserving fuel. These applications underscore the transformative role of predictive analytics in resource optimization, driving cost savings, operational efficiency, and sustainability in supply chain operations.

Predictive analytics enhances efficiency and reduces costs by enabling organizations to identify inefficiencies and optimize processes across supply chains.By leveraging advanced algorithms and real-time data, predictive tools can pinpoint operational bottlenecks and suggest cost-effective solutions.For instance, analytics can optimize inventory management by accurately forecasting demand, reducing the need for excess stock while ensuring product availability during peak periods (Waller & Fawcett, 2013).Transportation logistics is another area where predictive analytics delivers cost savings. By analyzing historical and live traffic data, companies can determine the most efficient delivery routes, minimizing fuel consumption and reducing delivery times (Okeke, et al., 2024). For example, dynamic rerouting during peak traffic hours can significantly lower operational expenses and enhance customer satisfaction.In manufacturing, predictive analytics reduces downtime through predictive maintenance, which anticipates equipment failures before they occur (Ijiga, et al., 2024).This ensures uninterrupted production and avoids the high costs associated with unplanned repairs.These applications demonstrate how predictive analytics drives efficiency and cost reduction, solidifying its role as a cornerstone of modern supply chain management.

Green logistics and circular supply chains are pivotal strategies in achieving sustainability within modern supply chain systems.Green logistics emphasizes minimizing environmental impacts through efficient transportation, eco-friendly packaging, and reduced emissions.Companies adopting these practices often implement fuel-efficient delivery routes and optimize warehouse energy use to align with sustainability goals (Sarkis et al., 2011).For instance, firms like DHL have introduced carbon-neutral shipping options, leveraging green logistics to meet customer demand for environmentally responsible services.Circular supply chains, on the other hand, focus on creating closed-loop systems where resources are reused, recycled,or repurposed, reducing waste and conserving raw materials (Ijiga, et al., 2024).This model deviates from the traditional linear supply chain, where products typically end up in landfills post-consumption (Genovese et al., 2017).For example, companies have developed programs to recover and recycle used devices, extracting valuable materials like aluminum and cobalt for reuse in production.These approaches underscore the integration of environmental consciousness into supply chain management, ensuring resource optimization while fostering long-term sustainability. By adopting green logistics and circular supply chain practices, organizations can simultaneously enhance their operational efficiency and contribute to global environmental preservation efforts.Visualization tools such as Tableau further enhance interpretability, providing actionable insights in user-friendly formats (Ijiga, et al., 2024).These tools and techniques empower organizations to integrate data into decision-making processes, driving efficiency, resilience, and innovation in supply chain operations and beyond risk.Predictive analytics is defined as the process of using historical data, statistical algorithms, and machine learning techniques to identify patterns and predict future outcomes. This approach supports data-driven decision-making by enabling organizations to anticipate trends and make informed choices that align with strategic objectives (Chen et al., 2012).Predictive analytics goes beyond descriptive analytics by focusing on the “what might happen” instead of “what has happened.” Several tools are widely used in predictive analytics to facilitate decision-making.

In manufacturing, predictive analytics optimizes production schedules by analyzing machine performance and maintenance needs. Real-time monitoring of equipment enables predictive maintenance, reducing downtime and extending the lifespan of machinery. This ensures that production resources are utilized to their fullest potential (Gupta & George, 2016).Simulation is another critical application, where digital twins create virtual environments to test different scenarios without disrupting physical operations. By simulating demand surges or equipment failures, supply chain managers can assess the potential impact and develop preemptive strategies (Qi & Tao, 2018). This capability is especially valuable in industries like automotive manufacturing, where production delays can lead to significant financial losses.Predictive modeling amplifies the value of monitoring and simulation by using historical and real-time data to forecast future trends and risks.For instance, predictive models can anticipate seasonal demand fluctuations, enabling businesses to optimize inventory and staffing levels. Together, these functionalities empower supply chains to shift from reactive management to proactive decision-making, ensuring operational excellence and adaptability.

A vast amount of work has been done on the application of IoT into the manufacturing context, explaining the role of IoT in analysing and improving the SC through real-time monitoring, predictive maintenance and data analysis novelties. Based on the literature Wan et al. (2016) give a detailed description on IoT applications in manufacturing where IoT technologies improve efficiency and productivity by providing real time data acquisition.They stress on the application of IoT in maintenance and quality checking, showing how the applied IoT can assist in predicting when and equipment is likely to fail and hence necessary steps can be taken to prevent the same from happening besides enhancing the checks on the quality on the products on the production line.Lee, Bagheri, and Kao (2015) discuss on the topic of smart manufacturing, where IoT stands as an important aspect of developing smart systems to interconnect. Issues of applying IoT in their organisations are examined, and they mention various areas of IoT application such as; they highlighted the improvement in the operations of the production lines through IoT, efficient sensors in production, and business decisions that incorporate data gained from IoT, leading to better production outcomes. According to Khan and Salah (2018), there is a detailed explanation of different IoT technologies and how they may be applied in industries. In their survey, the authors list such evident advantages as the improved control of assets, the increased level of process automation, and better supply chain visibility within the company, but also they state some issues concerning data protection and the integration of the iPulse system with legacy systems.Xu, He and Li in their paper on integrating IoT data with cloud based analytics also look at the complementarity of the two technologies (Xu, He, Li, 2014).They introduced policies for the processing and storage of the big data that can be used by manufacturers not only for predicting the failure points of their machines but also for improving their processes.Carvalho et al. (2019) present a SLR about predictive maintenance in manufacturing through IoT support. As you will learn today, It includes matters like condition monitoring, fault detection, and its effect on time and cost of maintenance.Gilchrist (2016) looks at the main problems that manufactures experience when adopting the IoT in smart production environment, and covers aspects such as communication, data integration, skills issues and change management. Wamba et al. (2015) discuss the use of IoT and big data analytics in the manufacturing industry. Their study presents knowledge regarding the analysis of data from IoT and its application for decision-making to increase the supply chain's value and effectiveness.Other works that are related to this research are Monostori et al, (2016) in his paper on the integration of IoT with cyber-physical systems in creating environment of smart manufacture.They stress how CPS improves such aspects as real-time monitoring and decision-making in view of IoT.The research work that deals with the overview of the design principles and security concerns of industrial IoT is made by Madakam, Ramaswamy, and Tripathi in 2015. IoT in manufacturing uses them to stress the requirements of having secure protection layers to safeguard critical information and guarantee the dependability of IoT structures.In their paper, Lee et al. (2013) look into the use of IoT in managing anticipatory maintenance in mechanical structures. Their study proves how firms can save fortunes, and also improve the durability of their equipment through administering IoT based systems that identify decaying conditions of the equipment.Khan et al. (2024) also have continued the series of papers on IoT applications, namely, real-time environment monitoring, health monitoring, business sustainability, and challenges for enterprises.Altogether, these studies underline the IoT's capability to revolutionize the manufacturing supply chain management of products, which presents an essential framework to help manufacturers comprehend IoT's advantages, limitations, and prospects.These elaborate reviews and research findings do suggest that IoT becomes a criterion necessity in today's manufacturing industries and the driver of the supply chain revolution. The following sections will further describe certain characteristics of IoT applications and concerns in the manufacturing context as well as vision towards the future.

Enterprise Resource Planning systems are comprehensive software designed to integrate and manage all organizational business functions.This set of software includes applications for human resources, financial and accounting, sales and distribution, project management, material management, supply chain management (SCM), and quality management.ERP systems are also integrated enterprise resource planning (ERP) systems. Similarly, the concentration of information inside the organization through a centralized database is the primary focus of ERP systems. Information flows between the modules of an ERP system, which are software components of an Information System (IS) that share a central database and contain functionalities for sales and marketing, development and product design, field service, production, inventory control, distribution, process design, management, and procurement industrial facilities management, quality, manufacturing, human resource, finance and accounting, and information services, among other areas of management (Ali & Miller, 2017).ERP systems build productivity by strengthening an organization's capability while creating accurate and timely information across the firm and its supply chain.This can be accomplished by improving an organization's ability to generate this information. Successful adoption of ERP systems can result in lower inventory, a shorter cycle for product creation, improved customer service, increased efficiency (productivity), profitability, and effectiveness through improved customer service.Business organizations are investing in information systems to increase performance, and they are turning to ERP systems to deal with changing environments and overcome the limits of legacy systems.This is done with the benefits and functionalities of these systems in mind.Performance improvements can be attributed to the deployment of the ERP system.These systems brought enormous benefits to organizations, such as increased productivity, improved access to accurate and timely information, enhanced workflow, reduced dependence on paper, knowledge sharing, tight control, and the ability to automate business processes by coordinating and integrating information across departments (Roy et al., 2019). Moreover, these benefits are clear evidence, so larger firms with extensive data are attracted to these solutions.ERP can deliver on its name. Ignore planning and resources as they are insignificant. Keep the enterprise in mind. This is ERP's goal. The goal is to centralize all company departments and functions into a single computer system to meet their unique needs.Creating a software package that caters to the demands of finance, human resources, and warehousing personnel is challenging (Roy et al., 2020). Each department often has a computer system tailored to its specific work processes.ERP integrates all departments into a single software package using a single database, enabling easier information sharing and communication.Proper software installation can benefit firms using this integrated approach (Alhirz & Sajeev, 2015).Example: a customer order.Customer orders typically undergo a paper-based journey through the organization, including keying and rekeying into various departments' computer systems.Ín-basket idling delays and missed orders while keying into several computer systems lead to errors (Maddali et al., 2021). The company needs more visibility into order status due to the inability of the finance department to access the warehouse's computer system to verify shipment.ERP replaces separate computer systems in finance, HR, production, and warehouse with a unified software program with modules that closely resemble the prior systems.Finance, manufacturing, and warehouse software are now integrated, allowing finance to check warehouse software for order status. ERP software from most suppliers allows for module installation without purchasing the entire package. Many firms install ERP finance or HR modules and abandon the rest.

The best chance enterprise resource planning (ERP) has of showing its worth is if it can serve as a battering ram to improve how our firm takes an order from a customer and converts it into an invoice and revenue.This process is more often known as the order fulfillment process.ERP is frequently called "back-office software" for this exact reason. It does not handle the up-front selling process(although most ERP companies have lately introduced CRM software to do this); instead, it takes a client order and gives a software road map for automating the various processes to complete it.This procedure is known as an enterprise resource planning system (ERP).When a customer service representative enters an order for a customer into an enterprise resource planning (ERP) system, the representative has access to all of the information necessary to fulfill the order (for example, the customer's credit rating and order history from the finance module, the company's inventory levels from the warehouse module, and the shipping dock's trucking schedule from the logistics module) (Chauhan & Singh, 2017).People working in these various departments are all privy to the same information and can make changes to it.When one department is finished processing an order, it is automatically sent to the following department via the enterprise resource planning (ERP) system.We need to determine where the order is at any given time to log in to the ERP system to locate it. With any luck, the order process will proceed as follows: a bolt of lightning that travels through the corporation, causing customers' purchases to be fulfilled more quickly and with fewer errors than before (Maddali et al., 2020). ERP can do the same feats of magic on the other central business procedures, such as employee benefits and financial reporting. At the very least, that is the ideal goal of ERP. The reality is significantly more difficult.Let us return for a moment to the messages currently in those inboxes.It is the procedure could have been more effective. However, doing so was easy.The finance department performed its job, the warehouse did its job, and it was not their fault if anything went wrong. If something goes wrong outside the department's walls, somebody else must fix it.No longer, however. Since the implementation of ERP, customer service professionals are no longer merely typists who enter data. Putting a person's name into a computer and then pressing the return key.The ERP screen enables the persons in the business world.It varies depending on the credit rating of the customer obtained from the finance department and the product inventory levels originating from the warehouse. Will the client make a payment? to the minute? When it comes to shipping the order, can we meet the deadline? These are choices that the customer makes. These questions that customer service workers have never been required to address before, and the responses of customers in addition to every other division of the organization.However, the customer is one of many factors here.Service representatives need to get up and start working. People currently working in the warehouse,mainly those keeping inventory in their brains or on scraps of paper, must put that information in a more organized way (Nwankpa et al., 2016).On the web. If they do, customer service representatives will see low inventory levels displayed on their screens and inform the clients that the requested item is unavailable. Accountability and responsibility are both required.Moreover, communication has never been put to the test in a manner like this one.People have a natural aversion to change, yet ERP forces them to alter how they perform their tasks.That is why it is so difficult to quantify the benefits of ERP.The hardware is significantly more crucial than the software.Alterations made to the standard operating procedures of businesses. If we employ ERP to enhance, we will benefit from the software in how our people take orders, create goods, ship them, and bill for them.Those are all things that our people do. If all we do is install the software without making any adjustments to ys that people carry out their responsibilities, we could not see any value in it at all; in fact, the brand-new software may make us move more slowly by merely installing fresh software in place of the outdated program that everyone already knew software that no one uses (Kaluvakuri & Lal, 2017).The functionality of ERP software also sets it apart.Despite being organized in functional modules like financial accounting or sales, the primary ERP solutions all take a process-oriented perspective of organizations.Users generally do not need to realize which practical module they are in because typical business procedures are supported seamlessly across functions (Garg & Khurana, 2017).

Enterprise resource planning (ERP) systems have the potential to become the repository for a client order from the time a customer service representative accepts it until the loading dock ships the item and finance sends an invoice for it.Companies can more easily maintain track of orders and coordinate manufacturing, inventory, and shipping across many different sites all at the same time if the information necessary to do so is contained within a single software system rather than being dispersed across a large number of distinct software systems that are unable to communicate with one another (Kaluvakuri & Amin, 2018). Manufacturing organizations, particularly those with an appetite for mergers and acquisitions, frequently discover that numerous business units across the company create the same widget using different processes and computer systems (Kaluvakuri, 2022).This is especially true in companies with an appetite for mergers and acquisitions. ERP software typically comes packaged with several pre-defined ways for automating various stages of the production process.Time may be saved, productivity can be increased, and the number of employees needed can be decreased if those procedures are standardized, and a single integrated computer system is used.

ERP) software systems are nothing more than representations of the standard business practices employed by most organizations.While most packages contain everything imaginable, every sector has its peculiarities that set it apart.The vast majority of ERP systems were developed by discrete manufacturing companies (companies that make physical things that can be counted), which instantly left all process manufacturers (companies that measure their products by flow rather than individual units) in the lurch (Maddali et al., 2018).Integrating ERP packages with other company software requires case-by-case testing, which is typically underestimated.A manufacturing company may use add-on applications ranging from e-commerce and supply chain to sales tax computation and bar coding. All need ERP integration. We are better off with integrated ERP. add-ons. Expect unpleasant results if we construct links manually. As with training, ERP integration testing must be process-oriented.Veterans advise against entering false data and transferring it between applications. Execute a purchase order through the system, from order entry to delivery and payment receipt. Preferred with employee participation who will eventually do those occupations (Mustafa & Abbas, 2017).

Data-driven business optimization is an emerging approach that leverages data analytics and technological advancements to improve organizational efficiency, reduce operational costs, and foster sustainable growth. At its core, data-driven optimization involves collecting, processing, and analyzing data to gain actionable insights that inform strategic decisions and operational improvements (Ohakawa, et al., 2024, Okeke, Bakare & Achumie, 2024, Olorunyomi, et al., 2024). For U.S. small enterprises, which often operate within resource-constrained environments, adopting data-driven methods can be transformative, providing opportunities to enhance competitiveness and resilience.Data-driven optimization encompasses a broad range of activities and tools aimed at extracting value from data.This includes integrating structured and unstructured data from internal sources such as sales, inventory, and customer records, as well as external sources like market trends and competitor analysis. By applying advanced analytical techniques, such as predictive modeling and machine learning, small enterprises can uncover patterns, forecast future trends, and identify inefficiencies that would otherwise remain hidden (Achumie, Bakare & Okeke, 2024, Ezeafulukwe, et al., 2024, Gil-Ozoudeh, et al., 2023). This capability enables small businesses to transition from reactive to proactive decision-making, empowering them to anticipate challenges and capitalize on emerging opportunities.The benefits of data-driven optimization for small enterprises are multifaceted. One of the most significant advantages is enhanced operational efficiency. Through data analytics, businesses can identify bottlenecks in workflows, streamline processes, and optimize resource allocation, leading to cost savings and increased productivity. For example, by analyzing sales data, small retailers can optimize inventory levels, ensuring that high-demand products are consistently stocked while reducing overstock of less popular items. Similarly, manufacturers can use predictive maintenance tools to identify potential equipment failures before they occur, minimizing downtime and avoiding costly repairs.In addition to operational efficiency, data-driven approaches provide small enterprises with a strategic edge in navigating competitive markets. By leveraging insights from customer data, businesses can develop personalized marketing strategies that enhance customer engagement and loyalty (Adeyemi, et al., 2024, Ezeafulukwe, et al., 2024, Gil-Ozoudeh, et al., 2024).

Understanding customer preferences and behavior allows for the creation of tailored product offerings and targeted promotions, which can drive revenue growth and improve market positioning. Furthermore, small enterprises can use competitor analysis to benchmark their performance, identify gaps, and implement strategies that differentiate them from larger competitors.The transformative potential of data-driven business optimization is closely tied to the role of data analytics in driving business transformation. Data analytics serves as the backbone of this approach, enabling organizations to process vast amounts of data and distill it into meaningful insights. One of the key aspects of this process is the generation ofactionable insights that inform decision-making across all levels of the organization. For small enterprises, this capability is particularly valuable as it provides the clarity needed to allocate limited resources effectively and prioritize initiatives that deliver the highest impact.Data analytics also enhances decision-making by reducing reliance on intuition and subjective judgment. Instead, decisions are based on objective data, leading to greater accuracy and consistency. For example, a small service-based business might use analytics to determine the optimal pricing strategy by analyzing customer behavior and market demand (Adewumi, et al., 2024, Ewim, et al., 2024, Gil-Ozoudeh, et al., 2022, Samira, et al., 2024). Similarly, a small e-commerce enterprise could use data to identify the most effective marketing channels, ensuring that advertising budgets are allocated to platforms with the highest return on investment.Beyond decision-making, data analytics fosters adaptability and innovation within small enterprises. In today’s rapidly changing business environment, the ability to adapt to new trends and disruptions is critical for survival. Data analytics equips businesses with the tools needed to monitor market dynamics in real time, enabling them to respond swiftly to changes in customer preferences, competitor strategies, or economic conditions (Ochuba, Adewumi & Olutimehin, 2024, Okeke, Bakare & Achumie, 2024, Tula, et al., 2004). For instance, during periods of market volatility, small enterprises can use analytics to identify emerging trends and adjust their offerings accordingly, ensuring that they remain relevant to their target audience.Innovation is another area where data analytics plays a pivotal role. By uncovering hidden opportunities and facilitating experimentation, analytics can drive the development of new products, services, and business models. For example, a small food and beverage company might use customer feedback data to identify demand for a new product line, while a tech startup could leverage usage data to refine and enhance its software solutions. This iterative process of innovation, informed by data, allows small enterprises to continuously improve and stay ahead of competitors.The integration of data analytics into business operations also promotes a culture of continuous learning and improvement. As small enterprises adopt data-driven practices, they gain a deeper understanding of their operations, customers, and markets. This knowledge serves as a foundation for making incremental improvements and achieving long-term growth

(Adekoya, et al., 2024, Ekpobimi, Kandekere & Fasanmade, 2024, Gil-Ozoudeh, et al., 2024). Moreover, the insights generated through analytics can be shared across the organization, fostering collaboration and aligning teams around common goals.While the benefits of data-driven business optimization are clear, it is important to acknowledge the challenges that small enterprises may face in adopting this approach. Limited financial resources, technological expertise, and access to advanced tools can pose barriers to implementation. However, these challenges can be mitigated through strategic planning and the adoption of scalable solutions. For example, cloud-based analytics platforms offer cost-effective alternatives to traditional data infrastructure, enabling small enterprises to harness the power of data without significant upfront investment. Similarly, partnerships with technology providers and training programs can help build the necessary skills and capabilities within the organization.In conclusion, data-driven business optimization represents a powerful framework for enhancing operational efficiency and strategic growth in U.S. small enterprises. By leveraging data analytics to generate insights, make informed decisions, and drive innovation, small businesses can overcome resource constraints and navigate competitive markets with confidence (Ajiga, et al., 2024, Ekpobimi, Kandekere & Fasanmade, 2024), Gil-Ozoudeh, et al., 2024. This approach not only improves productivity and profitability but also positions small enterprises for long-term success in an increasingly data-driven economy. As technology continues to evolve, the potential for data-driven optimization to transform small enterprises will only grow, offering new opportunities for growth and resilience.

The conceptual framework for data-driven business optimization comprises three critical components: data acquisition and management, analytics and insights generation, and strategic implementation. Together, these components create a systematic approach to leveraging data for enhancing operational efficiency and driving strategic growth in U.S. small enterprises.Data acquisition and management serve as the foundation of the framework, focusing on the collection, storage, and maintenance of data from various sources. Small enterprises can gather data internally from operations, sales, customer interactions, and inventory systems, which provide valuable insights into business performance. External data sources, such as market trends, competitor activities, and customer preferences, further enrich the dataset, enabling a comprehensive understanding of the business environment (Akinsulire, et al., 2024, Ekpobimi, Kandekere &Fasanmade, 2024, Mokogwu, et al., 2024). The integration of internal and external data creates a robust repository that supports more accurate and actionable analysis.Building a reliable data repository requires careful attention to data quality and integrity. Poor-quality data can lead to flawed insights and suboptimal decisions, undermining the optimization process. Small enterprises must prioritize data validation, standardization, and regular updates to ensure that their datasets remain accurate and relevant. Additionally, leveraging scalable data storage solutions, such as cloud-based platforms, can help small enterprises manage their data efficiently while minimizing costs. These platforms also enable secure data access and sharing, fostering collaboration across the organization.Once a robust data repository is established, the next step is analytics and insights generation. This component utilizes advanced tools and techniques, including predictive modeling, machine learning, and data visualization, to uncover meaningful patterns, trends, and inefficiencies. Predictive modeling enables businesses to anticipate future outcomes, such as demand fluctuations or potential risks, allowing for proactive planning. Machine learning algorithms enhance the ability to process large datasets and detect complex patterns that may be missed through manual analysis (Adeyemi, et al., 2024, Ekpobimi, Kandekere & Fasanmade, 2024, Olorunyomi, et al., 2024). Data visualization tools, on the other hand, simplify the interpretation of insights, making it easier for decision-makers to understand and act on the findings.Analytics not only provides a snapshot of current performance but also identifies opportunities for improvement and innovation. For example, a small retail business may analyze customer purchase behavior to determine which products are most popular and adjust its inventory accordingly. Similarly, a manufacturing enterprise could use analytics to identify inefficiencies in production processes and implement changes to improve efficiency. By continuously monitoring and analyzing data, small enterprises can stay agile and responsive to changing market conditions.The final component of the framework is strategic implementation, which involves applying the insights gained through analytics to optimize workflows, reduce costs, and drive growth. This step translates data-driven insights into actionable strategies that align with the organization's goals. For instance, if analytics reveal that a particular marketing campaign is underperforming, the business can reallocate resources to more effective channels. Similarly, insights into operational inefficiencies can lead to process improvements that enhance productivity and reduce waste.Strategic implementation also involves formulating proactive strategies for growth based on identified trends and opportunities. Small enterprises can leverage insights to develop new products or services, enter emerging markets, or tailor their offerings to meet the needs of specific customer segments. For example, a service-oriented business might use customer feedback data to design personalized service packages, while a technology startup could refine its product features based on user behavior analytics (Aminu, et al., 2024, Ekpobimi, Kandekere & Fasanmade, 2024, Nwobodo, Nwaimo & Adegbola, 2024). By aligning strategic initiatives with data-driven insights, small enterprises can maximize their return on investment and strengthen their competitive position.In addition to these applications, strategic implementation requires fostering a culture of data-driven decision-making within the organization. This involves equipping employees with the skills and tools needed to interpret and act on data insights effectively. Training programs, workshops, and user-friendly analytics platforms can empower teams to integrate data-driven practices into their daily workflows. Moreover, fostering collaboration between departments ensures that insights are shared and leveraged across the organization, amplifying the impact of data-driven strategies.Collectively, the components of data acquisition and management, analytics and insights generation, and strategic implementation form an interconnected framework that empowers small enterprises to harness the full potential of their data. By systematically addressing each component, businesses can build a robust foundation for optimizing operations and achieving strategic growth (Adewumi, et al., 2024, Ekpobimi, 2024, Folorunso, et al., 2024, Oyeniran, et al., 2023). This comprehensive approach enables small enterprises to navigate the complexities of today's business landscape with confidence and agility, paving the way for long-term success.

The role of emerging technologies in data-driven business optimization is increasingly crucial for small enterprises aiming to enhance operational efficiency and drive strategic growth. Among the most transformative of these technologies are Artificial Intelligence (AI) and Machine Learning (ML), as well as the Internet of Things (IoT). Together, these technologies provide the tools and capabilities for small businesses to streamline their operations, enhance decision-making, and stay competitive in an increasingly data-driven market.Artificial Intelligence (AI) and Machine Learning (ML) have become pivotal in automating various aspects of business analytics and decision-making. AI refers to the simulation of human intelligence processes by machines, while ML is a subset of AI that focuses on the ability of computers to learn and improve from experience without being explicitly programmed. For small enterprises, AI and ML can dramatically reduce the time and effort required to analyze data and generate insights (Achumie, Bakare & Okeke, 2024, Ebeh, et al., 2024, Folorunso, et al., 2024). Traditionally, businesses relied heavily on human intervention to interpret data, which often led to slower decision-making processes. With AI and ML, businesses can automate these tasks, enabling real-time analysis of large datasets and quick, data-driven decisions. For instance, AI algorithms can sift through customer data to predict purchasing behaviors, automate inventory management, or provide personalized marketing recommendations, all without human oversight.One of the key advantages of incorporating AI and ML into business operations is the enhanced accuracy and scalability they offer. Machine learning algorithms, for example, can analyze vast amounts of data far more efficiently and accurately than human analysts. These technologies can identify subtle patterns and trends in data that might otherwise go unnoticed. For instance, a small manufacturing company could leverage machine learning to detect anomalies in production lines that could indicate potential malfunctions or inefficiencies, allowing them to make adjustments before issues escalate into costly problems (Adewumi, et al., 2024, Ebeh, et al., 2024, Folorunso, et al., 2024, Samira, et al., 2024). Moreover, AI and ML algorithms can continuously improve their accuracy over time by learning from new data, further enhancing decision-making capabilities.Another significant impact of these technologies is their ability to scale operations. As small enterprises grow, managing large volumes of data becomes increasingly complex. AI and ML systems can handle this complexity, enabling businesses to scale their operations without the need for proportional increases in manual labor or resources. By automating data processing and decision-making, small enterprises can expand their reach, improve their services, and adapt quickly to changes in the market, all while maintaining or even increasing efficiency.The Internet of Things (IoT) is another emerging technology that plays a critical role in data-driven business optimization. IoT refers to the network of interconnected devices that can collect and exchange data. For small businesses, IoT offers a vast array of opportunities to enhance operational efficiency through real-time data collection and monitoring. By embedding sensors and connected devices into key business operations, small enterprises can gather live data on everything from inventory levels and machine performance to environmental factors like temperature and humidity. For example, a small logistics company can use IoT sensors to track the condition of shipments in transit, ensuring that products remain within specified temperature ranges and reducing waste or loss due to spoilage.Real-time data collection enables businesses to respond immediately to emerging issues. In the context of retail, small businesses can leverage IoT devices to monitor foot traffic, customer behavior, and product interactions within physical stores. This data can then be analyzed to optimize store layouts, staff allocation, and inventory management, ultimately improving the customer experience and boosting sales (Ajiga, et al., 2024, Ebeh, et al., 2024, Folorunso, et al., 2024, Segun-Falade, et al., 2024). Similarly, manufacturers can use IoT-enabled machinery to monitor the health of equipment in real-time, predicting when maintenance is required and reducing the likelihood of costly downtime. This continuous flow of real-time data fosters a more agile and responsive business environment, where small enterprises can make informed decisions on the fly.The impact of IoT on operational efficiency is profound, as it offers businesses the ability to automate and optimize processes that were previously reliant on manual intervention. For example, IoT can streamline inventory management by providing accurate, up-to-the-minute data on stock levels, automatically triggering restocking orders when inventory falls below a certain threshold. This eliminates the need for periodic stock checks and reduces the risk of human error, ensuring that businesses maintain optimal inventory levels at all times.In addition to operational efficiencies, IoT can also provide valuable insights into customer behavior and preferences. For small enterprises, understanding customer needs and expectations is critical for crafting personalized experiences and staying competitive. With IoT, businesses can track customer interactions with products or services in real time, gaining insights into what products are in demand, which features are most appealing, and how customers are navigating their offerings (Arinze, et al., 2024, Ebeh, et al., 2024, Mokogwu, et al., 2024, Sanyaolu, et al., 2024). This data can then be used to refine marketing strategies, develop new products, and create tailored customer experiences that drive loyalty and satisfaction.The combination of AI, ML, and IoT offers small enterprises the opportunity to enhance both operational efficiency and strategic growth. AI and ML can automate the process of analyzing vast amounts of data, generating accurate insights that drive decision-making. IoT, on the other hand, provides the real-time data necessary for businesses to adapt to changing conditions quickly and effectively. Together, these technologies enable small enterprises to operate more efficiently, respond to market changes with agility, and create new growth opportunities based on actionable insights.However, it is important to note that the adoption of these technologies is not without its challenges. Small businesses may face barriers such as the initial cost of implementing AI, ML, and IoT systems, as well as the need for specialized skills to manage and interpret the data generated by these technologies (Akinsulire, et al., 2024, Ebeh, et al., 2024, Nwaimo, Adegbola & Adegbola, 2024). Additionally, there are concerns about data privacy and security, especially when dealing with large volumes of customer and operational data. Small enterprises must take the necessary steps to protect sensitive information, ensuring that they comply with regulations and industry standards regarding data security.Despite these challenges, the potential benefits of emerging technologies for small enterprises are immense.The integration of AI, ML, and IoT into business operations provides a clear path toward greater operational efficiency, enhanced decision-making, and sustained strategic growth. As technology continues to advance, small enterprises that adopt these technologies will be better positioned to thrive in an increasingly data-driven and competitive marketplace. By embracing these innovations, small businesses can streamline their operations, improve customer experiences, and unlock new avenues for growth, ultimately securing a more competitive and sustainable future.

The implementation of a data-driven business optimization framework in small enterprises presents several challenges that must be addressed to ensure successful integration and long-term growth.While the potential benefits of leveraging data to enhance operational efficiency and strategic growth are clear, small businesses often encounter barriers to adopting emerging technologies, such as a lack of technological expertise, data privacy and security concerns, and the high costs associated with advanced tools and systems.These challenges require effective mitigation strategies to help small businesses overcome the obstacles they face.One of the most significant barriers to the adoption of data-driven optimization is the lack of technological expertise.Small enterprises often operate with limited resources, and investing in new technologies such as Artificial Intelligence (AI), Machine Learning (ML), and the Internet of Things (IoT) requires specialized knowledge and skills (Adeyemi, et al., 2024, Ebeh, et al., 2024, Nwaimo, et al., 2024, Samira, et al., 2024). Many small businesses may not have in-house IT departments or data scientists to manage the implementation of complex data analytics systems, which can create a gap in their ability to effectively utilize the data they collect.Without the proper expertise, small businesses risk either underutilizing or misusing the technology, which can lead to suboptimal results and wasted investments.To mitigate this challenge, small enterprises can pursue capacity building and training initiatives. Investing in the upskilling of existing employees can help bridge the knowledge gap and make them more adept at using emerging technologies.This could involve sending employees to workshops, online courses, or certifications related to data analytics, AI, and ML, thus improving the organization's ability to harness the full potential of data. For example, small businesses could partner with local universities or training providers that offer customized programs tailored to the specific needs of small enterprises (Odunaiya, et al., 2024, Okeke, Bakare & Achumie, 2024, Olorunyomi, et al., 2024).Additionally, businesses could engage external consultants or hire temporary specialists to guide them through the adoption process, ensuring a smoother transition.Another challenge that small enterprises face when adopting a data-driven optimization framework is data privacy and security concerns. As businesses increasingly collect and store sensitive customer and operational data, ensuring that this data is protected from breaches and misuse becomes paramount (Adeyemi, et al., 2024, Bakare, et al., 2024, Folorunso, 2024, Oyeniran, et al., 2023). Small businesses may lack the resources to implement comprehensive data security measures and could be more vulnerable to cyberattacks. Data privacy regulations, such as the General Data Protection Regulation (GDPR) or the California Consumer Privacy Act (CCPA), further complicate the issue, as small businesses must ensure that they are in compliance with these regulations when handling customer data. Failing to do so can lead to legal consequences, financial penalties, and reputational damage.To mitigate these risks, small businesses should adopt a proactive approach to data privacy and security. This includes implementing robust security measures, such as encryption, secure access controls, and regular security audits, to protect sensitive data. Additionally, small enterprises should work with legal and compliance experts to ensure that their data collection and processing practices align with relevant regulations (Adekoya, et al., 2024, Cadet, et al., 2024, Nwaimo, Adegbola & Adegbola, 2024). Small businesses can also invest in cybersecurity insurance, which can provide financial protection in the event of a data breach. It is important for small enterprises to make security and privacy a central part of their data strategy, ensuring that they not only protect customer trust but also avoid potential legal and financial ramifications.The high cost of adopting advanced technologies also presents a significant barrier to data-driven business optimization for small enterprises. AI, ML, and IoT technologies often require significant upfront investment, both in terms of purchasing software and hardware and in terms of the labor required for implementation.Additionally, maintaining and upgrading these technologies can add ongoing costs that many small businesses find difficult to justify, particularly when budgets are already stretched thin. This can discourage small businesses from pursuing data-driven optimization, as the costs may appear prohibitive compared to the potential returns.To address these financial barriers, small businesses can explore cost-effective adoption strategies. One of the most effective strategies is to begin with scalable, cloud-based solutions. Many cloud service providers offer data analytics, AI, and ML tools that are accessible on a subscription basis, which eliminates the need for large upfront investments in software and infrastructure (Adewumi, et al., 2024, Cadet, et al., 2024, Nwobodo, Nwaimo & Adegbola, 2024).These solutions are also scalable, meaning that businesses can start with a basic package and upgrade as their needs grow.Additionally, small businesses can take advantage of open-source tools and software, which provide robust functionalities without the associated licensing fees. By leveraging these low-cost or free resources, small enterprises can access powerful technologies without significant financial risk.Another strategy for mitigating the costs of adopting data-driven business optimization is to start small and focus on specific, high-impact areas of the business.For example, a small retail business might begin by using data analytics to optimize inventory management or improve customer targeting, rather than attempting to overhaul the entire business with AI and IoT systems from the outset. By concentrating on a few key areas, small enterprises can build confidence in their ability to use data effectively and generate a return on investment before expanding the use of these technologies to other parts of the business (Ajiga, et al., 2024, Cadet, et al., 2024, Nwaimo, Adegbola & Adegbola, 2024, Soremekun, et al., 2024).This approach helps businesses manage their budgets while still reaping the benefits of data-driven optimization.Another aspect of cost-effective adoption is the potential for small businesses to collaborate with industry peers or external partners to share the costs of implementing data-driven technologies.For example, small businesses could join industry consortia or business incubators that provide access to shared resources, such as data analytics platforms or training programs.Collaborative efforts can also lead to shared learning opportunities, where small enterprises can learn from one another's experiences in adopting new technologies and optimize their approaches accordingly.Partnerships with universities, technology vendors, or even other small businesses can also provide opportunities for innovation and cost savings through shared technology adoption (Akinsulire, et al., 2024, Bello, et al., 2023, Nwaimo, et al., 2024, Samira, et al., 2024).Finally, small businesses should ensure that they are measuring the return on investment (ROI) of their data-driven optimization initiatives to make sure that they are achieving tangible benefits.By focusing on key performance indicators (KPIs) that align with the company's overall business goals, small businesses can track progress and demonstrate the value of their investments in data technologies.This can help secure ongoing buy-in from stakeholders and provide a clearer path to scaling data-driven optimization efforts in the future.In conclusion, while there are several challenges to adopting a data-driven business optimization framework, including technological expertise gaps, data privacy concerns, and high implementation costs, small enterprises can mitigate these obstacles through strategic approaches (Adeyemi, et al., 2024, Bakare, et al., 2024, Folorunso, 2024, Oyeniran, et al., 2023).Capacity building, cost-effective technology adoption, and collaboration are key strategies for overcoming these challenges and successfully integrating data-driven practices.By addressing these barriers head-on, small businesses can unlock the full potential of data analytics, AI, ML, and IoT, thereby enhancing their operational efficiency and fostering long-term strategic growth.

Data-driven business optimization is increasingly becoming a game changer for small enterprises across various sectors.By leveraging analytics, artificial intelligence (AI), and other emerging technologies, small businesses in the United States are enhancing operational efficiency and driving strategic growth (Akinsulire, et al., 2024, Bakare, et al., 2024, Iwuanyanwu, et al., 2024, Oyeniran, et al., 2022). These technologies offer solutions to optimize operations, improve decision-making, and increase profitability. Case studies across different sectors—retail, manufacturing, and services—demonstrate the transformative potential of adopting data-driven strategies.In the retail sector, small businesses face numerous challenges related to inventory management, such as stockouts, overstocking, and the inability to predict demand accurately. These inefficiencies not only lead to lost sales but also tie up capital in excess inventory.One small retail business implemented a data-driven inventory optimization system that integrated historical sales data, market trends, and predictive analytics (Adeyemi, et al., 2024, Bello, et al., 2023, Nwaimo, Adegbola & Adegbola, 2024). By utilizing machine learning algorithms, the business was able to forecast demand more accurately, ensuring that they ordered the right amount of inventory at the right time.This improved stock turnover and reduced the costs associated with storing excess inventory.Furthermore, the integration of data analytics allowed the business to identify underperforming products and optimize their inventory mix. By aligning their inventory levels with customer preferences and seasonal trends, the retailer was able to reduce waste and enhance profitability. The data-driven approach also facilitated real-time inventory tracking, which improved stock replenishment cycles. As a result, the small business was able to reduce operational costs, enhance customer satisfaction through the availability of desired products, and increase its market share.In the manufacturing sector, small enterprises often struggle with maintenance issues that disrupt production schedules and lead to costly downtime.One small manufacturing business, focusing on automotive components, adopted a predictive maintenance system to optimize its operations (Adewusi, et al., 2024, Bello, et al., 2022, Mokogwu, et al., 2024, Oyeniran, et al., 2023). Using data collected from sensors embedded in machinery and equipment, the company was able to predict when machines were likely to fail.This system was based on machine learning algorithms that analyzed historical maintenance data, current operational conditions, and performance metrics.By implementing predictive maintenance, the company was able to reduce unplanned downtime by identifying and addressing potential issues before they became critical.This approach not only saved the business significant costs associated with emergency repairs but also optimized production schedules, ensuring that manufacturing processes were running smoothly and efficiently. Additionally, the predictive maintenance system provided insights into the lifespan of components, enabling the business to plan for replacements or upgrades in advance (Ogedengbe, et al., 2024, Okeleke, et al., 2024, Olorunyomi, et al., 2024, Oyeniran, et al., 2024).This proactive approach to maintenance helped the company reduce operational disruptions, improve equipment lifespan, and increase overall production efficiency.The service sector is another area where small enterprises have successfully leveraged data-driven optimization to enhance their competitive edge.A small customer service company specializing in telecommunications was facing challenges related to customer satisfaction, particularly regarding response times and service quality.By adopting data analytics tools, the company was able to gain deeper insights into customer feedback, interactions, and service performance metrics (Akinsulire, et al., 2024, Bello, et al., 2023, Iwuanyanwu, et al., 2024, Oyeniran, et al., 2022). The business utilized sentiment analysis and customer feedback data from various channels, such as social media, call center logs, and surveys.The analysis revealed key patterns in customer complaints, allowing the company to identify pain points in the service delivery process.For example, the data indicated that long wait times were a major factor in customer dissatisfaction. By identifying this trend, the company was able to implement changes in staffing levels during peak hours and optimize scheduling to ensure faster response times.In addition, data analytics enabled the company to personalize its customer service offerings by identifying customers' unique needs and preferences.This led to more targeted communications, improving overall customer experience and increasing retention rates.By continuously monitoring and analyzing customer feedback, the company was able to fine-tune its service offerings and proactively address issues before they escalated, ultimately leading to higher customer satisfaction and loyalty (Adewumi, et al., 2024, Bello, et al., 2023, Iwuanyanwu, et al., 2022, Oyeniran, et al., 2023).In another case within the service sector, a small hotel chain in the United States utilized data analytics to optimize its operations and improve guest satisfaction.The business integrated data from various sources, such as guest feedback surveys, booking patterns, and online reviews, to gain insights into guest preferences and identify areas for improvement.Using this information, the hotel was able to adjust its pricing strategies, offer personalized discounts, and customize guest experiences, resulting in increased bookings and positive reviews. Furthermore, data analytics enabled the hotel chain to streamline its operations by predicting peak demand periods, optimizing staff schedules, and reducing waste.This data-driven approach helped the hotel reduce operational costs, improve service delivery, and increase profitability. As a result, the business saw an increase in repeat customers and improved its market position within a competitive industry (Ajiga, et al., 2024, Bakare, et al., 2024, Iwuanyanwu, et al., 2024, Oyeniran, et al., 2024).In the transportation sector, a small logistics company adopted data-driven optimization to enhance route planning and reduce fuel consumption.By using GPS data, real-time traffic information, and predictive analytics, the company was able to identify the most efficient routes for its fleet of delivery vehicles.The data allowed the company to optimize delivery schedules, reduce fuel consumption, and minimize delays. As a result, the business not only reduced its operational costs but also improved customer satisfaction by offering faster and more reliable delivery services.Data-driven optimization also played a role in streamlining the human resources operations of small enterprises. A small tech firm implemented a performance analytics system that tracked employee productivity, engagement, and retention.By analyzing the data, the company was able to identify high-performing employees and areas where additional training or support was needed.This led to better talent management, reduced turnover, and improved team performance (Aminu, et al., 2024, Bakare, et al., 2024, Iwuanyanwu, et al., 2022, Oyeniran, et al., 2023). In addition, the company utilized predictive analytics to forecast staffing needs based on project timelines and workload forecasts, ensuring that the right people were available at the right time.The case studies from these various sectors highlight the diverse ways in which data-driven business optimization can enhance operational efficiency and drive strategic growth for small enterprises.Retailers can optimize inventory management, manufacturers can improve production efficiency through predictive maintenance, and service providers can enhance customer satisfaction by utilizing insights from data analytics. In each case, the integration of data-driven strategies allowed small businesses to address specific pain points, reduce operational costs, and enhance their overall performance.Moreover, these case studies underscore the importance of using data not just for immediate problem-solving but also as a tool for long-term strategic growth.By continually collecting and analyzing data, small enterprises can gain a competitive edge, stay agile in the face of market changes, and better meet customer demands (Arinze, et al., 2024, Bakare, et al., 2024, Iwuanyanwu, et al., 2024. Oyeniran, et al., 2023). As data technologies continue to evolve, the potential for small businesses to enhance their operational efficiency and achieve sustained growth will only increase.Therefore, small enterprises must prioritize the integration of data-driven practices to remain competitive in an increasingly data-centric business environment.In conclusion, small enterprises across the retail, manufacturing, and service sectors are successfully adopting data-driven optimization strategies to improve their operations and accelerate growth. From inventory management to predictive maintenance and customer satisfaction improvements, data analytics offers small businesses a wealth of opportunities to enhance their competitive edge (Olorunyomi, et al., 2024, Onyekwelu, et al., 2024, Oyedokun, 2019, Oyeniran, et al., 2022). As these case studies demonstrate, the effective use of data not only solves immediate operational challenges but also positions small enterprises for long-term success in the marketplace.

2.2 Local Studies

In the Philippine context, According to Dela Cruz and Santos (2022) application-based systems developed in local institutions help improve data management and process automation. Their study also showed the use of modern automation systems automation. Their study also showed the use of modern automation systems. Similarly, Dela Cruz and Santos (2020) examined inventory management practices among small and medium enterprises (SMEs) in the Philippines. The researchers found that effective inventory control helps businesses improve stock monitoring, reduce overstocking and stockouts, and increase profitability. The study emphasized the importance of using inventory management systems, conducting regular audits, and training employees to ensure efficient operations. Overall, the findings showed that proper inventory management significantly enhances business performance in local enterprises.

According to Janice E Carrillo, Michael A Carrilla, (Jan 2024): A company's success or failure is highly dependent on its inventory management. Meeting consumer demand is a critical business function, and inventory is the glue that holds the purchasing, supply, and production processes together. The advantages of having inventory, such as improved customer service and protection from supply risks, are often outweighed by the significant costs of having too much inventory. Not only that, but new research has demonstrated that ineffective inventory management has far-reaching consequences that impact the performance and market value of both firms in the long run. The fundamentals of efficient inventory management are laid forth in this chapter. Along with more contemporary supply chain and e-commerce strategies, we examine traditional methods of inventory management. In addition, we go over some of the technology that really makes inventory management a breeze. The chapter concludes with a summary of the most important managerial issues for inventory management.

Therefore, the study investigated how inventory management affects financial performance. Parilla et al. (2022) assert that one of the crucial management dilemmas for various companies in the Philippines is how they manage their inventory. Poor customer service, dwindling resources that might prohibit things from being sold, and unused gear and equipment are all problems with inventory management. Higher rates of inefficiency characterize this, declining productivity, rising operational expenses, bolstered working capital, significant declines in reaching customer service goals, and a precarious bottom line. Poor stock management puts firms in a logistical and financial bind by reducing their efficiency, negatively impacting their financial performance.

Effective inventory management is an essential part of retail business operations, particularly for mini grocery stores that rely on accurate product tracking and timely restocking. Various studies have emphasized that efficient inventory systems reduce financial losses, improve customer satisfaction, and ensure operational stability. Manual methods of tracking items using paper logs or spreadsheets often lead to data inconsistencies, slow decision-making, and inventory mismanagement.Laudon and Laudon (2020) explained that computer-based information systems can significantly improve efficiency by automating data collection and processing, enabling organizations to make better decisions through accurate and real-time information. In the context of small-scale businesses, automation ensures that routine operations such as stock updates and report generation are streamlined. Similarly, Pressman and Maxim (2020) discussed how systematic software engineering practices enhance the accuracy, usability, and maintainability of business applications, which serve as guiding principles in system development like Sort Out. Data analytics has also been identified as a key factor in modern inventory management. Kumar et al. (2021) highlighted that the integration of analytics into retail systems allows managers to forecast demand, monitor sales trends, and optimize stock levels. Descriptive analytics, in particular, converts collected data into meaningful summaries that support decision-making. Turban, Sharda, and Delen (2018) also stressed that data-driven systems enhance business intelligence by identifying consumption patterns and generating insights for inventory control.The concept of rule-based algorithms has become increasingly relevant in automated systems.

According to Sommerville (2019), rule-based systems operate using logical conditions to execute specific actions, ensuring consistent and reliable results. This principle is used in Sort Out to generate alerts automatically when a product's stock level or expiration date meets predefined thresholds. Local studies in the Philippines also confirm the benefits of digitizing small business operations. Researchers found that automation helps small retailers reduce human error, enhance inventory visibility, and maintain competitiveness in a growing digital marketplace. According to Reyes and Dela Cruz (2020), small retail stores in the Philippines continue to suffer from losses due to manual stock tracking and the absence of expiry monitoring.Their paper advocated for the integration of low-cost digital inventory systems tailored for perishable goods and small business contexts.

Mini grocery stores continue to rely heavily on manual inventory tracking, which often leads to inaccurate stock records, overlooked expirations, and difficulty forecasting product demand. These problems affect store operations and cause financial losses, especially for small retailers with limited manpower. According to Bandara et al. (2019), manual inventory processes create inconsistencies that accumulate over time, making it difficult for small businesses to make data-driven decisions. Similarly, Almeida and Sales (2020) emphasized that human error in recording stocks is one of the main contributors to mismanagement in retail operations.

Digital transformation in micro-businesses has become increasingly necessary. Prasetyo et al. (2021) found that automated systems reduce operational delays and improve accuracy in retail inventory handling. Furthermore, Cruz and Tinio (2018) highlighted that local grocery stores benefit significantly from real-time tracking and automated reporting tools because they streamline routine tasks. In addition, Dizon and Reyes (2020) noted that adopting rule-based systems can enhance decision-making by providing systematic notifications, such as low stock alerts and expiration reminders. Analytics-driven systems have also shown positive impacts on retail performance. Hewamalage et al. (2019) demonstrated that descriptive analytics can support small retailers in understanding sales trends and inventory movement, helping them make more informed restocking decisions. Meanwhile, Morales and Santos (2021) stressed that systems utilizing analytical outputs help store owners anticipate shortages, reduce waste, and maintain optimal stock levels.

2.3 Related Literature

Operational efficiency, customer happiness, and profitability are all impacted by inventory management, which is a fundamental component of supply chain management. This research looks at modern methods of inventory management, specifically at tactics that strike a compromise between operational efficiency and cost-effectiveness. The research's overarching goals are to determine what constitutes optimal inventory management practice, examine how methods like Just-In-Time (JIT), Economic Order Quantity (EOQ), and ABC analysis affect company performance, and draw attention to the ways in which technology might enhance inventory control.

The research assesses the efficacy of inventory management in lowering costs, minimizing stockouts, and optimizing resource allocation through the use of case studies, data analysis, and interviews with industry experts. Automation, realtime inventory tracking, and predictive analytics are highlighted as key components of the modern inventory system. Inventory management is the process by which a business keeps track of its stock, including production, ordering, storage, and sales. The business is responsible for managing the warehousing and processing of all materials, from raw materials to finished products. A company's ability to maintain profitability, respond to trends, avoid supply chain management failures, and remain organized is proportional to the quality of its inventory management.

According to Pressman (2019), a proper software engineering process is essential to ensure the quality and reliability of a system. Apart from that, studies also show that the use of automation helps in improving data accuracy, system security, and operational efficiency of organizations.It is critical for supply chain partners to enhance cooperative relationships, decrease inventory, increase forecasting accuracy, and supply chain efficiency through the use of CPFR, a collaborative method for managing inventory in the supply chain. A definition of CPFR, an explanation of the CPFR method, examples of advantages, challenges to implementation, and solutions to these problems are all included in the article(Pei Zhang, A Qian Yuan, (Aug 2014).

According to Dr. Nivetha P , Mr. Abimanyu, (Dec 2024) Businesses cannot afford to waste money, enhance operations, or lose consumers due to ineffective inventory management. Automation, artificial intelligence (AI), and data analysis have replaced antiquated manual processes as the backbone of current commercial inventory systems. Inventory management systems have grown in significance and influence over the years, and this research delves into all of that and more. As an added bonus, it explains how these technologies help businesses remain competitive while also discussing the pros and cons of utilizing them. Management of inventory, company operations, optimization of resources, contemporary business, automation, artificial intelligence (AI), and customer happiness are all keywords.

The inventory tracking system also known as the inventory monitoring system, according to Zoho Corp (2024), assists you in keeping an eye on the whereabouts and movements of your company's inventory. The majority of inventory tracking systems are a component of inventory management software, such as Zoho Inventory, while there are standalone systems designed for this purpose. On the other hand, according to Nicole(), inventory tracking is the act of keeping track of all the goods and SKUs that you own, are transporting to and from your warehouses, and the amounts that are available everywhere. Maintaining inventory levels is essential for warehouse and order fulfillment processes. Inventory is tracked using two primary methods: digital systems and manual systems.There are two main systems of tracking inventory which are a manual-based system and a digital-based system. The manual based system involves the use of pen and paper to physically count, trace, and check inventories from time to time be it periodic, perpetual, or otherwise, whereas, the digital-based system deals with the use of software or advanced inventory management systems to count, trace and check inventories from time to time. Adam (2024) stated that organizations typically maintain sophisticated inventory management systems capable of tracking real-time inventory levels popular POS systems like Square, Vend or Lightspeed offer inventory systems that let you do a whole lot more.

Businesses need to manage their inventories well to succeed. By carefully planning and controlling their inventory, organizations may optimize overall operational efficiency and accomplish particular goals by maintaining ideal inventory levels. By focusing on streamlining and standardizing inventory management procedures, it increases the efficiency of the company's inventory (Hänninen, 2024). According to Mamuda and Adamu (2023), SMEs must use efficient inventory management techniques to improve performance and boost their competitiveness. An organization's performance may be attributed to the reduction of operational issues through efficient inventory management. Despite the accepted value of inventory management in organizations, many SMEs still find it difficult to set up a thorough and efficient inventory management system (Hänninen, 2024). Several studies have been carried out by scholars in academic on inventory management, prominent among them are Olaide and Omodero (2023), Mamuda & Adamu (2023), Iliemena, Odukoya, and Aniefor (2022), Chizoba, Clara, and Juliet (2022), Eze & Uchenu (2021), Sonko & Akinlabi (2020), Olowolaju and Mogaji (2020), Folajimi et al. (2020) Ugwu and Nwakoby (2020) Ogidiolu, Akinosun, and Ajakore (2019), etc, all these studies and to the best of the researchers' knowledge, did not investigate the inventory management practices with the dimensions of inventory control techniques, inventory storage system, and inventory tracking system specifically in Lagos state, hence the gap this study fulfills. SMEs who fail to practice inventory control techniques may tend to buy and hold unnecessary goods which might affect their performance. Also, SMEs who do not put adequate storage systems in place may lose customers which would influence sales, profits, customer satisfaction, market share, etc. Finally SMEs who do not track their inventory might lead to stock pilferage, buffer stock position, expiry of products, loss of customers, loss in profit etc. Inventory is a crucial asset for many organizations, according to Olaide and Omodero (2023), who referenced Gokhale & Kaloji (2018). This is because inventory is often a substantial asset reported in fiscal statements and provides a source of returns through item sales in the near future. Inventory is also defined as “the stock of any item or resource used in an organization” by Okolocha, Anuri, and Anugwu (2022) citing Davis et al. (2003). Inventories are stockpiles of semi-finished, finished, and raw materials that businesses hold to ensure a smooth production process (Otuya and Eginwin, 2017). Kilonzo et al. (2016) define inventory as idle physical products or stock with a high economic value that organizations hold to package,process, or prepare for sale.

While MSMEs in the Philippines have consistently faced challenges such as limited access to finance, technology, and skills, information gaps, and issues with product quality and marketing have demonstrated resilience by continually generating employment opportunities, adding economic value, and playing a significant role in export activities (Bolido, 2020). The Philippines presents a compelling local case study. Mendoza (2015) found out that while Filipino MSMEs perform reasonably well in terms of activity, their profitability presents challenges. This suggests that access to financial services could be a vital element in improving profit margins within this sector. Additionally, Jou (2022) highlights complexities in local policy, noting that financial considerations often play a role on how local councils support MSME development. Moreover, effective financial management practices further amplify the impact of financial resources on MSME performance. This includes prudent budgeting, cash flow management, cost control measures, and investment strategies aligned with business goals (Das and Mahapatra, 2021). MSMEs that employ sound financial management practices are better equipped to weather economic fluctuations, mitigate financial risks, and capitalize on growth opportunities (Kumar, 2013). Furthermore, sound financial management is crucial for product innovation, which in turn enhances MSME performance (Ferine, 2023). A range of studies has consistently highlighted the critical role of financial resources in driving the performance of MSMEs (Omer et al., 2018; Sarkar, 2016). Beck et al. (2014) found a positive correlation between access to finance, strong financial management.practices, and firm performance in developing countries. Similarly, Berger and Udell’s (2004) study in the United States demonstrated that access to credit positively impacts small business growth.

While this definition may differ from other authors, the management literature predominantly attributes human capital to prior education and relevant experience (Taboada and Moya, 2014; Harris et al., 2012; Chen et al., 2006). Investing in human capital is crucial for companies as it enables them to bolster competitiveness, productivity, and profitability. Every individual brings a unique set of skills, knowledge, and experiences to the table. By prioritizing human capital investment, companies facilitate the homogenization of knowledge, thereby amplifying financial performance and strengthening competitive advantages (Francisca et al., 2023). Employee experience, particularly prior industry experience, plays a crucial role in MSME success. Employees with a strong background in the relevant sector possess a deeper understanding of industry dynamics, challenges, and best practices. This translates to several key benefits for MSMEs. Firstly, experienced employees can meet company expectations more readily. They require less training and onboarding, and can quickly adapt to existing workflows, contributing meaningfully from the outset (Muda and Rahman, 2016). Secondly, their experience equips them with the necessary skills and knowledge to perform their duties effectively, leading to fewer errors and efficient task completion. Finally, industry-specific experience allows employees to anticipate and address client needs with greater accuracy, boosting customer satisfaction and loyalty (Jayabalan et al., 2020).

Organizations are continuously looking for strategies to improve operations and gain competitive advantage. Tracking the performance of maintenance is a key management issue for many organizations. Wilson et al (2000) argue that a properly executed maintenance program is a strategic tool that could ensure continued generation of benefits. TPM and similar strategies promise to improve performance though they require increased commitment to training, resources and integration (Swanson, 2001). A study made by Méchain et al. (2020) found that maintenance of a large fleet in TOCs requires careful planning and resources management. The planning and resources involved largely depend on the maintenance strategy selected. A maintenance strategy is a strategic plan used to cover all aspects of maintenance management, setting the direction of the annual maintenance program, containing firm action plans to achieve a desired outcome for the organisation (Heizer et al., 2015). It can be simplified as a firm plan specifically intended for rolling stock preservation. Research made by Eisenberger and Fink (2017) reported that efficient operation and maintenance needs to be increased due to the development of the industry.

Manufacturers offering product-service solutions on average appear to have lower profits than those that do not (Neely 2008)Tuli, Kohli, and Bharadwaj (2007) find that suppliers and customers indeed see solutions differently: whereas suppliers tend to view solutions as simply a bundle of products and services, customers emphasize the importance of the relational processes of solution design and delivery. Processes can be defined as “the procedures, tasks, mechanisms, activities and interactions that support the co-creation of value” (Payne, Storbacka, and Frow 2008, p. Mo85). These solution processes draw on a set of resources (Ulaga and Reinartz 2011), the tangible and intangible entities available to the firm which enable it to produce a market offering that has value for some market segments (Hunt and Madhavaram 2012); Implicit in our research question is the further requirement to explore what value from solutions comprises. Empirically, value has received less attention than quality. While some prior work examines value in business markets (Ulaga and Eggert 2006), the constructs by which the value of solutions is judged have yet to be explored. Our second objective is therefore to understand what constitutes value for solution customers. In this relational context, this value perception is of interest not just at the moment of exchange but throughout the solution’s relational processes. Customers assess the quality not just of the processes and resources of the supplier but also of customer and joint ones—a significant extension of the quality concept (Golder, Mitra and Moorman 2012) which may apply more broadly in both business and consumer markets. Second, we find that customers link their solution quality perceptions to individual value-in-use constructs as well as collective ones. This significantly extends the dominant view of value in business markets as a function of organizational goals (Ulaga and Eggert 2006);equally, it contrasts in the opposite direction with the dominant typologies of consumer value which relate exclusively to the goals of the individual.

These goals may be preventative, such as avoiding high prices, or promotional, such as looking good to others (Chitturi, Raghunathan, and Mahajan 2008). A goal-based conception of value-in-use hence integrates the notions of benefits and sacrifices, as a sacrifice such as a high price corresponds to low achievement of the goal of minimizing the price paid (Macdonald et al. 2011). Similarly, customers may have a goal to minimize their effort (Mathwick, Malhotra, and Rigdon 2001). With this goal perspective, the notion of trade-offs of benefits against sacrifices is more precisely described as the customer balancing and prioritizing between all goals (Epp and Price 2011). Further work has extended the application of goal theory to suggest that value arises not only through product usage processes but at any point in the customer journey (Lemke, Clark, and Wilson 2011). This has led to the suggestion of such alternative terms as value-in-context (Vargo and Lusch 2016), value-in-social-context (Edvardsson, Tronvoll, and Gruber 2011), and experiential value (Mathwick, Malhotra, and Rigdon 2001). While we adopt the term value-in-use to avoid unnecessary terminological multiplicity, its definition needs to allow for goals to be met at any point in the relational process of a solution; hence our definition of value-in-use provided earlier as all customer perceived consequences arising from a solution that facilitate or hinder achieving the customer's goals. This definition distinguishes value-in-use from value-in-exchange, the product or service attributes promised by the supplier and expected by the customer at the time of purchase (Grönroos and Voima 2013). This does not automatically translate into value-in-use, which is also dependent on other resources, such as a customer's own skills, and other processes, such as peer-to-peer processes (McColl-Kennedy et al. 2012). In this vocabulary, some studies referring simply to value examine value-in-exchange (e.g. Sweeney and Soutar 2001) while others examine value-in-use, such as Sirdeshmukh, Singh, and Sabol (2002)'s study of the value perceived through service delivery.In a solutions context, if value is "realized from integration of resources through activities and interactions with collaborators in the customer's service network" (McColl-Kennedy et al. 2012, p. 1), the question arises whether customer evaluation is focused not just on supplier processes but also on customer ones (Payne, Storbacka, and Frow 2008). This leads us to ask whether these customer processes, too, have a quality—a perceived excellence or superiority.

In response to this deficiency, Aren and Nayman Hamamci suggested a new categorization that includes transaction systems, short-term planning and inventory refilling structures, flow systems design, and network design and planning structures. Adding to Bowersox's original concept, Liu, Qian, and An classified logistics into two broad groups. In the first group, customers' actual physical labor is included to meet their diverse service needs. Logistics, shipping, and customer support will all be a part of this. The second kind includes any information or monetary transaction flows that follow or set off the physical ones. Numerous logistics technologies are used to manage, coordinate, and facilitate the flow of information between logistical functions. A feature of logistics IT, is crucial. One of logistics IT's key functions is to replace stock with data, as pointed out by Oláh, Karmazin, Pető, and Popp. Goods can be tracked in real time with the help of logistic data systems of business contacts if they are equipped with real-time communication capabilities. Partners in business need a unified messaging platform for sharing and transforming data in a variety of organizational contexts and formats. In a dynamic scheduling and routing system, real-time communication channels also allow for schedule changes to be made while vehicles are already on the road. The ability of these systems to communicate in real time is what has made last-minute adjustments to routing and scheduling possible, as well as constant tracking.

According to Di Noi et al. , determining which sourcing method will provide the greatest results requires first learning as much as possible about the raw materials or final goods themselves, as well as the company and nation from which they will be procured. Supply chains need decisions to be made about how to carry out certain functions. The supply or procurement division is responsible for this complex operation, since it should only be carried out by duly authorized employees. Adopting green sourcing choices with critical suppliers is one way to ensure long-term viability of a company's operations. Information is often regarded as the single most important factor shaping the supply chain. It facilitates communication between all parties in the supply chain, reducing friction and making cooperation less intimidating. According to Agarwal, data has been dubbed "the lifeblood of business" because of how crucial it is to making good decisions and acting in the right way. The same authors also stress the need of having easily accessible information inside the supply chain in order to facilitate the planning, implementation, and assessment of critical procedures. It is critical to understand not just price and procurement, but also the other facilitators of the supply chain, such as storage, transportation, and inventories. Suppliers, manufacturers, intermediates, logistics service providers, and end-users may all benefit from the fast, low-cost exchange of information made possible by today's supply chain information systems. Since the lack of control over the dissemination of information inside these supply chains may have disastrous consequences, it is crucial that this aspect of the networks be managed carefully. For instance, a supply chain with unethical participants might lead to the disclosure of confidential information to other parties. Information management's contribution to supply chain efficiency was the subject of Yin and Tian research. Whether a company's focus is on production or service, bringing attention to such data is crucial. The execution of formal confidentiality contracts between and among supply chain participants may be a component of information management techniques. Putting the data provided throughout the supply chain into relevant categories is a great assist in getting the data to the right people at the right time. For example, Chi proposes many types of data, including processing transactions, logistics planning and cooperation, order fulfillment, and delivery coordinating. Next, we'll concentrate on properly routing data for the supply chain's future success. When everyone in the supply chain has access to the same information, risks are mitigated but not eliminated. Equitable sharing of information among supply chain parties improves supply chain performance. For instance, Joireman, Liu, and Kareklas propose that continual communication with the firm's consumers is vital to maximizing green initiative objectives inside the supply chain. According to the theory put forward by Connor a company's ability to survive depends on its ability to capture demand and supply trends using internally designed business information systems.

According to Damoah, the success of the supply chain might depend on how well the pricing strategy accommodates both suppliers and customers. According to the research presented by Peter and Zeisel organizations may reap supply chain advantages by using pricing methods such as regional pricing, mark-up, and bundling techniques. If customers believe the prices

charged by a business are unreasonable, they may go elsewhere for similar products or services, or even switch to other businesses. The supplier, on the other hand, might go out of business if the low costs agreed upon by the purchasing company aren't reviewed on a regular basis. Nonetheless, this will undoubtedly have far-reaching consequences for the supply chain as a whole. In light of these drivers and enablers, it is critical to pinpoint supply chain management procedures that must be taken into account to guarantee efficient supply chain operations. Collaboration, supply- and demand- planning, inventory,
Manufacturing, distribution, and logistics are all on the list of supply-chain management techniques listed by Choe.Modern digital technologies – acronym SMACIT (Social, Mobile, Analytics, Cloud, and Internet of Things [IoT]) – are the generators of digital business transformation today. Companies have been doing their business in the same way for years, while, according to Wade (2015), with various digital technologies, they are facing the inevitable digital transformation of their business. Wade highlights seven categories of content that can be digitally transformed in a company: business model (how the organization earns), organizational structure, people, processes, IT capacities (how information ís managed), offer (what products and services the organization offers), and engagement model (how the organization deals with its clients and other actors). These categories are the most important elements of the organizational value chain related to digital transformation (Wade, 2015).

In this regard, researchers (Dijkman et al., 2015; Kiel et al., 2017) emphasize that product development processes and the creation of IT infrastructure are the most common cost drivers. On the other hand, researchers (Zhou et al., 2015; Cheah & Wang, 2017) have established that implementation of new technologies can lead to the development of more cost-efficient processes. Such development is commonly performed with stakeholders outside the boundaries of the incumbent firms (Sensi Zancul et al., 2016). Indeed, such development is critical to improve the business model (Gauthier et al., 2018). One challenge that needs to be managed when co-creating technology-based processes is to find a balance between the cost of collaboration and the financial benefits (Kuula et al., 2018). Incumbents, therefore, need to create a flexible risk management system that can capture the risk arising from shared development (Dellerman et al., 2017; Ehret, and Wirtz, 2017). Thus, there are substantial cost-saving effects to consider when fine-tuning the business model using AI-based solutions.The other element to consider is the revenue model. Research has established that new technologies such as AI can improve the revenue streams of incumbent firms (Rachinger et al., 2018) and create more transparent supplier/customer relationships (Zhou et al., 2015; Zheng & Wu, 2017). Novel technologies can be used to create more flexible revenue models (Kiel et al., 2017) and promote increased customer satisfaction (Kotarba, 2018). Thus, there are substantial revenue-enhancement effects to consider when fine-tuning the business model through AI-based solutions. In this study, we are primarily concerned about how manufacturing incumbents use AI to reduce the cost structure by using data analytics, improving risk assessment, and creating new revenue streams. A recent study suggests that manufacturing incumbents have used AI to seek cost advantages through efficiency improvements, while initiatives using AI to drive revenue growth have been far fewer (Björkendahl, 2020). For growth-driven innovation, these incumbents must initiate a fuller business model transformation because problems may need to be solved in novel and more creative ways. However, radical transformation exposes them to higher levels of uncertainty. This study will,therefore, investigate how AI enables manufacturing firms to capture value.

2.4 Synthesis of the Reviewed Literature

According to the foreign and local study, inventory management is crucial to a company's success. It can and may be time-consuming, error prone, and inefficient for SMEs with limited resources. Automation can increase inventory management efficiency and accuracy. Manual inventory management can cause time-consuming inventory counts,difficulty tracking inventory levels, inaccuracies, and greater expenses due to overstocking or stockouts. These may reduce production, profitability, and customer satisfaction. efficiency, inventory control, overstocking, stockouts, and data-driven decision-making. High implementation costs, technological complexity, and staff training are other drawbacks. The literature reviewed consistently demonstrates that automation, data analytics, and rule-based processing are fundamental in improving the efficiency and reliability of inventory management systems. Prior research confirms that businesses adopting digital tools experience reduced operational errors, improved decision-making, and better control over product movement and expiration.This combination not only enhances inventory management but also contributes to the growing body of research promoting technology adoption among small and medium enterprises.

Automatic inventory management solutions benefit SMEs the most, according to research. One researcher found that SMEs using an automated inventory management system had better inventory control, cheaper inventory holding costs, and higher revenues. In a manufacturing SME case study, an automated inventory management system reduced lead. According to the study, automated inventory management solutions can benefit SMEs.Firms must assess the costs and advantages of deploying such systems to ensure they meet their needs and goals. Businesses should consider technological complexity and personnel training before deploying automated inventory management solutions.Localized inventory automation represents a distributed approach to inventory management that leverages technology to optimize inventory levels and decisions at multiple locations throughout the supply chain network. This approach moves beyond traditional centralized inventory management models to create more responsive and flexible systems that can adapt to local market conditions while maintaining overall system optimization. The theoretical foundation for localized automation draws from distributed systems theory, control theory, and optimization methods to create coherent frameworks for managing complex inventory networks. The automation component of localized inventory systems involves implementing intelligent systems that can make inventory decisions with minimal human intervention, using algorithms that incorporate real-time data, predictive insights, and business rules to optimize inventory levels continuously. These systems typically employ optimization algorithms, control systems, and decision support tools that can process large amounts of data and make rapid decisions in response to changing conditions.

The automation enables faster response times and more consistent decision-making compared to manual inventory management processes. Network theory provides important insights into the design and operation of localized inventory systems, particularly regarding the optimal distribution of inventory across multiple locations and the coordination mechanisms necessary to maintain system-wide performance. The design of effective localized inventory networks requires careful consideration of trade-offs between local responsiveness and global optimization, inventory holding costs and service levels, and automation benefits and implementation cost. Manufacturing operations in emerging economies have increasingly adopted predictive analytics and localized inventory automation to address challenges related to supplier reliability, demand variability, and infrastructure limitations that are common in these market.Large multinational manufacturers operating in emerging markets have implemented sophisticated demand sensing systems that integrate multiple data sources including point-of-sale data, social media, sentiment, economic indicators to create more accurate demand forecasts for local markets.

Localized inventory automation complements predictive analytics by enabling distributed supply chain architectures that reduce dependence on centralized distribution models and improve responsiveness to local market conditions.This approach involves implementing automated inventory management systems at multiple locations throughout the supply chain, enabling real-time adjustment of inventory levels based on predictive insights and local demand patterns. The combination of these technologies creates supply chain systems that are both intelligent and adaptive, capable of maintaining performance levels despite external disruptions and market volatility.Supply chain resilience has become a critical imperative for businesses operating in emerging economies, where volatility, infrastructure limitations, and resource constraints create unique challenges for traditional supply chain management approaches. This review examines the integration of predictive analytics and localized inventory automation as strategic solutions for enhancing supply chain resilience in these dynamic markets. The research synthesizes current methodologies, implementation frameworks, and performance outcomes across various sectors in emerging economies. Predictive analytics approaches, particularly machine learning algorithms, time series forecasting, and demand sensing technologies, have demonstrated significant potential in addressing the complexity of supply chain environments where traditional reactive models fail to capture market volatility and disruption patterns. Emerging trends indicate growing adoption of hybrid models that combine predictive analytics with localized automation strategies, leading to more adaptive and responsive supply chain architectures.

Several factors have contributed to the growing interest in these advanced supply chain technologies within emerging economies. The increasing availability of digital infrastructure, including mobile networks and cloud computing services, has made it feasible to implement sophisticated analytics and automation systems even in markets with limited traditional IT infrastructure. Additionally, the proliferation of e-commerce and digital payment systems in emerging markets has generated vast amounts of transactional data that can be leveraged for predictive analytics applications. The automation component of localized inventory systems involves implementing intelligent systems that can make inventory decisions with minimal human intervention, using algorithms that incorporate real-time data, predictive insights, and business rules to optimize inventory levels continuously. These systems typically employ optimization algorithms, control systems, and decision support tools that can process large amounts of data and make rapid decisions in response to changing conditions. The automation enables faster response times and more consistent decision-making compared to manual inventory management processes. Over the past few decades, managing inventory effectively is something companies have come to use to cut costs and still ensure adequate stock so as not to run into stock-outs in increasingly complicated supply chains. Traditional techniques used in inventory optimization cannot frequently achieve these multiple goals all at once, more especially with an increase in supply chain size and complexity. To address these difficulties effectively, researchers have applied some advanced approaches, including frameworks with simulation and multi-objective optimization methods. These methods combine complex computational tools with statistical techniques to attain a better balance of key inventory objectives.

Sustainability is an approach to address social, environmental, and economic challenges related to present business activities and needs. Moreover, sustainability is vital to enterprises' business strategy to potentially reduce cost, minimize risks, and new opportunities that promote social good. Thus, more enterprises give attention to sustainability as a holistic approach, taking an account of the three pillars in all aspects of business. However, research regarding the applications of artificial intelligence for sustainability in SMEs and practices are scarce. Also, there is an increasing attention to apply long-term strategies to achieve SDGs, likewise, sustainability practices among SMEs. While there are sustainability efforts by SMEs, the applications of artificial intelligence and extent of practices that contribute to sustainability have been neglected which SMEs contribute significantly. Thus, it is relevant to study the applications of AI for sustainability present in the Philippines SMEs. However, the development and deployment of AI systems also raise significant challenges needing attention to ensure the responsible and ethical use of AI. Some of the most pressing challenges include the potential loss of jobs because of automation, the lack of transparency in AI decision-making and the need for responsible data management. Additionally, there are concerns about the fairness and bias of AI systems, which can perpetuate existing social and economic inequalities. Also, the AI training and deployment may lead to high resource consumption.Likewise, lack of competence, integration of AI in sustainability goals, financial support, infrastructure, and cyberthreats are present challenges of using AI for business to achieve sustainability.

Previous studies suggest the role of AI for SMEssustainability initiatives. SMEs have started automating business processes, develop new business models, analyze risks and safety, targets new customers, improve working capital, material sourcing and logistics, and employee's productivity and behavior related footprint using AI. Likewise, conversational AIs were used to facilitate ordering products, obtain production information, and assist customers with product related feedback and concerns. In terms of project management, AI is used to automate project management activities such as estimation, resource management, KPIs, prediction, experimentation, and security risks in construction. Likewise, AI is used to build recommender systems for adaptive source and inventory, human resource management and logistics and delivery using efficient routing and navigation systems. These AI applications in business were found to contribute to achieving sustainability. On the other hand, the benefits of AI are numerous and far-reaching, and its impact is felt across business areas in SMEs. Some recent findings found that (AI) is transforming various SMEs, leading to improved efficiency, accuracy, and better decision-making capabilities. Also, AI increased revenue by improving employees' productivity through streamlined business processes and efficient use of data. Moreover, AI in

SMEs resulted in lower cost and improved quality of products by understanding real-time product controls.

Recently, SMEs in the Philippines have growing appreciation of innovative technologies that support business and sustainability. For instance, mobile payments have been adopted to provide an efficient payment collection system for SMEs involved in retail and service activities. Likewise, social media is evidently utilized in SMEs to target customer segments and improve customer experience.More so, SMEs were supported by startups with emerging technologies that focuses on addressing problems on transport and logistics, zero-waste products, repurposing, energy-saving solutions, and locally sourcing of eco-friendly raw materials. Thus, SMEs in the Philippines continue to grow by utilizing technologies that result in economic prosperity and environmental sustainability. However, SMEs recognized the challenges dealing with large enterprises on sustainability. For instance, SMEs introduced sustainable practices and initiatives to change customers behavior to a greener lifestyle activity while addressing the market needs. Furthermore, SMEs have developed sustainable practices on greening logistics,information technology (IT) infrastructure, work practices, and waste management.More recently, SMEs consider artificial intelligence to add value to business operations and customer experience in many routine tasks. Hence, artificial intelligence can pave the way to achieve sustainability in SMEs in the Philippines.

This study involves small and medium enterprises with evidence of utilization of artificial intelligence for business applications resulting in sustainability. The study initially reviewed a list of small and medium enterprises that have good innovation adoption rating and supported by differences in select small and medium enterprises in the Philippines. Results show that a few SMEs have AI applications in business to address sustainability. While there are on-going sustainability efforts, most SMEs are in the incremental and situational development levels. This study confirms that insufficient physical and technological infrastructure, availability of data, customers' privacy and security, insufficient legal frameworks, management support, and lack of AI adoption strategy are evident issues and challenges to progress AI for sustainability in SMEs. Thus, this needs much attention to enable more aligned sustainability efforts to take place among SMEs. However, the results of the study cannot be used to generalize all SMEs in the Philippines. As such, future work may focus on conducting a national study to provide comprehensive understanding of AI in SMEs. Likewise, cross-country studies may enrich understanding through comparative analysis of AI presence in SMEs of similar developing countries. On the other hand, AI is more likely to be implemented in several business areas that persuade individuals to shift to greener lifestyle and behavior. Thus, this future gap can be further explored.

Manufacturing has evolved throughout history because of technological revolutions that transformed both production methods and business operations. The first stage of industrial development began with the implementation of mechanized systems and large-scale production methods which later transitioned into automated processes that utilized computer-based manufacturing systems. Information and communication technologies have become globally adopted by manufacturing systems during the past three decades which enables better system connectivity and data sharing together with improved operational control capabilities. The present situation has established core components which enable modern manufacturing systems to function as the main method of industrial development in current times. Production systems which require only limited human oversight to operate because they continuously observe their environment and make decisions based on data analysis of their operations represent the concept of autonomous manufacturing. Conventional automated systems operate according to established rules which autonomous systems use to develop new methods for system advancement. They can respond to variations in material supply and demand patterns together with changes in equipment status and product quality requirements. The shift from automated systems that functioned based on predetermined rules toward intelligent systems which operate independently developed when AI together with machine learning and real-time analytics achieved their current advanced state. The control systems of Industry 4.0 establish connections between physical objects and Industrial Internet of Things (IIoT) systems which handle all data-based decision-making processes that occur in manufacturing operations. This framework establishes smart factories which operate through their networked equipment and system and software applications to realize improved production results and adaptable manufacturing capabilities and enhanced operational durability. The autonomous manufacturing vision achieves its goals through systems which enable machines to make independent decisions based on current circumstances. The market demands autonomous operations because product life cycles shorten and product customization grows and global competition increases. Manufacturers need to adapt rapidly to evolving customer demands while they strive to deliver high-quality products through their implementation of affordable production methods. The autonomous manufacturing system permits organizations to operate their facilities without interruption because its monitoring system predicts future events while its instant optimization functions reduce equipment downtime and operational waste and errors. Modern industrial enterprises need autonomous manufacturing as a strategic requirement because it has developed into an essential component of their operations. Intelligent control systems with advanced analytics functions together with cloud-based infrastructure create operational autonomy which organizations can now use as a business capability. The manufacturing industry requires autonomous manufacturing to support its ongoing digital transformation which serves as the primary development path for this industry.

The Intelligent Manufacturing Execution Systems created advanced shop-floor management systems which exceed traditional MES systems because they use intelligent systems and adaptive capabilities and can make decisions in real time. Intelligent Manufacturing Execution Systems use artificial intelligence and machine learning and real-time analytics to control manufacturing operations through autonomous systems which depend on data analysis instead of traditional Manufacturing Execution Systems which use predefined instructions to oversee production activities. Intelligent Manufacturing Execution Systems use core technology which transforms operational data into usable business intelligence. The system needs real-time production data which intelligent Manufacturing Execution Systems obtain through machine and sensor and operator and enterprise system data collection to analyze changing production conditions. The systems support functions which include adaptive production scheduling and predictive maintenance and automated quality control and anomaly detection capabilities. The manufacturing processes achieve enhanced performance through improved responsiveness and operational efficiency which increases their ability to handle unexpected events. The system with its intelligent manufacturing execution system uses contextual information to make operational decisions about production processes. Intelligent Manufacturing Execution Systems link with Enterprise Resource Planning and Product Lifecycle Management and supply chain systems to establish a comprehensive view of manufacturing processes. The integration enables decision-making based on actual shop-floor conditions and demand forecast information and inventory status and resource availability.

The combination of Intelligent Manufacturing Execution Systems with cloud-native analytics creates a fundamental transformation for monitoring, controlling, and optimizing manufacturing operations. The Intelligent MES system functions as the primary element of the production area because it gathers equipment data and sensor data and workforce details through real-time monitoring, while cloud-native analytics provides the essential computational power needed to analyze this data from various sites. The system establishes an entire manufacturing ecosystem through its two systems which work together to enable automatic decision-making. The integration system transfers shop floor information to cloud platforms which use advanced analytics and machine learning models and pattern analysis to build predictive models and generate usable insights. The intelligent MES system receives the insights back again to initiate automatic processes which include dynamic rescheduling and process parameter changes and predictive maintenance implementation. The closed-loop architecture of manufacturing systems enables machines to acquire knowledge through their past operations which they use to modify their behavior during changing operational circumstances. The integration of these systems establishes improved visibility and coordination throughout all business functions of the organization. The combination of cloud-native analytics with Intelligent MES enables remote monitoring of multiple production facilities through a single system while maintaining operational control through local processing.. The centralized intelligence system which operates together with decentralized execution processes enables manufacturing companies to achieve three operational benefits which include operational efficiency and operational reliability and operational standardization across their various manufacturing sites. Companies can develop manufacturing systems that automatically improve their performance through current manufacturing methods which allow them to shift away from traditional maintenance practices.

The full industrial transformation through Industry 4.0 needs manufacturing systems to change from their existing operation mode into advanced intelligent systems which process data for their functions. The Intelligent Manufacturing Execution Systems (MES) use cloud-native analytical systems to create a new manufacturing system which enables complete manufacturing independence. The Intelligent Manufacturing Execution System (MES) goes beyond traditional factory floor management by combining real-time data processing with artificial intelligence and machine learning for automatic decision-making and dynamic resource management and forecast-driven maintenance and quality control processes. The combination of cloud-native analytics with manufacturing enterprises enables businesses to obtain scalable computing resources and advanced data processing capabilities and smooth connection between their different production facilities. The research study demonstrates how cloud-native analytics platforms deliver operational control through intelligent MES frameworks which create automatic manufacturing systems for enhanced operational efficiency and responsiveness. The research shows how three main elements lead to people adopting Internet of Things technology and digital twin systems and real-time data analytics while protecting data security and enabling different systems to work together and businesses to function at their current level. The combination of intelligent MES systems and cloud-native analytics solutions enables organizations to boost their productivity while decreasing operational expenses through automated systems which improve their operational efficiency.

Autonomous manufacturing systems operate through digital technologies which create essential components needed for their three functions of sensing and analysis and decision-making and execution. The Industrial Internet of Things serves as the foundational element of this ecosystem because it establishes connections between machines and tools and sensors to gather operational data throughout the entire operational period. The IIoT system enables intelligent manufacturing execution systems and cloud-based analytics to achieve precise monitoring of production conditions through its advanced visibility features. Artificial Intelligence and machine learning technologies serve as the core technologies that transform unprocessed information into operational outcomes which organizations use for their work. The system enables predictive maintenance through its ability to diagnose equipment malfunction signs which occur during the initial phases of deterioration and adaptive quality control through its capability to identify production process irregularities and intelligent scheduling through its resource distribution optimization abilities. The use of digital twin technology allows autonomous systems to operate through virtual models of real-world physical assets and their corresponding operational activities. Manufacturers create production simulations through digital models which allow them to test optimization methods while assessing potential operational disruptions without any effects on their actual business activities. Organizations achieve fast decision-making capability through real-time data processing combined with edge computing technology because it enables them to perform analysis at their actual data collection sites. This technology combination establishes the required technological foundation which enables autonomous manufacturing systems to function through intelligent manufacturing execution systems and cloud-based analytical systems.

The systems achieve their goal of unplanned downtime reduction through their real-time data analysis system which operates independently to adjust production settings while they achieve maximum resource efficiency through waste management. The system uses predictive maintenance to boost equipment availability by solving problems before they result in production downtime.' The system offers two primary benefits because it can enhance product quality while it maintains product quality through its standardized production processes. The automated systems function in real time to track product performance while their superior quality control systems detect product anomalies which the system will automatically resolve before they develop into major problems. The process produces better results because it generates fewer defects and requires less rework while still meeting all essential quality control requirements. The system enables manufacturers to create custom products through its dynamic process adaptation feature which maintains efficiency across all products because it employs advanced manufacturing techniques. The system enables organizations to achieve autonomous manufacturing which leads to improved decision-making capabilities.The system enables manufacturers to operate their businesses normally during uncertain times by delivering rapid yet precise solutions to handle disruptions which include both supply.The autonomous manufacturing systems create essential operational advantages for businesses through their development of production systems which allow organizations to track their operational workflows and assess their functional systems. The system generates its main benefit through improved operational efficiency which operates throughout its entire system.

The establishment of autonomous manufacturing systems requires scientific expertise and organizational management and strategic planning to face multiple challenges which impede its development. The main challenge exists because companies need to implement their existing systems with intelligent manufacturing execution systems and cloud platforms which their business operations utilize. Manufacturing facilities face operational challenges and incurring high costs because their equipment and outdated software create problems for sharing data between different systems. System engineers need to develop solutions which will solve a basic technical problem that enables various systems to operate together without interruptions. The organization encounters major challenges because it must implement various requirements which protect both data security and privacy rights. The autonomous manufacturing systems use interconnected systems which depend on cloud-based data processing to create elevated risks of cyber attacks. The organization needs strong cybersecurity systems which meet regulatory requirements to protect production data and intellectual property while sustaining its business operations. The organization needs better security systems to encourage organizations to implement autonomous solutions throughout their operations. The process of implementing changes faces additional obstacles because of existing organizational structures and the needs of employees. Organizations need new data analytics and AI systems along with system integration capabilities because their current staff members lack these skills which are essential for autonomous manufacturing. The process of adopting new technologies becomes difficult for organizations because employees resist changes and the company lacks digital capabilities and the organization needs to spend money to implement new systems. The organization needs a complete execution plan together with employee training and dedicated management support to achieve successful implementation of an autonomous manufacturing system.

Autonomous manufacturing systems create fundamental transformations which affect all aspects of contemporary production system development and operation and enhancement activities. Manufacturing systems in Industry 4.0 environments utilize Intelligent Manufacturing Execution Systems and cloud-native analytics because these systems enable self-monitoring adaptive operations that depend on data to function. Intelligent MES create advanced shop-floor management systems which combine real-time intelligence with contextual decision-making and learning capabilities, while cloud-native analytics deliver the necessary infrastructure to handle large-scale industrial data processing needs. The research shows that intelligent MES systems combined with cloud-native analytics create an entire manufacturing solution which allows data to move between physical machines and analytical systems and return to the operational control room. The system integration allows automated operations to function because it enables the organization to track all maintenance predictions and scheduling modifications and quality control activities throughout the entire organization. The implementation of these goals depends on enabling technologies which include artificial intelligence, machine learning, IIoT, digital twins, and real-time analytics because these technologies enable systems to detect and understand their environment while executing tasks without needing human guidance. The development of autonomous manufacturing systems presents multiple obstacles which need to be overcome. The industry faces major difficulties because organizations must deal with four critical issues, which include connecting old technologies, protecting confidential information, filling employee competency voids, and managing substantial upfront costs. Organizations need to develop a detailed plan which requires them to execute their work in predetermined stages while receiving organizational support and deploying strong cybersecurity measures and conducting ongoing work.

Manufacturing companies require real-time data processing because their systems need automated decision-making capabilities which analytical tools can provide. Modern manufacturing systems create massive data streams which require immediate processing to enable control systems to manage their operating changes. Organizations can observe their system operations through intelligent manufacturing execution systems which they connect to cloud platforms for real-time monitoring and operational efficiency improvements. Manufacturers can process extensive data volumes with quick response capabilities because they use edge computing together with cloud-based analytical solutions. Edge processing enables immediate operational control on the shop floor while cloud analytics provides detailed insights that help organizations achieve sustainable development and conduct assessments across their entire facilities. The intelligent manufacturing execution systems of manufacturing systems together with their advanced analytics capabilities enable them to maintain fast response times and efficient operational processes.

The Industrial Internet of Things functions as the data collection system which supports autonomous manufacturing operations.IIoT establishes connections between machines and sensors and tools and devices throughout the shop floor to execute uninterrupted data collection at high precision. The interconnected systems transmit equipment performance data together with operational condition information and climate element details and manufacturing productivity metrics to create an accurate digital model of actual business operations. The combination of IIoT and intelligent MES systems enables organizations to better track their production processes and evaluate their operational activities. IIoT devices gather data which enables organizations to monitor their systems in real time and detect operational problems at an early stage and resolve these issues with great speed. The shop floor systems use IIoT technology to send data to cloud-native analytics platforms which create enterprise-wide operational insights and coordination. IIoT provides the necessary connectivity and responsiveness that autonomous manufacturing systems need to operate at their required levels of performance. The autonomous manufacturing systems create essential operational advantages for businesses through their development of production systems which allow organizations to track their operational workflows and assess their functional systems. The system generates its main benefit through improved operational efficiency which operates throughout its entire system.The systems achieve their goal of unplanned downtime reduction through their real-time data analysis system which operates independently to adjust production settings while they achieve maximum resource efficiency through waste management.The system uses predictive maintenance to boost equipment availability by solving problems before they result in production downtime.' The system offers two primary benefits because it can enhance product quality while it maintains product quality through its standardized production processes. The automated systems function in real time to track product performance while their superior quality control systems detect product anomalies which the system will automatically resolve before they develop into major problems. The process produces better results because it generates fewer defects and requires less rework while still meeting all essential quality control requirements. The system enables manufacturers to create custom products through its dynamic process adaptation feature which maintains efficiency across all products because it employs advanced manufacturing techniques. The system enables organizations to achieve autonomous manufacturing which leads to improved decision-making capabilities. The system enables manufacturers to operate their businesses normally during uncertain times by delivering rapid yet precise solutions to handle disruptions which include both supply.

TPM and TQM have a common and continuous goal of waste reduction. Some of the common themes of the two value systems include continuous improvement, employee empowerment, process focus, information gathering and analysis, and top management commitment. This implied commitment takes into account the interests of customers, employees, shareholders, competitor and society at large. The two approaches emphasize effective management of primary factors, such as, top management leadership, process management, employee training and empowerment. The result are achieved through secondary benefits, such as, lower costs, improved reputation and market share, increased employee motivation and satisfaction. The objectives of the two philosophies do not explicitly state improvement in profitability, nevertheless, with objective execution they will inevitably follow. One of the major differences between TPM and TQM lies in customer focus. While TQM explicitly emphasizes customer focus, TPM implicitly takes into account the customer dimension through waste reduction, productivity improvement, timeliness and planning of activities, gathering and analyses of data, and improvement of quality. Also, while there are many studies linking TQM programs and organizational performance, its close ally TPM has not received widespread attention. However, TPM is gaining recognition gradually and is spreading around the world. The recognition given to the TPM Excellence Awards, offered by the Japan Institute of Plant Maintenance (JIPM), substantiates its importance. JIPM awards TPM Excellence Awards to plants all over the world for successful TPM implementation. The assessment of applicants for the award is based on the improvements achieved through proper equipment maintenance; increased productivity, elimination of accidents, and creation of favorable work conditions.The relationship between TPM and business performance and show the extent of differences between firms implementing TPM and those without TPM.The study seek to gain insights into the success factors of a TPM program and examine the factors of successful TPM implementation to arrive at a better understanding of the implementation process.

Many manufacturers look to business solutions to provide growth, but success is far from guaranteed, and how solutions can create superior perceived value is not clear. It explores what constitutes value for customers from solutions over time, conceptualized as value-in-use, and how this arises from quality perceptions of the solution’s components. A framework for solution quality and value-in-use is developed through 36 interviews combining repertory grid technique and means-end chains. Significantly extending the extant view of quality as a function of the supplier's products and services, findings show that customers also assess the quality of their own resources and processes, and of the joint resource integration process.Contrasting strongly with prior research, value-in-use corresponds not just to collective, organizational goals but also to individuals' goals. Four moderators of the quality-value relationship demonstrate customer heterogeneity across both firms and roles within what the authors term the usage center. When shifting towards solutions, manufacturers require very different approaches to market research, account management, solution design and quality control, including the need for a value auditing process.

Ubiquity of Information Technology (IT) and internet technologies has boosted and tested logistics. To better control the flow of information, new technologies provide new tools. The use of IT as a productivity tool has the dual benefit of expanding potential and lowering expenses. It is generally agreed that companies can gain a competitive edge through the strategic use of Information Technology (IT) by cutting costs or setting themselves apart in some other way. Thanks to advancements in information technology, logistics are now used by many businesses as a strategic advantage. There are two main lines of inquiry that show the importance of information technology in the logistics process. For starters, there's the JIT logistics data stream. With the advent of the Just-In-Time (JIT) approach to business, the value of Information Technology (IT) in logistics has increased. Companies' increasing reliance on JIT has increased logistics' significance as an established mechanism for on-time, damage-free deliveries with minimal lead times. Many authors in the field of logistics have argued that adopting cutting-edge IT may help businesses become more competitive. There have been little empirical research examining the correlation between logistics information skills and actual logistical proficiency. Our framework in the field of information systems categorizes logistics decisions as sequential, with levels ranging from the tactical to the strategic. It has been established that logistics information systems serve as strategic and operational enablers throughout the firm's supply chain. IT's role in the supply chain has evolved from facilitating implementation and material-handling activities to facilitating decision-making and work scheduling. Using a system of categorization to examine how information technology has modified logistics is a practical and fruitful approach to the study of this topic. Information systems used in logistics have been studied and classified in many ways in the past. Facility placement, transportation, inventory management, information and material flow are the five main components of a logistics system. Facility placement, stock management, order taking, transportation scheduling, warehouse just some of the logistical issues that this program helps with. The facility's location, inventory management, logistics, production planning, and overall physical distribution were also highlighted as key factors in a separate study. Each category was handled as an independent entity rather than as part of a larger system.

RFID is being used by a wide range of manufacturers, including Dell, Seagate, Boeing, and Ford, to keep tabs on their work in progress. An RFID system consists of RFID transponders or tags, RFID antennas for interrogating and communicating with the tags, and RFID software for managing the system's data and interfacing with enterprise applications. RFID has a lot of untapped potential to boost supply chain efficiency and cut down on waste. A good use case for

RFID tags would be the automatic updating of inventory systems as goods are unloaded from trucks and brought into stores. These are some of RFID's many benefits over barcodes: RFID tags can read from farther away, store more information, don't need a clear line of sight between the reader and the tag, and can collect data from a plethora of different sources all at once. Before RFID can be implemented everywhere, some technical and commercial hurdles must be cleared. Interference, security, and accuracy are all examples of technical issues, while cost and a lack of standards are examples of business issues. Managers also face the difficulty of making a business case for RFID adoption to upper management. There are a number of challenges facing early adopters of these technologies, including consumer backlash over perceived invasions of privacy, the inherent unpredictability of the RFID system, and concerns over the system's impact on health, safety, and IT infrastructure.

This study explores how digital technologies support strategic sustainability transitions across the following five industries: smart cities, fashion, food retail, construction, and consumer goods. Building on the dynamic capabilities theory and sustainable business model innovation (SBMI), we develop a cross-sector evaluation framework that assesses five key technologies—AI, IoT, big data, blockchain, and 3D printing—along the following four dimensions: efficiency, circularity, scalability, and transformative potential. Through a qualitative comparative case study grounded in secondary data and digital observations, we reveal distinct adoption patterns, strategic functions, and barriers to implementation. The results show that while AI and big data drive short-term optimization, blockchain and 3D printing enable deeper business model transformation. This study contributes by integrating a digital strategy with sustainability frameworks and offering a decision-support matrix for managers to align technology investments with environmental goals.

Traditional inventory and procurement management approaches, designed primarily for formal sector operations, prove inadequate in addressing the unique characteristics of informal supply chains.These systems typically operate with minimal documentation, rely heavily on personal relationships and trust networks, and function within resource-constrained environments where technological adoption has historically been limited.Research indicates that inefficient inventory management in informal markets leads to stockouts affecting 40-60% of small traders monthly, while poor procurement practices result in cost overruns of 15-25% across various African markets. The emergence of artificial intelligence as a transformative technology presents unprecedented opportunities for modernizing inventory and procurement systems within informal supply chains. AI-driven solutions offer the potential to address longstanding challenges through advanced analytics, predictive modeling, and automated decision-making capabilities that can operate effectively within the constraints of informal market environments. Recent developments in mobile technology, cloud computing, and machine learning algorithms have made sophisticated AI tools increasingly accessible to small-scale operators in developing economies. The integration of AI technologies with existing informal supply chain infrastructure represents a paradigm shift toward more efficient, data-driven operations. Mobile-first AI solutions, designed specifically for resource-constrained environments, enable real-time inventory tracking, demand forecasting, and supplier optimization without requiring significant infrastructure investments. These systems leverage existing mobile networks and simple interfaces to provide powerful analytical capabilities that were previously accessible only to large formal enterprises.

Informal supply chains face numerous operational challenges that limit their efficiency and growth potential. Inventory management difficulties stem from limited storage capacity, inadequate preservation techniques, and the absence of demand forecasting capabilities. Many traders operate with minimal buffer stocks, making them vulnerable to supply disruptions and unable to capitalize on demand fluctuations. The lack of standardized measurement systems and quality control mechanisms further complicate inventory management processes.Procurement challenges in informal supply chains include fragmented supplier networks, limited access to quality products, and the absence of formal contract mechanisms. Price volatility, often driven by seasonal factors and market speculation, creates uncertainty that makes procurement planning difficult. The reliance on cash transactions and limited access to credit facilities constrains purchasing power and limits the ability to take advantage of bulk purchasing opportunities. Information asymmetries represent another significant challenge, with traders often lacking access to real-time market information, price data, and demand trends. This information gap leads to suboptimal decision-making, missed opportunities, and inefficient resource allocation. The absence of standardized communication systems and data sharing mechanisms prevents the development of coordinated responses to market changes and supply disruptions.

Modern AI-enabled inventory management systems for informal supply chains are built on mobile-first architectures that leverage smartphones and basic feature phones as primary interfaces. These systems integrate cloud-based processing capabilities with edge computing solutions to ensure reliable operation in environments with limited connectivity. The architecture incorporates lightweight machine learning models optimized for resource-constrained devices, enabling real-time inventory tracking and analysis without requiring high-end hardware. Data collection mechanisms in these systems utilize multiple input methods including manual entry, barcode scanning, image recognition, and IoT sensors adapted for informal market environments. Natural language processing capabilities enable voice-based interactions in local languages, overcoming literacy barriers that might otherwise limit system adoption. The integration of satellite imagery and geographic information systems provides additional context for inventory management decisions, particularly in rural and remote areas. Machine learning algorithms specifically designed for informal supply chains incorporate uncertainty quantification and robust optimization techniques to handle the inherent variability in these systems. These algorithms adapt to local market conditions, learning from historical patterns while remaining responsive to sudden changes in demand or supply conditions. The use of federated learning approaches enables system improvement across multiple locations while maintaining data privacy and security.

AI-driven demand forecasting systems for informal supply chains utilize ensemble methods that combine multiple data sources and modeling approaches to generate accurate predictions.These systems incorporate traditional sales data with external factors such as weather patterns, seasonal variations, local events, and economic indicators to create comprehensive demand models. The algorithms account for the high variability typical in informal markets while identifying underlying patterns that can inform inventory decisions. Predictive analytics capabilities extend beyond simple demand forecasting to include supplier reliability assessment, price trend analysis, and risk prediction. These systems analyze historical supplier performance, market conditions, and external factors to predict potential supply disruptions and price fluctuations. The integration of social media data and news feeds provides additional context for predictive models, enabling more accurate forecasting of market conditions and consumer behavior. Real-time adjustment mechanisms allow the system to continuously refine predictions based on incoming data and changing conditions. This adaptive approach ensures that inventory recommendations remain relevant and accurate even in rapidly changing market environments. The system provides confidence intervals and uncertainty measures alongside predictions, helping traders make informed decisions about inventory levels and procurement timing.

Integrated payment systems connect with mobile money platforms and digital financial services to facilitate secure and efficient transactions. These systems support various payment methods including mobile money, bank transfers, and digital wallets while maintaining transaction records for accounting and audit purposes. The integration with microfinance institutions and alternative lending platforms provides access to credit for inventory purchases and working capital needs. Smart contract capabilities, implemented through blockchain technology, automate payment processing and ensure contract compliance. These systems reduce transaction costs and eliminate the need for intermediaries while providing transparency and security. The contracts can include quality assurance requirements, delivery terms, and dispute resolution mechanisms tailored to informal supply chain environments. Credit scoring and risk assessment features evaluate trader creditworthiness and payment capacity to enable access to trade finance and supplier credits. These systems analyze transaction history, business performance, and external data sources to generate credit profiles that support financial inclusion. The integration with insurance products provides additional protection for both buyers and suppliers against payment defaults and delivery failures.

Automated inventory optimization systems utilize multi-objective optimization algorithms that balance competing priorities such as stock availability, carrying costs, and working capital requirements. These systems consider the unique constraints of informal supply chains, including limited storage capacity, cash flow restrictions, and the need for product diversity to meet varied customer demands. The optimization algorithms incorporate risk preferences and business objectives specific to individual traders while maintaining overall system efficiency. Dynamic pricing recommendations based on inventory levels, demand patterns, and market conditions help traders maximize profitability while maintaining competitive positioning. The system monitors competitor pricing, seasonal demand variations, and product lifecycle stages to suggest optimal pricing strategies. Integration with mobile money platforms enables automated pricing updates and promotional campaigns based on inventory optimization objectives.Replenishment automation features generate procurement recommendations based on optimized inventory levels, supplier performance, and cash flow constraints. The system identifies optimal order quantities, timing, and supplier selection while considering transportation costs, quality requirements, and payment terms. Safety stock calculations account for demand variability and supply uncertainty specific to informal market conditions.

AI-powered supplier discovery systems leverage network analysis and machine learning to identify potential suppliers within informal supply chains. These systems analyze transaction patterns, geographic proximity, and product availability to map supplier networks and identify new sourcing opportunities. Natural language processing capabilities enable the system to process unstructured information from various sources, including social media, market reports, and trader communications, to maintain comprehensive supplier databases. Supplier evaluation algorithms assess performance across multiple dimensions including reliability, quality, pricing, and delivery performance. These systems incorporate both quantitative metrics and qualitative assessments, using sentiment analysis of trader feedback and reviews to evaluate supplier reputation. The evaluation process accounts for the dynamic nature of informal supply chains, where supplier capabilities and performance can change rapidly based on external factors. Risk assessment models evaluate supplier stability, geographic risk factors, and market concentration to help traders diversify their supply base effectively. These systems identify potential supply chain vulnerabilities and suggest mitigation strategies, including backup supplier identification and supply base optimization. The integration of credit scoring and financial stability assessments provides additional insights into supplier reliability and long-term viability.

Advanced matching algorithms connect buyers and suppliers based on product requirements, location, timing, and other relevant criteria. These systems utilize graph neural networks and collaborative filtering techniques to identify optimal matches while considering preferences, constraints, and historical relationships. The matching process accounts for the informal nature of many supplier relationships, incorporating trust scores and relationship quality metrics alongside traditional business factors.Dynamic pricing mechanisms facilitate negotiations between buyers and suppliers by providing real-time market information and fair price recommendations. The system analyzes historical transactions, market conditions, and product characteristics to suggest pricing ranges that benefit both parties. Automated negotiation features enable preliminary agreements while maintaining human oversight for final decisions. Bulk purchasing coordination features aggregate demand across multiple small traders to achieve better pricing and terms from suppliers. The system identifies opportunities for collaborative procurement while managing logistics and payment coordination. This approach enables small traders to access benefits typically available only to larger buyers while maintaining their independence and flexibility.Comprehensive technology risk management strategies are essential for ensuring the reliability and security of AI-enabled inventory and procurement systems. System redundancy and backup mechanisms protect against hardware failures and data loss while ensuring continuous operation during technical difficulties. Regular security audits and penetration testing identify vulnerabilities and ensure ongoing system security. The implementation of blockchain technology for critical transactions provides additional security and transparency.Data validation and quality assurance processes prevent errors and inconsistencies that could lead to poor decision-making.The systems incorporate multiple validation layers and cross-checking mechanisms to ensure data accuracy. Machine learning models include uncertainty quantification and confidence intervals to help users understand the reliability of predictions and recommendations.Regular model retraining and validation ensure continued accuracy and relevance. Cybersecurity measures protect against various threats including data breaches, fraud, and system manipulation.The systems implement multi-factor authentication, encryption, and secure communication protocols. User education and awareness programs help prevent security breaches caused by human error. Incident response procedures ensure rapid response to security threats and minimize potential damage.

Financial risk management strategies protect both system operators and users from various financial risks associated with AI-enabled inventory and procurement systems. Credit risk assessment and management help prevent defaults and payment failures.The systems implement risk-based pricing and credit limits based on user profiles and transaction history. Integration with insurance products provides additional protection against financial losses. Market risk management features help users navigate price volatility and market fluctuations.The systems provide early warning capabilities for significant price changes and market disruptions. Diversification recommendations help users reduce concentration risk in their supplier base and product portfolio. Hedging strategies and financial instruments can provide additional protection against market risks.Operational risk management addresses risks associated with system failures, human errors, and external disruptions. The systems implement robust backup and recovery procedures to ensure business continuity.Training programs and user support services help prevent errors and ensure proper system use. Clear policies and procedures govern system operation and user responsibilities.

The future of AI-enabled inventory and procurement systems in African informal supply chains will be shaped by several emerging technological trends. Advances in edge computing and 5G networks will enable more sophisticated real-time processing capabilities while reducing latency and improving system responsiveness. The development of more efficient machine learning algorithms specifically designed for resource-constrained environments will enhance system performance without requiring significant hardware upgrades.Internet of Things (IoT) integration will expand beyond basic inventory tracking to include environmental monitoring, quality assessment, and supply chain visibility. Low-cost sensors and smart packaging will provide real-time information about product condition, location, and handling throughout the supply chain. The integration of satellite imagery and remote sensing technologies will provide additional context for inventory management and demand forecasting.Blockchain technology adoption will accelerate as infrastructure develops and costs decrease. Smart contracts will automate more aspects of procurement and payment processes while providing transparency and security. The development of blockchain-based identity systems will enable better supplier verification and credit assessment. Cryptocurrency adoption may provide alternatives to traditional payment systems, particularly for cross-border transactions.

Inventory management is a highly beneficial method of balancing supply with demand as oversupply of goods may necessitate returns or cancellations. These tasks are undertaken at costs, the cumulative effect of such avoidable costs in business may have serious loss consequences on the firm. Managing a firm’s inventory may be undertaken using different techniques. Meanwhile, prior to implementing any inventory management technique, firms often identify their supply sources and the demand channel. Many techniques have been advanced in the academic circle for inventory management. Just-in-time (JIT), Just-in-case stock control, First In, First Out (FIFO), Last In, First Out (LIFO), Weighted average cost (WAC), Economic Order Quantity (EOQ), Cross docking (CD), and Drop shipping are some of them. Each technique has individual costs and benefits, and firms have often adopted them as each aligns with their objectives, type of business partners, and other intrinsic factors. Demand forecasting and inventory management are undertaken using many techniques. In many literatures, demand forecasting is treated as one of the tools used for managing the inventory level in a firm. Although, demand forecasting can serve the purpose of inventory management as emphasized in some literature, it is also capable of playing a role independent of inventory operations. This literature has identified two methods used in supply chain forecasting, the quantitative method, which deals with the use of historical data to predict the future trend or a repetitive pattern, and the qualitative method, which relies on primary data generated as a forecast of the expected demand. However, the adoption of the demand forecasting and inventory management techniques depends on the available data, the specific field of use, among others. Simple moving average (SMA), Autoregressive integrated moving average (ARIMA), Multiple aggregation algorithm (MAPA), Life cycle modelling (LCM), and Adaptive smoothing (AS) are common techniques used for quantitative demand forecasting. In the alternative, market research, panel consensus, historical method, and the Delphi method are the common techniques used for qualitative demand forecasting.

While demand forecasting has been identified as a subset of inventory management by some studies, they are both veritable tools in supply chain operations. If the conclusion drawn by some studies that demand forecasting is a tool for inventory management is valid, then, it would be concluded that the role of artificial intelligence in inventory management is more superior. This means that the level of accuracy that AI can generate will be more beneficial to inventory management processes.Therefore, the need to enhance inventory management processes is at the heart of supply chain management, which has become an indispensable tool in modern business. Oliver Munro, an inventory management specialist, identified six best practices for inventory management: Proper categorization of inventory ,Adopting a sustainable inventory management software ,Benchmarking a safe level for stock ,Performing regular stock taking, Benchmarking the optimal lower and upper bounds for stock level and optimizing inventory control level. The last point is the most critical aspect of infusing AI technologies into the inventory management processes. Optimization is a way of enhancing the efficiency of the process by reducing the likelihood of human errors in the process and relying more on the efficacy of AI robotics and Machine Learning models for accuracy and efficiency. The use of AI in reading barcodes is a deciding factor between the manual spreadsheets process and full automation. With the ability of AI to seamlessly perform supplier management, inventory analytics and reporting, batch tracking and barcode scanning, as well as multichannel order management among others, it can be concluded that with time, business operations will require less humans and more robots. Whilst this outcome will provide for smooth inventory management at the levels of retail, warehouse, ecommerce, and manufacturing, the consequences of such on unemployment and low household incomes may attract the attention of policy makers.Despite numerous benefits adduced to explain the role of Artificial Intelligence (AI) in demand forecasting and inventory management, the integration harbors some inherent challenges, which may undermine its widespread adoption and complete installation. While the list of challenges is seemingly inexhaustive, the following challenges have been identified.

Artificial Intelligence (AI) has emerged as a disruptive force in supply chain management, offering innovative solutions to long-standing challenges. By enabling predictive analytics, automation, and optimization, AI enhances the agility and efficiency of supply chain networks. Unlike traditional methods, AI-driven systems analyse vast datasets in real time, providing actionable insights that improve decision-making and operational performance. One of AI's most transformative applications lies in predictive analytics. Machine learning algorithms process historical and real-time data to forecast demand patterns, helping businesses align production schedules with market needs. For example, AI-powered demand forecasting tools consider variables such as seasonal trends, economic indicators, and consumer behaviour, enabling more accurate planning and reducing the risks of overproduction or stockouts. AI also facilitates automation in supply chain processes,streamlining tasks such as order management, inventory tracking, and logistics coordination. Robotics and autonomous systems supported by AI optimize warehouse operations, improving productivity and reducing labor costs. Moreover, AI-driven optimization tools enhance supply chain efficiency by identifying cost-effective transportation routes and minimizing delivery times. In addition, natural language processing (NLP) technologies enable real-time communication across supply chain networks, fostering collaboration and improving visibility. These advancements make AI a cornerstone of modern supply chain management, addressing complexities with unprecedented precision and speed.

Predictive models analyse multiple variables—such as historical sales data, seasonality, consumer preferences, and external factors like holidays and weather conditions—to forecast demand accurately. For instance, an ML model might identify a trend indicating increased demand for certain products during the holiday season while simultaneously accounting for supply chain capacity. By aligning inventory levels with predicted demand, businesses can avoid overproduction, minimize stockouts, and reduce waste. Retailers such as Amazon and Target use predictive analytics to optimize their inventory management. These tools help them anticipate consumer buying patterns, ensuring that warehouses are stocked with the right products at the right time. For example, predictive systems in e-commerce platforms can trigger automated stock replenishment for fast-moving items based on real-time sales trends.
Predictive analytics plays a crucial role in identifying and mitigating risks that could disrupt supply chains. By analysing patterns in real-time logistics data—such as shipping delays, traffic congestion, or weather forecasts—ML algorithms provide early warnings of potential disruptions. This enables businesses to implement contingency plans before issues escalate.A prominent example is Walmart's use of predictive tools to manage supply chain risks during natural disasters. When hurricanes or snowstorms are forecasted, Walmart's predictive models analyse customer purchasing trends and logistical constraints to pre-stock essential goods in affected regions.This ensures the availability of critical supplies, enhancing both operational resilience and customer satisfaction.

Machine learning models play a crucial role in demand forecasting by processing historical data and identifying patterns. Regression models predict future demand based on relationships between dependent and independent variables, while decision trees and support vector machines (SVMs) classify and analyse large datasets for more granular insights. For example, Amazon utilizes ML algorithms to predict customer purchases based on browsing history, seasonality, and purchasing behaviour, ensuring optimal inventory placement across its vast distribution network. Artificial Intelligence (AI) has revolutionized demand forecasting, transforming it from a manual, reactive process into a dynamic, data-driven capability. By leveraging advanced techniques like machine learning (ML) and neural networks, AI enables businesses to analyse diverse datasets, uncover trends, and make accurate predictions. Unlike traditional forecasting methods that struggle with complex, high-dimensional data, AI excels in integrating multiple variables, ensuring precision and reliability. Neural networks, especially recurrent neural networks (RNNs) and long short-term memory (LSTM) models, further enhance demand forecasting accuracy. These algorithms are particularly effective for analysing time-series data, allowing businesses to predict both long-term trends and short-term fluctuations. For instance, Starbucks uses LSTM models to predict demand for beverages and snacks based on weather patterns, local events, and economic conditions. These predictions enable Starbucks to optimize staffing, procurement, and promotional strategies, ensuring maximum efficiency and customer satisfaction.

Excess inventory is a costly burden for businesses, tying up capital and increasing storage costs. AI tools address this challenge by identifying slow-moving or obsolete items through the analysis of sales patterns, market trends, and product life cycles. These insights enable targeted strategies, such as offering discounts, bundling items, or redistributing products to higher-demand locations. For instance, AI algorithms in an e-commerce platform can analyse customer preferences and suggest price reductions for surplus inventory, driving sales and clearing warehouse space. Reinforcement learning algorithms play a vital role in identifying optimal reorder points. These systems consider demand variability, lead times, and carrying costs to determine when to replenish inventory. Unlike static reorder models, AI algorithms adapt continuously to changes in supply chain conditions, ensuring balanced inventory levels. For example, an AI-powered inventory management system in a retail setting can adjust reorder points based on unexpected demand spikes during a promotional event or delays in supplier shipments. This dynamic adjustment minimizes the risk of stockouts while avoiding unnecessary inventory buildup. AI-driven inventory optimization automates and refines critical inventory processes, addressing inefficiencies inherent in traditional systems. These processes include determining reorder points, calculating safety stock levels, and managing excess inventory. Conventional methods often rely on static rules and manual inputs, which fail to account for dynamic market conditions, resulting in overstocking, stockouts, or misaligned inventory levels. In contrast, AI systems leverage real-time data and predictive analytics to dynamically adjust inventory parameters, optimizing performance. AI also excels in optimizing safety stock levels, which act as a buffer against demand uncertainty and supply chain disruptions. Machine learning models analyse historical demand patterns, supplier reliability metrics, and external factors such as seasonal trends or geopolitical events to calculate the ideal safety stock. By doing so, businesses can reduce excess inventory without compromising service levels, ensuring that products remain available even during unexpected fluctuations in demand.AI tools enable real-time monitoring of supply chain activities, providing end-to-end visibility. When disruptions occur, these tools suggest alternative strategies, such as rerouting shipments or reallocating inventory. For example, Maersk leverages AI to monitor global shipping routes and predict port congestion. This system reduces delays by identifying optimal rerouting options Additionally, AI systems enhance supplier risk management. By analysing historical data on supplier performance, delivery timelines, and financial stability, businesses can assess supplier reliability and diversify sourcing strategies to mitigate risks. AI enhances inventory management by predicting demand surges and ensuring optimal stock levels. Predictive analytics tools identify periods of high demand, enabling proactive inventory replenishment. This approach minimizes stockouts, enhancing customer satisfaction and reducing lost sales. A case study on Amazon revealed a 15% reduction in stockouts through AI-driven demand forecasting.Conversely, AI also reduces excess inventory by identifying slow-moving items and recommending redistribution strategies. For example, Zara employs AI models to predict demand and limit excess inventory, reducing waste and improving profit margins.

Statistical analyses underscore AI's transformative impact on demand accuracy. A study of retail chains using AI-driven forecasting tools reported a 20% improvement in forecast accuracy, leading to a 15% reduction in carrying costs.In the FMCG sector, AI systems implemented by Unilever resulted in a 25% decrease in stockouts and a 10% increase in sales efficiency.By aligning inventory with demand patterns, AI not only optimizes stock levels but also enhances overall operational efficiency, positioning businesses to meet consumer needs dynamically. AI significantly improves demand forecasting by analysing vast datasets, identifying trends, and predicting future demand patterns. Traditional forecasting methods often struggle with demand variability and fail to incorporate external factors like market trends and weather. In contrast, AI models such as neural networks and machine learning algorithms consider these variables, enabling accurate predictions.Machine learning algorithms process historical sales data,customer preferences, and external influences to generate detailed demand forecasts. For instance, Walmart uses AI-powered demand forecasting systems that reduce forecast errors by up to 30%, ensuring better alignment between inventory and demand patterns.The integration of Artificial Intelligence (AI) into supply chain management has revolutionized how businesses operate, offering unprecedented levels of efficiency, accuracy, and resilience. AI-driven predictive supply chain management addresses key challenges, such as demand variability, inventory inefficiencies, and disruptions, by enabling real-time data analysis and informed decision-making.One of the most significant benefits of AI is its ability to enhance demand forecasting. By leveraging machine learning algorithms and neural networks, businesses can predict market trends, align inventory levels, and reduce stockouts and overstocking. For example, AI tools process historical sales data, external variables, and real-time inputs to create precise demand forecasts. This results in improved customer satisfaction, reduced operational costs, and better inventory utilization.AI also drives operational efficiency by automating repetitive tasks and optimizing resource allocation. From warehouse automation to route optimization, AI-powered systems streamline processes, reduce human error, and improve speed and accuracy. In logistics, AI ensures optimal transportation routes, saving time and fuel costs, while robotic systems in warehouses increase productivity and reduce labor costs.Supply chain resilience has also been significantly enhanced through AI’s predictive and prescriptive capabilities. By analysing real-time data from IoT sensors and external sources, AI identifies potential disruptions and offers actionable solutions, such as rerouting shipments or reallocating inventory. This proactive approach ensures continuity during unexpected events, such as natural disasters or market shocks.Moreover, emerging technologies like blockchain, quantum computing, and edge computing further amplify AI's impact. Blockchain enhances transparency and security, quantum computing tackles complex optimization problems, and edge computing enables real-time decision-making. Together, these technologies create a robust, adaptive supply chain ecosystem.In summary, AI is a transformative force in supply chain management, addressing inefficiencies, reducing costs, and improving responsiveness. Its ability to adapt dynamically to market changes and disruptions positions businesses for sustained growth and competitiveness in a rapidly evolving global landscape. Artificial Intelligence has emerged as a cornerstone of modern supply chain management, revolutionizing how businesses respond to market demands, disruptions, and operational challenges. Its ability to analyse vast datasets, predict trends, and make real-time decisions positions AI as an indispensable tool for businesses striving to remain competitive in a fast-paced global economy.The transformative potential of AI extends beyond operational efficiency. By enabling smarter decision-making, AI empowers businesses to align their supply chains with broader sustainability and resilience goals. From optimizing energy consumption to mitigating risks during disruptions, AI-driven supply chains are not only more adaptive but also more environmentally responsible.As AI continues to evolve, its integration with emerging technologies such as quantum computing, blockchain, and edge computing will unlock new possibilities. These advancements will enable businesses to tackle complex challenges, enhance transparency, and process real-time data with greater precision. The vision of a fully autonomous supply chain seamlessly managed by AI from procurement to delivery is rapidly becoming reality.However, realizing this vision requires businesses to address challenges such as data quality, workforce readiness, and ethical considerations. By adopting strategic approaches and fostering a culture of innovation, organizations can overcome these barriers and unlock the full potential of AI.Therefore, AI represents the future of supply chain management. Its ability to drive efficiency, resilience, and sustainability underscores its critical role in shaping the next generation of supply chain ecosystems, ensuring long-term success in an ever-changing world.

While several dimensions of data quality, such as completeness, consistency, accuracy, and timeliness, have been discussed, the key question of data value stems from whether BDA can provide novel and valuable insights to exploit new business opportunities or defend competition threats. If the answer is yes, which is often difficult to determine a priori, then BDA can become valuable resource for business. Insights generated from it can be used to create business value in many areas, such as business process improvement, product and service innovation, customer experience and market enhancement, organization performance improvement, and the creation of symbolic value such as business image and reputation. For example, insights about business processes can be used to improve their efficiency, productivity, accessibility, and availability, and even to transform business processes information flows can be leveraged to create competitive advantage and dampen competition. BDA obtains insights about products and services by analyzing both internally generated data and external user-generated content (e.g.,online product/service reviews). Firms can use those insights to increase product/service differentiation, adjust the price of products/services, and develop innovative products and services. Likewise, insights about customers and markets can be used to improve customer satisfaction and loyalty, lock in customers and suppliers, and create a niche market. Finally, many firms have used BDA to obtain insights about. their organization performance, which can enrich organizational intelligence for decision making, develop a dynamic organizational structure to respond to market and environmental changes, make better capacity utilization, and increase return on assets.The amount of big data should not make it a big deal. It is the ability to actually do something meaningful and valuable with it.How big data may create value for firms and public agencies is the utmost important question that requires further research.Today, a new, more pragmatic view of big data is taking hold,dominated not by discussions about the volume, velocity, and variety of data, but by the value of data—the ability to generate actionable insights and apply them to business practices to accelerate innovation, drive optimization, and improve business performance. When used appropriately, BDA can yield previously unknown, valuable, and actionable insights to help refine business processes; develop business initiatives; discover service flaws or operational road blocks; streamline supply chains; better understand customers and predict market trends; and develop new products, services, and business models. The value creation mechanisms are key elements of sustainable business models.

It is critical to integrate the views and priorities of stakeholders and get them on board when determining value targets. The four distinct targets of BDA value creation: (1) organization performance such as the quality of decision making, (2) business process improvement (e.g., the greater efficiency of business processes through automation), (3) product and service innovation, and customer experience and market enhancement (e.g., improved consumer satisfaction, retention, and customer-firm relationships). These targets are interrelated, as many initiatives involve multiple targets. For instance, automated analytics of business processes can improve customer service innovation.First, BDA can create value by improving organizational decision making. This can be accomplished by providing broad and consistent access to data across an organization complemented with empowerment structures to act on the data, or through decision models that augment human decision making or are built into business processes. For instance, analyzing streaming data, such as real-time performance data and just-in-time inventory status, can have significant impacts on organizational performance (e.g., situational awareness, data breach, and fraud detection).Second, BDA can create value by improving the effectiveness, efficiency, and productivity of business processes, which leads to better execution and less time spent on process breakdowns. Continuous improvement of business processes is a challenging task that requires complex and robust supporting systems. The three types of business process analysis are validation, verification, and performance, all requiring large volumes of process and event data.These business process analyses can explicitly benefit from the results of BDA, such as process mining (e.g.,inductive mining, event detection and analysis, bottleneck analysis, deviation analysis). BDA can help identify the strengths and weaknesses of a business process.What used to be a gut-feeling decision or a crudely automated decision can now be empirically supported with large-scale analyses.Third, BDA can create value for product and/or service innovation. Data analysis based on customer clickstreams or purchasing patterns can enable customized web interfaces or promotional strategies. For example, user-generated content available on social media has become increasingly analyzed for gaining business insights.Customers often share their experiences with services and opinions on products and influence each other through online product reviews and ratings. By analyzing those online consumer product reviews and/or consumer discussion forums, a firm can identify the frequently reported product flaws/issues or consumer desired features of certain products or services, which provide insights for product/service innovation.Some companies nurture crowdsourcing innovation platforms to facilitate this.Monitoring of products at customers' sites (as GE does for its power turbines) can also facilitate both product and service innovation.Fourth, BDA can also deliver a better customer experience and more competitive services, resulting in higher customer satisfaction and retention. Firms may identify competitive intelligence or acquire a significant number of new customers through social media analytics. They can identify potential competitors through online comments and link social activities, profiles, and historical purchase history of existing customers to create a better and more comprehensive understanding of their customers through BDA.For example, customers are increasingly looking for more specific answers to questions such as "what do other customers like me think of this product or service?" Answering such questions requires not only an understanding of what customers are looking for but also identification of "similar customers" and what exactly they have stated about a product or service.That is an opportunity for big data value creation.Eventually, functional and/or symbolic value can be created. In the literature,many measures have been developed to assess functional value of information technologies, such as financial returns, market share, and so on. Symbolic value may also be assessed through perceptions of a company's innovative image, industry leadership, market influence, discourse and media discussion, satisfaction from diverse stakeholders (customers, government, and other related entities), and other reactions to BDA initiatives.

A number of contextual enablers (i.e., moderators) can catalyze the conversion of BDA investments to value. A BDA strategy that depicts the objectives and approach to BDA is the glue that holds disparate BDA initiatives together. Further, alignment of a strategy with the organizational infrastructure requires a strong leadership in a firm to promulgate a data-driven culture. Lack of such a culture could be fundamentally detrimental to identifying and generating potential value of BDA. Emotional factors of decision makers may also have effect in real option IT investment decisions.Becoming a data-centric organization will involve organizational and cultural changes and innovation.Strong data-driven cultures can yield predictions that are instrumental in determining where a company is going. The report gauged corporate cultures of fact-based decision making as an important indicator of big data's value potential. Firms should also set up processes, governance structures,and teams with complementary data skills. Active data governance refers to the overall management of the availability, usability, integrity, and security of data. A sound data governance in an enterprise should include a governing body or council,a set of data governance procedures, and a plan to follow and execute those procedures. When business questions arise, a data provenance assessment can help prioritize data that can offer new insights into these questions. A strategy roadmap can then harness existing and new data assets. BDA initiatives without clear business goals and strategies will fail.The paths from investments to assets to capabilities to mechanisms to targets and finally to value largely represents functional value creation.We recognize that companies can create diverse datasets and strong analytics capabilities and directly "sell" them in the marketplace to create value. Alternatively, assets and complementary resources may provide symbolic value to a firm through their propensity to signal that they are innovative and at the forefront of technological development.These patterns are indicated by the direct (nonmediated) path in the framework.A second note regarding the framework is recognition of the temporal aspect.Capability building and realization processes occur through a series of interrelated decisions that facilitate organizational learning. This learning allows for coevolutionary adaptation (i.e., the reverse arrow), which is a virtuous feedback cycle that enhances the ability of a firm to build and realize future BDA capabilities through experience, successes, and failures.

Integrating IoT into lean manufacturing greatly improves energy efficiency, matching perfectly with the goals of reducing waste and optimizing processes.IoT devices monitor energy use in real-time across various manufacturing activities.This constant monitoring helps spot where energy is being wasted, such as with inefficient machines or system leaks.Fixing these issues can lead to significant cost savings by reducing unnecessary energy consumption.In addition, the use of IoT supports sustainability goals. By lowering total energy consumption, manufacturers contribute to reducing their environmental impact and meet sustainability targets.Real-time data allows for modifications in operations that align with environmental standards and regulatory requirements. Furthermore, IoT enables process optimization by providing actionable insights into energy usage patterns.This allows manufacturers to make informed decisions about modifying machine settings, scheduling production times, and optimizing workflows to use energy more effectively.For instance, production schedules can be adjusted to periods with lower energy costs or machinery can be adjusted to improve efficiency.Largely, the integration of IoT in lean .Manufacturing helps achieve substantial energy savings, supports environmental sustainability, and improves operational efficiency.By leveraging real-time data, manufacturers can continuously refine their processes, reduce waste, and enhance their energy management approaches.

Predictive maintenance is a major benefit of integrating IoT technology into lean manufacturing.Using IoT sensors, manufacturers can keep a constant watch on their machinery and equipment.These sensors detect early signs of wear or potential failures, allowing issues to be addressed before they become major problems. Cost Savings: Predictive maintenance helps save money by preventing sudden breakdowns and emergency repairs.Since maintenance can be scheduled and planned in advance, it avoids the higher costs associated with unexpected equipment failures and production stoppages. Reduced Downtime: IoT sensors help by providing real-time updates on equipment health, which means maintenance can be planned for less busy times. This prevents unexpected breakdowns that can disrupt production and keeps everything running smoothly. Extended Equipment Life: By catching small issues early, predictive maintenance helps avoid bigger problems that could cause significant damage.This proactive approach means machinery lasts longer and runs more efficiently, reducing the need for frequent, costly repairs.In short, predictive maintenance, powered by IoT technology, shifts maintenance from a reactive to a proactive strategy. It enables manufacturers to make smarter decisions about when and how to fix equipment, leading to smoother operations, longer equipment life, and significant cost savings.

IoT-enabled inventory systems give you a clear, real-time view of your stock levels, locations, and conditions.This capability supports lean manufacturing in several ways such as following With real-time data, you can keep your inventory levels just right, avoiding too much stock that ties up money and storage space. This helps cut down on storage costs and keeps your operations lean and efficient. IoT systems send alerts when your stock gets low, so you can reorder before you run out. This helps ensure that you always have enough products on hand to meet customer demands without any interruptions. By tracking how inventory moves around your warehouse, IoT can help you organize your storage more effectively.This means you can set up your warehouse to make it easier and faster to find and retrieve items, saving time and effort. IoT enables smarter inventory management by providing real-time insights, making your manufacturing process more efficient and better aligned with lean principles.Bringing IoT into lean manufacturing greatly improves supply chain visibility, offering several key benefits that help streamline operations and boost efficiency. Here’s how IoT makes a difference such as following IoT data is crucial for spotting and managing risks before they become big problems. Sensors can detect issues like temperature changes or equipment failures that might affect product quality or supply chain stability. By catching these problems early, companies can act, such as rerouting shipments or adjusting inventory, to prevent major disruptions.This proactive approach helps maintain a stable and reliable supply chain.IoT devices keep a constant watch on products and materials as they travel through the supply chain. This real-time monitoring gives up-to-date information on where goods are and their current status.For example, sensors can track the progress of shipments, alerting managers if there are any delays or issues right away.This way, potential problems like delays or lost shipments can be handled quickly, keeping everything running smoothly.Improved Coordination is achieved through better visibility which helps suppliers, manufacturers, and distributors work together more effectively. With IoT, everyone involved has access to the same data, such as inventory levels and shipment statuses.This shared information helps everyone stay on the same page.For instance, if a manufacturer sees in real-time that a supplier’s delivery is late, they can adjust their production plans to avoid any disruptions, keeping the whole supply chain in sync.

The need for information about customers, competitors, and trends affecting the market is now stronger than ever. Furthermore, the information needs to be continuous and not tied to a planning cycle, because a timely detection of threats, opportunities,strategic problems, or emerging weaknesses can be crucial to getting the response right. There is an enhanced premium on the ability to predict trends, project their impact, and distinguish them from mere fads. That means resources need to be invested and competencies created in terms of getting information, filtering it, and converting it into actionable analysis. A strategy that fails to create customer value has no future. This value must resonate with a segment of customers and offer more benefits and/or lower costs than competitors. Creating that value for customers and ensuring that the company profits from it over time are central tasks in strategy. Innovation. Markets evolve and competitors imitate customer value.Therefore, it is important that the company creates new sources of value over time.The ability to innovate is key to win in dynamic markets as numerous research studies have shown.Innovation, however,turns out to have a host of challenges.There is the organizational challenge of creating a context that supports innovation. There is the brand portfolio challenge of making sure that the innovation fits among current offerings.There is the strategic challenge of developing the right mix of innovations that ranges from incremental to transformational to ensure the company can maintain profits while also preparing for the future.There is the execution challenge; it is necessary to turn innovations into offerings in the marketplace.There are too many examples of firms that owned an innovation and let others bring it to market. It is in addition to product categories and subcategories, countries,and product categories. Decentralization is a century-old organizational form that provides for accountability, a deep understanding of the product or service, being close to the customer, and fast response, all of which are good things.However, in its extreme form, autonomous business units can lead to the misallocation of resources, redundancies, a failure to capture cross-business potential synergies, and confused brands. A challenge is to adapt the decentralization model so that it no longer inhibits strategy adaptation in dynamic markets. Creating sustainable competitive advantages (SCAs). Creating strategic advantages that are truly sustainable in the context of dynamic markets and dispersed business units is challenging.Competitors all too quickly copy product and service improvements that are valued by customers. What leads to SCAs in dynamic markets? One possible cornerstone is the development of assets such as customer relationships, brands, and distribution channels, or competencies such as digital marketing skills or marketing analytics expertise. Another is leveraging organizational synergy created by multiple business units, which is much more difficult to copy than a single new product or service. Growth is imperative for the vitality and health of any organization.In a dynamic environment, stretching the organization in creative ways becomes an essential element of seizing opportunities and adapting to changing circumstances.Growth can come from revitalizing core businesses to make them growth platforms as well as by creating new business platforms.

A business is an organizational unit with a defined strategy and a manager with sales and profit responsibility.The organizational unit can be defined by a variety of dimensions, including product line, country, channels, or segments.An organization will thus have many business units that relate to each other horizontally and vertically.There is an organizational and strategic trade-off in deciding how many businesses should be operated.On one hand, it can be compelling to have many units because then each business will be close to its market and potentially capable of developing an optimal strategy.Thus, a strategy for each country or each region or each major segment may have some benefits.Too many business units become inefficient, however, and result in programs that lack scale economies and fail to leverage the strategic skills of the best managers. As a result, there is pressure to aggregate businesses into larger entities.Business units can be aggregated to create a critical mass, to recognize similarities in markets and strategies, and to gain synergies.Businesses that have similar market contexts and business strategies will be candidates for aggregation to leverage shared knowledge. Another aggregation motivation is to encourage synergies among business units when the combination is more likely to realize savings in cost or investment or create a superior value proposition.There was a time when firms developed business strategies for decentralized business units defined by product, countries, or segments.These business strategies were then packaged or aggregated to create a firm strategy.That time has passed.There also now needs to be a firm strategy that identifies macro trends and strategy responses to these trends as a firm, allocates resources among business units, and recognizes synergy potentials.

The strategic assets and competencies that underlie the strategy are the critical resources that produce sustainable competitive advantage (SCA) for a firm. According to resource-based theories of the firm, these resources produce competitive advantage because they can be converted into sources of value for customers; they are rare, not easily imitated; and good substitutes for the resource do not exist, which keeps competitors from offering the same value;

and the firm can leverage them to its advantage.A strategic asset is a resource that the firm owns or controls that can be leveraged in the design or implementation of a firm's strategy.Assets include general resources such as financial assets, human assets (leaders and employees), physical assets (plant and equipment),legal assets (patents and trademarks) as well as marketing assets in the form of strong brand reputations, customer relationships, and powerful knowledge of markets.Competencies leverage these assets to perform activities important to the firm's strategy.They do so through organizational processes, which act as recipes for actions.Without the right assets,these processes are not likely to have much impact. At the same time, the assets, whether they are smart employees, patents, or strong brands, will not help the company unless they are leveraged repeatedly through strong processes. These processes help companies outpace competitors and also make it difficult for them to easily imitate the firm's strengths because it is hard to observe all of the complex activities involved in a competency.Therefore, companies must ensure that they have both the strong assets and processes for leveraging those resources for marketing excellence.If exercised well, these competencies can, over time, add to the strength of a company's asset base by improving customer relationships and brands.The ability of assets and competencies to support a strategy will in part depend on their strength relative to competitors.To what extent are the assets and competencies unique or rare in the marketplace? If unique now, how easily can they be imitated by competitors? Many assets require a long-time to develop and so competitors trying to build the strength of Amazon's strong customer relationships may be hopelessly behind. Competencies are often difficult to imitate because competitors cannot easily understand the recipe that companies use to create such outstanding processes. Multiple product markets offer additional synergies that can be a source of SCA.Synergies can come in many forms. Two businesses can reduce costs by sharing a distribution system, sales force, or logistics system, as when Gillette acquired Duracell (and later was itself acquired by P&G). Synergy can also be based on sharing the same asset, as it aims to provide a comprehensive analysis of current applications, methodologies, and outcomes associated with predictive analytics and localized inventory automation implementation in emerging economy supply chains. By examining both theoretical foundations and practical implementations, this work seeks to identify best practices, common challenges, and future research directions that will guide the continued development of resilient supply chain systems in developing markets.

The integration of digital twin technology, predictive analytics, and sustainability-driven strategies has emerged as a transformative approach in global supply chains, delivering unparalleled efficiency, resilience, and sustainability.These technologies collectively enable real-time monitoring, simulation, and predictive modeling, fostering proactive decision-making and adaptability in dynamic markets.The research highlights that digital twins offer a comprehensive, real-time virtual representation of supply chain processes, allowing businesses to identify inefficiencies, optimize resource allocation, and simulate potential disruptions.This capability reduces downtime and improves operational continuity.Predictive analytics enhances these benefits by analyzing historical and real-time data to forecast trends, identify risks, and suggest optimized strategies. For instance, predictive maintenance minimizes equipment failures, while demand forecasting aligns production with market needs, reducing waste and storage costs.The synergy between these technologies ensures not only cost savings but also improved customer satisfaction through faster and more reliable service delivery.Moreover, the integration supports sustainability goals by enabling organizations to reduce their environmental adopting circular supply chains andgreen logistics practices use these technologies to achieve greater energy efficiency and minimize emissions, contributing to global sustainability initiatives.Conclusively, the findings emphasize that organizations leveraging these technologies gain a competitive advantage by enhancing agility, reducing costs, and fostering innovation.This integration represents a paradigm shift in supply chain management, where advanced tools and sustainable practices drive value creation and long-term success in a rapidly evolving global environment.These financial supports can lower the entry barriers, particularly for small and medium-sized enterprises, enabling broader participation in technology-driven sustainability efforts.Industry-wide collaboration is essential for establishing standardized protocols and interoperability among digital tools.Shared frameworks for data integration will streamline the adoption process, ensuring seamless communication across systems.This can be achieved through alliances between technology providers, manufacturers, and logistics companies to develop and adopt universal standards.

Resource constraints, including insufficient technical expertise, further exacerbate these challenges. Companies can overcome this by fostering partnerships with technology providers and academic institutions to access specialized knowledge and training. Addressing these technical and organizational barriers through strategic planning ensures a smoother transition towards sustainable and efficient supply chain operations enabled by advanced technologies.IoT has drastically changed the modern world and has affected many industries, one of which is manufacturing. IoT can be defined as the interrelated web of embedded devices, automobiles, gadgets and other tangible assets installed with sensors, applications, and connection standards which make these objects capable of sensing, collecting and transmitting info. With respect to manufacturing, IoT enables the monitoring and collection of data integrated with automatic control of manufacturing and production processes, and thus, increases the efficiency of organizations and their productivity.This shift is revolutionizing various industrial production lines with smart factories becoming the new reality in supply chain management revolutionizing various industrial production lines with smart factories becoming the new reality in supply chain management

SCM in manufacturing defines the systematic, strategic coordination of the traditional supply chain operational resources while concurrently optimizing the related activities in a cost-effective way with the purpose of increasing the end-user value while satisfying consumer demand. This paper aims to establish that effective SCM is a vital ingredient in an Manufacturing environment to respond to market requirements, cut cost, and above all satisfy the consumer.The incorporation of SCM and IoT stands as a major innovation since it gives manufacturers new visibility, control and optimization capabilities.Many authors agree that IoT is now a critical factor affecting modern production lines. This trend is clear as globalization progresses and consumer requirements develop; manufacturers are called to optimize their processes, minimize loss, and produce high-quality goods with unleveled amounts. This is due to the ability of IoT technologies to offer the necessary tools to meet these demands. They include Real time data acquisition and analysis, Predictive maintenance and optimal automation. These capabilities enable manufacturers to closely monitor equipment, to anticipate failures and to maximise equipment effectiveness, hence lowers and overall production time is improved.Another advantage of the internet of things in manufacturing is that real-time information can be obtained from any point in the production process.Through the use of sensors and connectivity to machinery and equipment, operations permanently can be monitored, ensuring early identification of problems, thereby allowing proper decision making. It is at this level that one can easily spot issues within the process and find ways of enhancing the process speed as well as quality. In the same way, IoT makes it possible to gather large amounts of data to assess, in a recurring manner, subtle behaviors that form the base of knowledge creation and perpetual improvement.Based on the above-discussed definition of IoT, in the field of supply chain management it has an enabling character as it awards visibility, traceability, and responsiveness. Conventional supply chains have a disadvantage of minimal information flow in real time hence resulting in time delays, high costs and poor efficiency. IoT offers solutions to the above challenges by way of tracking the movement of goods, assets and shipments in the supply chain at a real-time platform.Thanks to the Internet of Things devices such as sensors and trackers, manufacturers can know the status location and condition of products from the suppliers, through the factories and on to the consumers.It also creates a better understanding and enables a better synchronized overall situation within the flow of supply chain activities with the aim of also decreasing lead time and disruption.For instance, IoT can be of great help to manufacturers in monitoring the stock level, use and rate of stocks, and the over nominal requirements.It thus helps those planning for business to accurately forecast the required quantity of stock and thus minimize the chances of stock out or having an excess of unnecessary stock.In addition to this, IoT also play an important role in supply chain tracing since it is otherwise difficult to keep a record on how goods are handled and moved in the supply chain since this is critical with respect to regulatory compliance and quality control.

With the introduction of IoT the manufacturing units laid the foundation of advanced factories and new ways to manage the supply chain. IoT is the complex of interconnected physical objects equipped with sensors, software, and other technology for the purpose of making such objects to monitor and, if necessary, interact between themselves. In the manufacturing businesses, IoT enables real-time tracking, data gathering, and management; hence, it has never been easier to achieve the set goals and objectives. IoT gives an opportunity to have manufacturing process visibility in real time, that helps manufacturers monitor the condition of used machinery and equipment in a perpetually manner.This capability is very important in areas such as highlighting constraints, managing the resource calendar and controlling quality of products.Devices connected to the Internet of Things can listen to even the slightest variations that can indicate the failure of equipment and alert them in advance to fix them to reduce on time they take to be repaired and the durability and strength of the apparatus. Such a proactive strategy of equipment maintenance not only optimizes processes and productivity continuously but also reduces the number of halts and the expenses that result from them.Moreover, IoT helps in the analysis of the data which is at the heart of the fourth industrial revolution essential for ensuring constant enhancements in any manufacturing sector.Since the IoT devices are capable of producing huge volumes of data, pattern and trends of the manufacturing process can be identified from the data hence aiding the manufacturers in making pertinent decisions. For example, data analytics can discover flaws in the production process, provide advices on how to eradicate them, and even estimate the potential changes in the performance level.This level of insight enables the manufacturers to be competitive by continuously enhancing their operations, and market responsiveness.Another relevant advantage of IoT in manufacturing processes is the opportunity to improve the supply chain management. Smart tags are devices which connect the internet, facilitate live tracking of the position, state as well as condition of products supplied chains.It also facilitates the integration of supply chain activities whereby manufacturers can increase visibility and thus decrease the lead-time and risks involved. For instance, active monitoring of the shipment enables one to note that a certain shipment is delayed and may need to be redirected to different channels.Further, IoT can enhance the efficiency of a firm's inventory management as it grants actual and timely information about the stock, its usage and demand rates.

This information helps the manufacturers in holding moderate inventory, thus managing avoidable shortages or excess stock, financial cost of holding inventory, and overall effectiveness of the supply chain is improved.IoT also has a large impact on compliance to standards and regulations that may be in place.In addition, IoT helps reduce and improve the traceability and accountability of the movement and handling of goods.It is particularly relevant for industries that are subjected to a high level of regulation like the health sector especially manufacture of drugs and the production of foods and beverages.Technology integration with IoT guarantees that manufacturers can have an overview of products in their supply chain from raw materials through to the finished products, in a bid meeting quality as well as safety standards.Most industries are facing many problems with the integration of IoT in manufacturing.Some technical factors include problems with interaction of Smart IoT and old network connection and incompatibility of new Smart IoT devices with other devices.This is particularly because many manufacturers still use a combination of both the newer and older machinery in which it is hard to integrate new devices. Also,because of the huge volume of data likely to be produced by IoT devices, there is a need for efficient data handling and analysis skills which can sometimes be challenging and time-consuming to integrate.Another research area that was established was the role of the organization in integrating IoTs into their operations.It is important due to the fact that, more often than not, Internet of Things solutions introduce change, which should be managed in an organisation.There can be problems with employees’ acquisition of new skills and knowledge concerning new technologies and approaches implemented, as well as with people’s reluctance to change for the better and switch to modern techniques instead of relying on usual procedures.Also, data security and data privacy are critical, as IoT devices are invasive and can be hacked easily by unauthorized individuals.This is evident in the aspect of security where manufacturers have to ensure that their information as well as operations are safeguarded against any threats.Accordingly, IoT is capable of revolutionalizing manufacturing in today’s world by intelligentizing the operation, supply chain, and making it absolutely compliant with the set standards.As is apparent from the various challenges that have been discussed above, IoT integration is thus an essential part of the manufacturing environment for any manufacturers who are keen on maintaining their competitiveness in a progressively intertwined and fluid global environment.Below are the next sections of the investigation: the difficulties of IoT implementation with present systems integration and the trends for IoT’s and supply chain logistics in the future.

Facing brilliant advance techniques of IoT, future of IoT in manufacturing and supply chain is set to show appreciable improvements and progress.AI/ML is one of the most emerging trends related to IoT and has already becoming prominent, with more and more devices integrated into IoT being empowered by Artificial Intelligence and Machine Learning. AI and Machine learning are thus capable of enhancing the performance of IoT through the provision of detailed analysis, forecast, and self-organization. For example, using AI in controlling IoT systems can analyze large amounts of data to forecast equipment breakdowns and accuracy of manufacturing timetables and product quality assurance with minimum error because they eliminate occasions of human intervention.One of the encouraging trends in the near future is the usage of edge computing – the computation process takes place nearer to the data source rather than in the data center.Edge computing helps to deal with low latency, improve the real-time decision-making process, and overcome the problem with the limited availability of bandwidth for transferring a vast amount of data.This is especially helpful in the manufacturing businesses as they require quick responses to their orders. The ability of managing data at the edge means that manufacturers can optimize their resource use and operating methods by relying on analytics in real time.The low data transfer delay, the Internet of Things devices will have improved reliability of connections, which will lead to the development of new applications such as remote control in real-time, autonomous transportation system, and smart manufacturing systems.The optimization of the IoT systems due to increased connectivity by the new generation instances of systems between the supply chain's production and distribution.This will lead to increases in the flexibility and agility of the supply chain and the chain being able to recover quickly from shocks.Another emergent idea that can be considered as a promising direction for the IoT and supply chain management is a blockchain system. Blockchain can offer the efficiency of maintaining the record of transactions which make it possible to improve the logistics of the supply chain as far as transparency and responsibility are concerned.By implementing the integration of blockchain with IoT, it is possible to produce the history and movement of the product, thus controlling its authenticity.This is especially useful in functions and sectors like pharmaceuticals and food and beverages since safety and the legal requirements have to be met at all times.

Digital twins' development is also expected to drastically change manufacturing and supply chain processes.Digital twins can be defined as the representation of an actual asset, process, or system in the digital environment to solve the real-time performance evaluation issue. Manufacturing companies can monitor and control the displayed equipment and supply chain processes efficiently with the help of the digital twins technique.Digital twins are used to provide insights for predictive maintenance, process improvements, and for modeling scenarios; thus, improving the manufacturing performance by minimizing downtime and fluctuations in the chain of production.Manufacturing is shifting toward sustainability more lately and IoT could be of great use in pursuing sustainable goals.The IoT technologies act as a way of measuring consumption of energy, avoiding wastage and ensuring that resources are properly utilized. For instance, energy uses in IOTs can be recorded in real-time and this helps the manufacturers to recognize chances to saveEnergy and put into practice more conserve power. Furthermore, through IoT, circular economy supply chains can be created thus enabling the design and creating of goods that can be recycled and reused instead of being disposed of in environmental degrading ways.Thus, the application of IoT in manufacturing and supply chain management in the future is based on the integration of current and emerging technologies and useful applications. AI and ML are ready to set new standards in manufacturing, while the technology of edge computing and 5G networks are already on the way to revolution the manufacturing sector, blockchain is also stretching its arms out to contain manufacturing in its fold, development of digital twins to make manufacturing smarter and focused and the force of sustainability already making its force felt in the manufacturing sector.Thus, the primary goal is to consider and adopt these trends, which will help the managers of the manufacturing companies to construct stronger, flexible, and sustainable supply chain networks appropriate for the global digital environment.

The integration of IoT solution on manufacturing supply chain management has emerged as a compelling force for increasing more supply chain effectiveness, transparency and quickness.This research has also pointed out the level of change that is possible with IoT and increase in predetermined measures, which include the stock turnover and the time taken to meet orders.Through the provision of essential information or data needed as well as support for predictive maintenance, IoT aspects have supported manufacturing businesses to enhance their operations, minimize the instances of downtimes, and produce quality products.Still, such advantages indicate that the further implementation of IoT is not without issues that manufacturers have to tackle to realise the opportunities of using the technology. Lack of compatibility and integration with ecosystems, unwillingness of organizations to modify their business models, and security problems are among the main barriers to the IoT integration.These are the barriers that manufacturers have to overcome, for which they need proper data management systems, comprehensive training of employees, proper cybersecurity measures, etc. So, the solutions meeting these challenges need more of the strategic approach, which lies in the union of advanced technology application and adaptation of organizational change with the necessary focus on constant learning.This study's analysis of the trends in the IoT present in the work highlights a promising future for manufacturing supply chains. Edge computing and combination with IoT will provide a much higher level of real-time data processing and connection, AI and machine learning with IoT ensures intelligent analysis and automated decision-making. It is believed that those technologies will continue to fuel more innovation and improvement in creating more flexible, adaptive and robust supply chains effectively for manufacturers.Summing up, it can be stated that IoT is one of the key technologies that will define the future state of managing manufacturing supply chains.The fact it can offer real-time visibility, prognostics, and automation of numerous business processes can greatly improve such characteristics as productivity and competitive advantage.Nevertheless, for coming around IoT to its real potential, the manufacturer has to solve the technical, organizational, and security issues connected with its implementation. If managed effectively, these challenges and encouraged investing in technologies of the new industrial revolution, today's manufacturers can find themselves standing prepared for a connected and digital future.Further research should expand upon the concept of IoT and its progress in the future based on integration's guidelines, existing technologies, and limitations.

IoT when adopted in manufacturing has numerous challenges of integration especially with the current existing manufacturing systems in use and are some of the areas that manufacturers need to overcome for them to fully optimize on the IoT transformative technology.A technical issue that seems to endure is the issue of working heavily with IoT devices and their compatibility with previous systems.Most factories work with a range of tools of different vintages and some of those tools could be very old. Integrating and dynamically communicating with these old systems and the newer more advanced IoT devices can be a technical challenge as well as demanding a heavy investment on the hardware and software upgrades. Planning and design problems can result in such a narrow separation of data that important information does not exchange between the systems, therefore potential benefits of the IoT integration may not be realized.Yet another technical aspect has to do with the analysis and especially the storage of the large quantity of data that IoT devices produce. Manufacturing environments with the incorporation of IoT generate massive data that need to be gathered, accumulated, and processed in real-time.This means that organizations should have sound data management procedures and sophisticated analyzing tools.The set up and use of these systems may prove expensive and may require professional manpower that may not be easily accessible.Further, the constant stream of data also requires an effective and fast network connection to transfer the data, which can prove to be very costly for many manufacturers.The difficulties that emerge at the organizational level also have a great influence on the successful implementation of IoT. It has been noted that the management of IoT creates a need to change organizational culture and processes, which is often faced with employee’s resistanceOther limitations of IoT application for manufacturing systems include ; security and privacy issues. Any connected device can be attacked through the IoT network and this makes IoT devices more susceptible to cyberattacks hence leading to data loss, disruption of operations and financial losses. Businesses themselves need to establish strong IT security solutions to safeguard vital information and guarantee business processes’ reliability.This includes the use of encryption when transferring data, ensuring proper connection to the network and its secure connection and ensuring the latest versions of the software are being used to prevent the current and emerging security threats.

Moreover, manufacturers have to put up the guidelines of data management to guarantee that data is collected, stored, and used are legal and compliant with policies and regulations.Therefore, we can identify the following issues among the most important ones: Deploying IoT solutions can be very capital-intensive because it involves procuring new gadgets and networks, equipment, and renovation. To most manufacturers, especially SMEs, these costs are likely to prove high. Manufacturers should consider the following issues while making strategies for IoT projects: Before they start IoT projects, it is appropriate for them to determine the break-even point taking into consideration the cost of projects in the short-term and the benefits of implementing IoT projects in the long-run. Gradual approach in deploying IoT and proper planning on how to implement IoT solutions to different parts of the business can prevent exposing the company to huge losses by focusing more on the value that is being added to the company as opposed to the costs incurred.However, there are still works in progress when it comes to standards by which IoT systems shall interconnect, which in a way can hamper integration.This implies that IoT devices generated by different manufacturers may not easily integrate to support each other because there is no inter-company or industry-wide set standard.This can result in vendor lock-in where manufacturers are restricted by the value chain from a particular vendor hence restricting options and also costing the manufacturers a lot of money to change to another vendor.These are the challenges and IoT urged the cooperation with other industries and creation of standards across the industries. Altogether, it can be concluded that even though the integration of IoT with the existing manufacturing systems is challenging, the consideration of these issues is critical to optimize the advantages of IoT for manufacturers.These strategic challenges include; technical compatibility, data management, organizational change, security, cost, as well as shifting standards. Specific actions that engage these challenges successfully transform a manufacturer’s supply chain into a dynamic digital enterprise ready to tap into the sophisticated global systems of supply.

Organizations tend to steer clear of enterprise resource planning (ERP) systems despite the significant benefits associated with their use.This is owing to the complicated deployment process and greater failure rate.The ERP project may fail in its entirety or part.Completing projects using information technology and information systems is a critical difficulty due to the uncertainties relative to the intricacies of technology.In contrast to the expansion of ERP systems, it is alleged that the implementation of these systems has failed at a greater rate, estimated to be between 60 and 90 percent and that there is an inability to appreciate the advantages that were promised while not reaching the high hopes and expectations that were associated with the deployment of ERP systems.An unanswered debate of whether its acceptance or rejection is warranted persists despite the rapidly accelerating pace of technological advancement and the pervasiveness of technological innovations in people's lives, whether in their personal or professional spheres. In the past, millions of dollars have been invested in information technology (IT), such as enterprise resource planning (ERP) systems, to enhance employee performance or effectiveness, boost workplace productivity, or gain a competitive edge.Nevertheless, these advantages will only be realized if the workers in these organizations make effective and appropriate use of information technology (IT) to carry out their organizational responsibilities. ERP addresses several sectors. Therefore, describing functions to describe ERP takes much work. Two ways ERP helps multiple industries. ERP can handle many industries (e.g., manufacturing and commerce) or offer pre-configured enterprise-specific solutions. PeopleSoft provides industry-specific solutions to communication, federal government, financial services, healthcare, higher education, manufacturing, public sector, retail, service industries, transportation, and utilities. ERP is for international procurement, production, sales, and administration.ERP must be able to meet regional needs. This includes country-specific chart-of-accounts, pre-formatted quotes, delivery notes, invoices, and HR procedures like payroll. All transactions must support numerous currencies.Finally, its use of frequency and repetition may be differentiating.ERP facilitates routine business procedures like procurement, sales order processing, and payment processing, but not marketing, product creation, or project management. ERP software is technological, too.While technical features may not differentiate ERP from other applications, they can differentiate it from similar tools like integrated, centralized software with rigorous platform requirements. Additionally, technical features significantly impact the functionality and potential of this product.

ERP makes the manufacturing process run more smoothly and improves the visibility of the order fulfillment process within the organization.ERP helps the manufacturing process flow more smoothly.This can result in decreased inventories of the materials needed to manufacture items (inventory of work in progress), and it can also help users better schedule deliveries to clients, resulting in decreased inventories of finished goods at warehouses and shipping docks. Both of these outcomes can be beneficial.We need ERP software in addition to supply chain software if we want to improve the flow of our supply chain significantly.Human Resources might need a more streamlined, uniform mechanism for tracking employees' time and communicating with them about benefits and services, and this is especially true in firms with numerous business units.That can be fixed with ERP. In a rush to find solutions to these issues, businesses frequently lose sight of the fact that enterprise resource planning.ERP makes the manufacturing process run more smoothly and improves the visibility of the order fulfillment process within the organization.ERP helps the manufacturing process flow more smoothly.This can result in decreased inventories of the materials needed to manufacture items (inventory of work in progress), and it can also help users better schedule deliveries to clients, resulting in decreased inventories of finished goods at warehouses and shipping docks. Both of these outcomes can be beneficial.We need ERP software in addition to supply chain software if we want to improve the flow of our supply chain significantly.

Further, few researchers evaluate ERP system performance. Only a few articles explored ERP systems' effects on productivity and performance.ERP implementation and performance shows that most studies measure the impact of ERP systems on organizational performance rather than individual performance, using outcome variables like product quality, benefits, financial performance, organizational service enhancement, market value, process efficiency, shareholder return, competitive advantage, and executive benefits.Over time, organizations have adopted new technologies faster, and many have invested heavily in ERP systems to integrate all business activities into a uniform system and improve efficiency.Users must comprehend its fundamental principles to maximize ERP system efficiency and user effectiveness.In conclusion, user performance is crucial to assessing ERP system deployment. The implemented system's benefits depend on its system utilization.ERP systems in post-implementation and user performance studies should be addressed as a necessity for empirical research.More research in different environments is needed to define the interaction between ERP systems and their users to help practitioners and researchers better understand this application.Thus suggesting future research into users' aspects related to ERP system adoption and use to determine how these systems affect efficiency, effectiveness, and creativity.It is often believed that ERP success relies on hiring top talent from both business and IT departments.Due to the program's complexity and significant business changes.Truthfully, the percentage is substantially more significant. Most performance issues stem from changes in appearance and functionality.Lack of familiarity with new methods might lead to panic and business disruptions.Data from the ERP system is often coupled with external data.Those with extensive analysis needs should budget for a data warehouse and anticipate significant effort to ensure seamless operation.Users need help with a dilemma and updating ERP data daily in a large corporate data warehouse is challenging, and ERP systems need help identifying changes, making selected updates difficult.One pricey option is custom programming. Companies often discover the usefulness of systems after some time, allowing them to focus on improving affected business processes.

Through the integration of data analytics, emerging technologies such as artificial intelligence (AI) and machine learning, and IoT, small businesses can leverage valuable insights to streamline operations, improve decision-making, and gain a competitive advantage in a rapidly evolving marketplace. The cases explored across various sectors—retail, manufacturing, and services—highlight the diverse applications of data-driven strategies and the significant impact they have on improving efficiency, reducing costs, and driving growth. By utilizing data to optimize workflows, predict trends, and better meet customer demands, small enterprises can position themselves for long-term success.As the adoption of data-driven approaches becomes more widespread, the future of small enterprises will likely see greater scalability of these frameworks. The ability to collect, manage, and analyze data efficiently will enable businesses to expand their operations while maintaining or even improving operational performance. Furthermore, the integration of advanced technologies, such as AI-driven predictive analytics and IoT-based real-time monitoring systems, will continue to push the boundaries of what small enterprises can achieve. These technologies will enable more accurate forecasting, faster response times, and the ability to innovate continuously in ways that were once reserved for larger organizations.However, while the potential for growth is significant, the challenge remains in overcoming barriers such as the cost of technology adoption, data privacy concerns, and the need for specialized skills. Future research should focus on developing scalable solutions that I address these challenges, making advanced data analytics more accessible and affordable for small enterprises. Exploring the potential for integration with next-generation technologies, such as blockchain for secure data sharing and enhanced transparency, could further empower businesses to leverage data for optimization.Overall, the framework for data-driven business optimization represents not just a strategy for operational improvement but also a key enabler of innovation and strategic growth. As small enterprises continue to embrace these tools and approaches, they will not only enhance their competitive position but also contribute to the broader economy by fostering a more agile and responsive business landscape.

According to scholars, such practices limit the ability of business owners to track cash flow, manage resources efficiently, and ensure regulatory compliance. For instance, failure to maintain accurate financial records can result in tax disputes or penalties, further straining already limited resources. Moreover, without proper records, business owners struggle to evaluate the true performance of their enterprises, which hinders long-term planning and growth.In addition to financial implications, inadequate record-keeping affects the ability of SSEs to compete in a rapidly evolving business environment. With the rise of digital technologies and increasing demand for transparency, businesses that lack proper documentation risk being excluded from formal economic systems.This exclusion can lead to missed opportunities for partnerships, grants, and government incentives aimed at supporting small businesses. Recognizing these challenges, there is a growing need to emphasize the importance of record-keeping among SSEs.This study seeks to explore how effective documentation can enhance the growth and sustainability of these enterprises.By addressing the gaps in record-keeping practices, it aims to I provide actionable recommendations for improving business performance and fostering economic development in the region.Small-scale enterprises (SSEs) are essential contributors to economic growth, especially in Delta State, Nigeria. They provide employment, stimulate local economies, and serve as incubators for entrepreneurial development. However, despite their potential, many SSEs in Nigeria face challenges that hinder their growth and sustainability. Among these challenges, the lack of systematic record-keeping stands out as a critical issue.Effective record-keeping is fundamental to business success, as it provides the foundation for financial management, performance evaluation, Stock inventory and long-term planning and business sustainability. It enables business owners to track income and expenditures, monitor cash flow, and evaluate profitability. However, in Nigeria, a significant number of small business owners rely on memory or informal methods to manage their operations. This reliance on unsystematic approaches leads to several issues, including Inefficiencies in Operations: Without accurate records, business owners struggle to understand the financial health of their enterprises. They may overestimate profits or underestimate losses, resulting in poor decision-making and inefficient resource allocation Inaccuracies and Financial Losses: Memory-based management is prone to errors, leading to inaccurate financial reporting. This can result in undetected revenue leakages, stock mismanagement, and unrecorded debts. Over time, these inaccuracies accumulate and negatively impact the business.Limited Access to Credit: Many financial institutions require detailed financial records as part of loan applications. Businesses in Agbor that fail to maintain proper documentation are often excluded from accessing credit facilities.This limits their ability to invest in growth opportunities, such as expanding operations or acquiring new equipment.Non-Compliance with Tax and Regulatory Requirements: Inadequate record-keeping also affects tax compliance. Businesses without proper documentation may either overpay or underpay taxes, leading to penalties or disputes with tax authorities. This non-compliance further jeopardizes their sustainability. Challenges in Strategic Planning: Proper records provide insights into business performance, allowing owners to identify strengths, weaknesses, and opportunities for growth.

Record-keeping refers to a vital aspect of business management that involves the systematic documentation of business activities, financial transactions, and operational processes. It is the practice of organizing, storing, and maintaining business records such as sales, expenses, inventory, receipts, and payroll information. The concept of record-keeping is not limited to mere data entry but extends to the accurate classification and retrieval of information for analysis, reporting, and strategic decision-making. According to previous studies, record-keeping is essential for the smooth operation and sustainability of businesses, especially small-scale enterprises (SSEs). It enables entrepreneurs to track their financial performance, manage cash flow, and monitor the effectiveness of business strategies. By maintaining organized records, businesses are better equipped to understand their profit margins, manage debts, and identify areas for cost reduction. Furthermore, effective record-keeping serves as a safeguard against financial mismanagement and fraud, which are common problems in informal small businesses. operations.Record-keeping encompasses various types of documents and records, such as following these include all documents related to revenue, expenses, investments, profits, and losses. Financial records are necessary for monitoring business performance and ensuring compliance with taxation laws. Without such records, businesses may struggle to understand their financial health or meet regulatory requirements. Some scholars highlight that proper

Financial record-keeping is critical for securing loans and other forms of financial assistance. Keeping track of the goods and materials a business owns is essential for inventory management. This includes tracking stock levels, purchase orders, sales, and reordering requirements. Previous studies emphasize that inventory records prevent stock shortages or surpluses and help businesses avoid unnecessary financial losses through overstocking.For businesses with employees, maintaining accurate records of wages, benefits, and working hours is crucial for ensuring compliance with labor laws. Employee records also help in managing human resources and assessing workforce performance. Previous studies note that human resource management is critical for maintaining smooth operations in small businesses. Businesses must also maintain records to comply with legal obligations, as business registration, tax filings, licenses, and permits. Failing to keep up-to-date legal records can result in penalties, fines, or the loss of business licenses. As noted by scholars, tax compliance is one of the areas where small businesses are most vulnerable due to inadequate record-keeping.The importance of record-keeping extends beyond just compliance and operational efficiency. It is also critical for strategic business planning. When accurate records are available, business owners can make informed decisions about pricing, marketing, and expansion. According to scholars,businesses that do not keep accurate records often face difficulties in assessing their strengths and weaknesses, which can lead to missed opportunities for growth and development. Despite the clear advantages of record-keeping, many small business owners, particularly in developing economies like Nigeria, do not implement effective systems for documentation. Research by previous studies suggest that small business owners in Nigeria often rely on informal, manual systems, or their memory for keeping track of business activities.This lack of formal record-keeping systems is a significant barrier to business growth, as it leads to inefficiencies, inaccurate reporting, and an inability to measure business performance accurately.The failure to maintain proper records is also linked to the lack of financial literacy and awareness among small business owners. Many entrepreneurs lack the knowledge or understanding of accounting principles, which makes it difficult for them to implement robust record-keeping practices.Additionally, the cost of hiring professional accountants or investing in software tools for record-keeping can be prohibitive for many small businesses. As a result, businesses often resort to rudimentary methods such as handwritten ledgers or simple spreadsheets, which are prone to errors and can be difficult to analyze over time.In conclusion, record-keeping is a fundamental element of business management that provides a structured approach to handling financial and operational data. Effective record-keeping ensures that small businesses are better prepared for decision-making, financial planning, and compliance with legal regulations. Previous studies argue, without proper documentation, small businesses are at risk of financial mismanagement, inefficiencies, and an inability to leverage opportunities for growth. Therefore, developing systematic record-keeping practices is a crucial step for improving the performance and sustainability of small-scale enterprises.

Record-keeping plays a crucial role in the management and sustainability of small-scale enterprises (SSEs). Accurate and organized records provide business owners with the necessary tools to track performance, make informed decisions, and ensure financial transparency. In the context of SSEs, particularly in developing economies like Nigeria, the importance of maintaining systematic records cannot be overstated. As noted by scholars, poor record-keeping practices are a primary reason for the failure of many SSEs, as they hinder effective decision-making, lead to financial mismanagement, and limit access to critical resources.One of the most significant benefits of proper record-keeping is the ability to monitor and evaluate business performance. By maintaining accurate financial and operational records, business owners can assess their income, expenses, and overall profitability. This data helps in identifying trends, spotting potential issues, and measuring the effectiveness of strategies. For instance, by reviewing sales records and inventory logs, a business owner can determine which products are performing well and which are not, enabling them to adjust marketing or inventory strategies accordingly.Scholars emphasize that the absence of this crucial information makes it difficult for small business owners to assess their business performance and make necessary adjustments. Without detailed records, decisions are often based on intuition rather than data, which can result in missed opportunities or costly mistakes. Accurate records provide the foundation for sound decision-making, enabling businesses to remain competitive and responsive to market changes.It is a significant challenge for many SSEs, particularly in developing countries where financial institutions require proof of financial stability before providing loans or credit facilities. Banks and other lenders typically require detailed financial records, such as income statements, balance sheets, and cash flow statements, to assess the creditworthiness of a business. Previous studies revealed that businesses that lack formal record-keeping systems are less likely to secure funding, as they cannot demonstrate their financial position or repayment capacity. Scholars argue that the lack of proper financial documentation often leads to the rejection of loan applications, depriving small business owners of the resources they need to grow their businesses. Conversely, businesses that maintain accurate records are more likely to gain the trust of financial institutions and access credit, which can be used to expand operations, invest in new technologies, or meet working capital needs.Record-keeping also plays a vital role in ensuring that SSEs comply with legal and tax regulations. In many jurisdictions, businesses are required to file tax returns, register income, and report expenses in a transparent manner. Without accurate records, it becomes impossible for business owners to fulfill these legal obligations, which can result in penalties, fines, or even the closure of the business.Previous studies highlight that one of the reasons many SSEs fail to comply with tax regulations is due to poor record-keeping practices. Business owners who do not maintain detailed financial records often underestimate their tax liabilities or misreport their income, leading to legal issues. Accurate documentation ensures that businesses remain compliant with tax authorities, thereby avoiding penalties and maintaining their legitimacy in the eyes of regulators.The ability to identify and seize growth opportunities is a key factor in the sustainability of SSEs. Effective record-keeping allows business owners to analyze trends and patterns within their operations, helping them to spot areas for expansion. For instance, by keeping track of sales data, a business can identify peak periods for certain products and plan marketing campaigns or production schedules to capitalize on these trends Some scholars explain that record-keeping provides the insight needed to make long-term plans and investments.

By reviewing historical performance, businesses can forecast future demand, invest in new product lines, or even expand into new markets. Without a solid record-keeping system, SSEs are at a disadvantage when it comes to recognizing growth potential or planning for future challenges.Accurate records promote financial accountability, which is vital for the long-term success of any business. Record-keeping helps business owners track their cash flow, ensuring that they do not overspend or incur unnecessary debt. By keeping detailed records of expenses and income, entrepreneurs can better manage their budgets and make more responsible financial decisions.As noted by scholars, poor financial management due to inadequate record-keeping often leads to the downfall of small businesses. Without an understanding of their financial position, business owners may fail to recognize when they are operating at a loss or struggling with liquidity, leading to poor financial health. Proper documentation mitigates these risks, promoting financial transparency and improving business sustainability.Accurate record-keeping also aids in managing risks, which are inherent in any business environment. By maintaining detailed records, businesses can track potential threats, such as rising operational costs or fluctuations in demand, and take proactive measures to mitigate these risks. Record-keeping helps businesses spot early warning signs of financial trouble, allowing them to adjust their operations or seek external support before issues escalate.Previous studies argue that businesses with poor records are often ill-prepared to handle financial downturns, market shifts, or unexpected expenses. On the other hand, businesses that maintain solid records are better equipped to respond to challenges and minimize the impact of unforeseen risks.Record-keeping is critical to the success and sustainability of small-scale enterprises (SSEs), yet many business owners face significant challenges in maintaining accurate and systematic records. These challenges, which include a lack of knowledge, limited resources, time constraints, and cultural and educational factors, often result in inefficient business practices and poor financial management. These barriers prevent small business owners from fully realizing the benefits of proper documentation and ultimately undermining their chances of success.One of the primary challenges facing SSEs and other regions is the lack of knowledge and financial literacy among business owners. Many entrepreneurs, especially those in the informal sector, lack a formal education in accounting or business management, which makes it difficult for them to understand the importance of accurate record-keeping. According to previous studies, many small business owners in Nigeria fail to adopt proper record-keeping systems because they are not well-versed in basic accounting principles such as budgeting, financial reporting, and cash flow management.

This lack of knowledge is a critical barrier; as it means business owners often rely on informal, rudimentary methods of record-keeping, such as relying on memory or using simple hand-written ledgers, which are prone to errors. Some scholars note that without proper financial education, business owners are unable to appreciate the significance of systematic documentation, which leads to poor decision-making, inadequate financial planning, and an increased risk of business failure. For many small businesses, especially in developing limited financial resources present a significant challenge to effective record-keeping. Business owners often prioritize immediate operational needs, such as purchasing inventory or paying employees, over investing in tools or resources for maintaining proper records. According to previous studies many SSEs operate with minimal capital, and as a result, they may not be able to afford accounting software, professional accountants, or even basic office supplies needed to maintain organized records.Additionally, the lack of resources can limit access to training programs or workshops that could help entrepreneurs understand the importance of record-keeping and how to implement better practices.Many small businesses operate in the informal sector, the cost of maintaining formal record-keeping systems, such as hiring professional accountants or purchasing advanced accounting software can be prohibitive for many entrepreneurs. As a result, businesses often rely on manual methods or makeshift solutions, which are inefficient and prone to inaccuracies.Time constraints are another major challenge for small business owners who are often involved in every aspect of their business, from managing operations to marketing and customer service. With such a wide range of responsibilities, business owners may find it difficult to dedicate sufficient time to maintain accurate records. According to scholars,many entrepreneurs simply lack the time or inclination to implement proper record-keeping practices, as they are focused on the day-to-day survival of their businesses.Moreover, small business owners may not see the immediate benefits of investing time in record-keeping, leading them to delay or neglect this important task. Without clear motivation, they may perceive record-keeping as a burdensome task that takes time away from generating revenue. Previous studies highlight that this time scarcity often results in a "fire-fighting" approach to business management, where immediate concerns take precedence over long-term strategic planning, including maintaining accurate financial records. Traditional business methods often rely on informal practices and oral communication, which can undermine the need for written documentation. Many business owners, especially those in the agricultural and retail sectors, have grown accustomed to conducting transactions verbally or through informal agreements, which can be challenging to document or track accurately. This reliance on oral agreements and verbal communication creates a gap in formal record-keeping systems.Cultural attitudes toward formal education also play a role in shaping business practices.Many small business owners may not prioritize formal education or training in business management, particularly in rural areas where literacy rates may be lower. As noted by previous studies, the lack of educational opportunities for many entrepreneurs, especially in the informal sector, can hinder their understanding of the importance of record-keeping and other business management skills. Without the necessary education and training, business owners may view record-keeping as unnecessary or overly complicated.Furthermore, some cultural attitudes may also influence attitudes toward financial transparency and accountability. In certain communities, informal cash transactions may be seen as a more convenient or trusted way to conduct business, reducing the perceived need for formal records. This cultural preference for informality contributes to a lack of standardized documentation practices and reinforces inefficient management methods.

The technological divide is another challenge that affects small business owners and in rural areas. While advanced record-keeping tools such as accounting software and cloud-based storage systems are available, many small business owners lack access to technology due to infrastructure limitations, high costs, and limited digital literacy. According to previous studies, the lack of reliable internet access and affordable technology makes it difficult for businesses to adopt more efficient digital record-keeping methods.As a result, many small business owners rely on traditional, manual methods of recording transactions, which are not only time-consuming but also prone to errors.The absence of digital tools or online platforms for record-keeping prevents businesses from taking full advantage of modern technologies that could help streamline their operations and improve accuracy.Lastly, regulatory and administrative challenges contribute to the difficulties small business owners face when implementing record-keeping systems. In some cases, the complexity of regulatory requirements, including tax laws and business registration processes, can overwhelm entrepreneurs who are already burdened with day-to-day business operations. As noted by scholars, businesses that lack proper records often struggle to comply with tax obligations, resulting in fines or penalties. At its core, the RBV Resource-Based View emphasizes the importance of resources that are valuable and can be used to create competitive advantage. For SSEs, one of the most important resources is the accurate and timely information provided by effective record-keeping practices. Financial records, sales data, inventory management logs, and employee information are all critical elements that businesses use to assess their performance, manage operations, and make strategic decisions.According to previous studies, resources that are valuable, rare, and difficult to imitate give businesses a competitive edge. In the case of SSEs, accurate records provide business owners with insights into cash flow, profitability, customer behavior, and resource allocation. This information, if properly utilized, can lead to better business decisions and more efficient operations. For instance, detailed financial records can help small businesses manage costs, avoid financial mismanagement, and optimize pricing strategies — all of which contribute to long-term business sustainability. Moreover, RBV suggests that a firm's resources must be organized and managed in a way that allows them to be fully utilized. In this case, record-keeping is not merely about collecting data, but about how businesses process and use that information to improve their performance. According to scholars, small businesses that manage their records systematically are more likely to make informed decisions, which can lead to better business outcomes. On the other hand, those that fail to effectively organize and utilize their records may struggle to compete with businesses that are more resource-efficient and better equipped to track their operations.According to this theory, businesses prefer some forms of funding over others,with internal funds coming first, then debt, and equity as a last option to reduce the expenses brought on by knowledge asymmetry.According to the pecking order theory of corporate finance, businesses would rather finance. themselves in a particular sequence: internal funds (retained earnings) come first, followed by debt, and equity comes last. The higher expenses and risks of external funding, especially equity, which is more vulnerable to information asymmetry—where management knows more about the business than outside investors—are the main causes of this preference.

In the context of SSEs, record-keeping is a valuable resource that contributes to decision-making, financial planning, and strategic management. According to previous studies, a firm's growth is often limited by its ability to manage and utilize its resources effectively. In the same vein, accurate and timely record-keeping allows SSEs to make informed decisions, manage growth, and remain competitive in a challenging business environment. Record-keeping is thus considered a "strategic resource" that enables businesses to identify growth opportunities, manage risks, and optimize operations. Scholars assert that businesses that maintain well-organized records can identify trends, forecast future demand, and plan accordingly. For instance, by analyzing sales data, a business owner can identify peak sales periods and adjust their inventory or marketing strategies to maximize profits. In this way, record-keeping contributes not only to the efficiency of day-to-day operations but also to long-term strategic planning.Furthermore, according to previous studies, businesses with robust record-keeping systems can effectively manage cash flow, ensuring that they have sufficient resources to meet their financial obligations. By having accurate records of income and expenditures, SSEs are better able to avoid liquidity problems and ensure financial stability. As a result, record-keeping becomes a fundamental resource for achieving business longevity and success. The RBV theory also highlights the importance of efficient resource allocation for achieving competitive advantage. Record-keeping facilitates effective resource allocation by providing business owners with the necessary data to assess how resources are being utilized across different aspects of the business. For example, inventory records allow business owners to monitor stock levels, avoid overstocking or under stocking, and manage working capital more efficiently.

According to some scholars, all businesses often face resource constraints, making it critical to allocate resources efficiently. Proper record-keeping enables business owners to track resource utilization and make data-driven decisions about how to allocate their limited resources. For instance, by reviewing financial records, a small business can determine which areas of their operations are the most profitable and which require more investment, ensuring that resources are allocated where they are most needed.Additionally, record-keeping allows small businesses to better manage their workforce, ensuring that labor resources are utilized efficiently.Employee records, such as attendance, hours worked, and performance reviews, can help business owners make informed decisions about staffing needs and payroll management, contributing to a more effective and efficient operation.In line with the RBV theory, sustainability is another crucial element for SSEs seeking to maintain a competitive advantage overtime.Record-keeping is vital for ensuring the long-term sustainability of small businesses. By maintaining accurate records, business owners can identify potential risks, track financial health, and make adjustments to their strategies as needed. Previous studies argue that businesses that fail to keep accurate records may struggle to identify risks, leading to poor financial management, regulatory non-compliance, and even business failure.Furthermore, businesses that consistently track and analyze their performance are better positioned to adjust to changing market conditions and consumer preferences, ensuring that they remain competitive. As such, record-keeping is not only about managing current resources but also about planning for the future. According to previous studies, firms that can systematically manage their resources and adapt to changes in the market environment are more likely to achieve sustainable growth and competitive advantage.The importance of proper record-keeping for small-scale enterprises (SSEs) has been widely documented in various studies, with empirical evidence consistently supporting the correlation between systematic record-keeping and business success. Several researchers have explored how businesses that engage in proper record-keeping practices tend to experience higher financial growth, improved operational efficiency, and enhanced sustainability. This section reviews relevant empirical studies, focusing on the role of record-keeping in the success of small businesses in both developed and developing economies.Finally, the RBV theory emphasizes that firms can use their resources to foster innovation and knowledge management, both of which are essential for growth. Record-keeping serves as a repository of knowledge and information that can be accessed and used to generate new ideas, improve business processes, and innovate products or services. According to scholars, firms with access to valuable knowledge resources can develop unique capabilities that allow them to out perform competitors.In this regard, record-keeping not only helps businesses track performance but also enables them to learn from past experiences. By reviewing historical records, small businesses can identify successful strategies, avoid past mistakes, and innovate in ways that improve their competitiveness. Previous studies argue that small businesses that use their records for learning and innovation are better equipped to adapt to market demands and technological changes.

They found that small businesses that maintained accurate financial records had a better understanding of their cash flow, enabling them to make informed decisions regarding resource allocation and investment. These businesses were also better positioned to secure loans and attract investors due to their transparent financial practices. The study concluded that businesses with poor record-keeping practices often struggled with financial mismanagement, which hindered their ability to expand or sustain operations in the long term. They observed that businesses that employed accurate record-keeping practices were able to track their profits and losses effectively, allowing them to identify inefficiencies and make necessary adjustments to improve profitability. The study highlighted that improper record-keeping, on the other hand, led to financial mismanagement, resulting in business failure and bankruptcy. These findings underscore the importance of record-keeping for financial performance and business growth.The role of record-keeping in supporting effective business decision-making has also been well-documented. According to the study, accurate records allow business owners to make data-driven decisions regarding operations, marketing strategies, and expansion. Their study found that small business owners who maintained proper records had a clear understanding of their business's strengths and weaknesses, enabling them to capitalize on opportunities and address challenges promptly.In their empirical research on small businesses in the retail sector, the study found that proper record-keeping practices helped business owners make informed decisions about inventory management, product pricing, and customer relations. Businesses with efficient record-keeping systems were able to track sales trends, predict customer demand, and avoid stockouts or overstocking, all of which contributed to higher customer satisfaction and profitability. The study emphasized that the absence of proper records resulted in poor decision-making, missed opportunities, and operational inefficiencies.The role of record-keeping in supporting effective business decision-making has also been well-documented. According to the study, accurate records allow business owners to make data-driven decisions regarding operations, marketing strategies, and expansion. Their study found that small business owners who maintained proper records had a clear understanding of their business's strengths and weaknesses, enabling them to capitalize on opportunities and address challenges promptly.In their empirical research on small businesses in the retail sector, the study found that proper record-keeping practices helped business owners make informed decisions about inventory management, product pricing, and customer relations. Businesses with efficient record-keeping systems were able to track sales trends, predict customer demand, and avoid stockouts or overstocking, all of which contributed to higher customer satisfaction and profitability. The study emphasized that the absence of proper records resulted in poor decision-making, missed opportunities, and operational inefficiencies.

A key aspect of record-keeping is its role in facilitating access to external financial resources, such as loans, grants, and investor funding. The author conducted a study on the challenges faced by small businesses in accessing finance and found that inadequate record-keeping practices were a major barrier to securing external funding. Their research revealed that financial institutions and investors were more likely to support businesses that had a clear, verifiable financial history, which could only be provided through accurate record-keeping.Furthermore, the study examined how proper record-keeping influences the ability of small businesses to access credit from banks and other financial institutions. Their findings indicated that businesses with well-maintained records, such as income statements, balance sheets, and tax returns, were more likely to receive loans and favorable credit terms. On the other hand,businesses that lacked proper documentation were often rejected for loans, as financial institutions perceived them as high-risk borrowers.Sustainability is another area where proper record-keeping plays a crucial role. Scholars argue that the sustainability of small businesses is directly linked to the ability of business owners to track their performance and adapt to changes in the market. Their study found that businesses that used accurate records to assess their financial health were better equipped to make strategic decisions that ensured long-term sustainability. For example, record-keeping enabled businesses to monitor their cash flow, plan for future growth, and manage risks, all of which are essential for long-term survival.Similarly, some studies found that small businesses that maintained proper records were more likely to survive in competitive markets. They observed that businesses with organized financial records could better withstand economic downturns or financial challenges, as they were able to manage their resources more effectively. The study concluded that record-keeping is a key determinant of business sustainability, as it allows small businesses to plan for the future and make informed decisions about expansion and diversification. Another significant benefit of record-keeping is its role in ensuring tax compliance. According to previous studies, businesses that maintain accurate records are better positioned to comply with tax regulations, avoid penalties, and ensure that they are paying the correct amount of tax. Their study found that poor record-keeping often led to tax evasion or underreporting of earnings, which could result in legal issues and fines.In a similar vein, some scholars highlighted that record-keeping is essential for maintaining proper tax records, which are necessary for businesses to fulfill their legal obligations. They found that businesses that were proactive in keeping accurate tax records were more likely to avoid issues with the tax authorities and were perceived as more trustworthy by regulators.In addition to its impact on business performance, record-keeping also contributes to the development of entrepreneurship. Some scholars found that small businesses that adopted efficient record-keeping practices were more likely to grow and scale their operations. Their study suggested that business owners who engage in systematic record-keeping are more entrepreneurial in nature, as they have a better understanding of their business environment, financial status, and potential growth opportunities. By maintaining accurate records, entrepreneurs are able to assess their progress and identify areas for improvement, which fosters innovation and business development.

The business environment is dramatically changing. Companies today face the challenge of increasing competition, expanding markets, and rising customer expectations. This increases the pressure on companies to lower total costs in the entire supply chain, shorten throughput times,drastically reduce inventories, expand product choice, provide more reliable delivery dates and better customer service, improve quality, and efficiently coordinate global demand, supply, and production. As the business world moves ever closer to a completely collaborative model and competitors upgrade their capabilities, to remain competitive,organizations must improve their own business practices and procedures. Companies must also increasingly share with their suppliers, distributors, and customers the critical in-house information they once aggressively protected. And functions within the company must upgrade their capability to generate and communicate timely and accurate information.To accomplish these objectives, companies are increasingly turning to enterprise resource planning (ERP)systems.ERP provides two major benefits that do not exist in non-integrated departmental systems: a unified enterprise view of the business that encompasses all functions and departments; and an enterprise database where all business transactions are entered, recorded, processed, monitored,and reported. This unified view increases the requirement for, and the extent of, interdepartmental cooperation and coordination. But it enables companies to achieve their objectives of increased communication and responsiveness to all stakeholders.

Performance measures that assess the impact of the new system must be carefully constructed. Of course, the measures should indicate how the system is performing. But the measures must also be designed so as to encourage the desired behaviors by all functions and individuals. Such measures might include on-time deliveries, gross profit margin, customer order-to-ship time, inventory turns, vendor performance, etc. Project evaluation measures must be included from the beginning. If system implementation is not tied to compensation, it will not be successful.For example, if all managers will get their raises and bonuses next year even if the system is not implemented, successful implementation is less likely. Management, vendors, the implementation team, and the users must share a clear understanding of the goal. If someone is unable to achieve agreed-upon objectives, they should either receive the needed assistance or be replaced. When teams reach their assigned goals, rewards should be presented in a very visible way. The project must be closely monitored until the implementation is completed. The system must be forever monitored and measured. Management and other employees often assume that performance will begin to improve as soon as the ERP system becomes operational. Instead,because the new system is complex and difficult to master, organizations must be prepared for the possibility of an initial decline in productivity. As familiarity with the new system increases, improvements will occur. Thus, realistic expectations about performance and time frames must be clearly communicated. Inventory revaluation is a huge concern when cost accounting systems are changed. In the case of an ERP implementation, this is unavoidable (unless the cost accounting process of the legacy system is somehow duplicated). To avoid any significant revaluation issues, a process was developed to export real-time data from the converted database and compare the cost variances.

Data accuracy is absolutely required for an ERP system to function properly. Because of the integrated nature of ERP, if someone enters the wrong data, the mistake can have a negative domino effect throughout the entire enterprise. Therefore, educating users on the importance of data accuracy and correct data entry procedures should be a top priority in an ERP implementation. ERP systems also require that everyone in the organization must work within the system, not around it. Employees must be convinced that the company is committed to using the new system, will totally change over to the new system, and will not allow continued use of the old system. To reinforce this commitment, all old and informal systems must be eliminated. If the organization continues to run parallel systems, some employees will continue using the old systems. Multi-site implementations present special concerns. The manner in which these concerns are addressed may play a large role in the ultimate success of the ERP implementation. The desired degree of individual site autonomy may be a critical issue which depends on two factors: the degree of process and product consistency across the remote sites, and the need or desire for centralized control over information, system setup, and usage. One of the objectives of an ERP implementation may be to increase the degree of central control through the implementation of standardized processes. Alternatively, the implementation may be undertaken in order to provide the remote sites with capabilities that allow them to fine tune their processes to their unique situations. Another complexity in dealing with multi-site implementations is the degree to which the culture of the organization differs between sites. The fundamental issue here is one of corporate standardization versus local optimization.Corporate standardization brings with it simplified interfaces among diverse parts of the organization, ability to move people and products between sites with minimal disruption, and relative ease in consolidating data across the entire organization.On the other hand, local optimization may result in more effective and efficient operation and may reduce costs.Perhaps the most difficult decision to be made in a multi-site implementation is the question of cutover strategy.The organization must choose between an approach where the implementation takes place simultaneously in all facilities or a phased approach by module, by product line, or by plant with a pilot implementation at one facility. With a large outlay of cash up front for software, hardware, and the project team, the company may want a simultaneous implementation in order to recoup its investment as quickly as possible.

This review aims to digital twin technology, predictive analytics, and sustainable project management to enhance global supply chain efficiency, resilience, and environmental sustainability. Digital twins provide real-time virtual representations of physical supply chain systems, enabling predictive analytics to identify potential disruptions and optimize decision-making processes.By combining these advanced technologies with sustainable project management practices, such as circular supply chains and green logistics, organizations can proactively address risks while reducing their carbon footprint.The focus on data-driven insights and scenario analysis facilitates informed risk mitigation and resource optimization.The study will enable the researcher to understand the current state of the art regarding the role of users in terms of performance. Understanding will be based on the premise that users are in a position to evaluate the benefits of these systems for organizations that have already implemented or are in the process of implementing ERP systems.

CHAPTER 3

METHODOLOGY

3.1 Research Design

This study used a descriptive and developmental research design to create an Automated Resource Management System. The descriptive part was used to understand how the organization manages its resources, find out what problems users face and figure out what the organization needs. About designing, building, testing and putting the proposed Automated Resource Management System into action. This approach allowed us to use the information we gathered and the requirements we found to create a working automated system that can manage resources, make things more efficient and help with decision making for the Automated Resource Management System.

We used interviews, observations and surveys to get the information we needed. We did interviews to get ideas from users and people involved with the Automated Resource Management System about their experiences. What they expected. We watched how the organization works and how it manages its resources. We also gave out surveys to get feedback and make sure we understood what users needed for the Automated Resource Management System. The information we gathered was used to design and build the system.

The Software Development Life Cycle guided our study. It gave us a plan for planning, analyzing, designing, building, testing and putting the Automated Resource Management System into action. The Software Development Life Cycle made sure each step was done correctly and that the final system met the needs of its users. Our study was also based on Information Systems Theory, which says that technology, data, processes and people should work together to help the organization achieve its goals and improve how it manages resources for the Automated Resource Management System.

The way we did our research is important. It includes the methods, techniques, tools and processes we used to gather, analyze and understand the information to achieve our goals for the Automated Resource Management System. Having a plan for our research made sure our findings about the Automated Resource Management System were valid, reliable and accurate. In this research the plan gave us a way to identify problems with resource management, gather what users needed and build an automated solution for the Automated Resource Management System. By following this approach we made sure the Automated Resource Management System we proposed was based on facts and could help the organization while making resource allocation, monitoring and use better.

3.2 System Development Methodology

The study used the Agile Development Model. This model is a way of working where people talk to each other and listen to users.The Agile Development Model is a repeating process. It involves people working together like developers and stakeholders.

Requirements Analysis: The first step was to figure out what the organization really needed.The study looked at problems with managing resources by hand. We did interviews with people. Watched how things were done.

We're also reviewing existing processes. Some problems were found, like not being able to get things done, having data issues taking a time to track resources. The Agile Development Model helped find these problems.

System Design: Next we made a plan for the system. We designed a database to store data in a way. A user-friendly interface was created.A plan for the system was also made.The System Design phase was crucial.It used the Agile Development Model. The goal was to make the system easy to use, able to grow and reliable.

Development: Then the system was built. The Development phase used the Agile Development Model. It was done in steps with each step improved. Users were asked for feedback to make the system better. The Agile Development Model helped minimize mistakes. It ensured the system was used throughout the Development phase.It really helped with managing resources. It was used to find problems and make the system better.

3.3 System Architecture

An Automated Resource Management System is a website that helps organizations manage their resources better. It is like a place where administrators and authorized users can see, assign and track resources in real time. This system makes resource management easier by automating tasks, reducing mistakes and making operations more efficient.
The system has important features such as It allows users to assign resources to tasks, departments or people and see what is available and being used. It also helps keep track of equipment, supplies and resources. The system controls who can see and use functions and information based on their role in the organization. The system also has a dashboard that shows summaries, statistics and pictures of how resources are being used.This helps management make decisions about planning and assigning resources.
With real-time data organizations can see trends, use resources better and be more productive.You can access the Automated Resource Management System through the internet using a browser.You do not need to install any software.This makes it easy to manage resources with an internet connection.The system also has a database that keeps information up-to-date and available to authorized users. In short the Automated Resource Management System is an efficient way to manage resources. It helps organizations use resources better work effectively and make better management decisions by automating resource management and providing accurate information. The Automated Resource Management System makes it easy to manage resources and improve operations.

3.4 Tools and Technologies Used

The Automated Resource Management System was built with some useful tools and technologies that help it do its job properly. We wanted the Automated Resource Management System to be efficient, reliable and easy to use so we chose these tools carefully.
These tools are very good at what they do. They are easy to use.

Frontend: HTML, CSS and Bootstrap 5

We used HTML, CSS and Bootstrap 5 to build the frontend of the Automated Resource Management System.
HTML helped us create the structure and content of the web pages for the Automated Resource Management System. The web pages are made up of HTML, which's a great way to organize things. The CSS made the Automated Resource Management System look better. It makes the Automated Resource Management System look nice and visually appealing. Bootstrap 5 gave us some built-in parts that made the Automated Resource Management System work well on devices and screen sizes. It helps the Automated Resource Management System to work well on all kinds of devices.
These technologies, like HTML, CSS and Bootstrap 5 helped us make the Automated Resource Management System more interactive and easier to use.

Backend: PHP

The Automated Resource Management System uses PHP on the server-side PHP helps us process what the users ask for. The Automated Resource Management System relies on PHP to process user requests PHP is very flexible. That is why it is perfect for making web applications and making sure the Automated Resource Management System works smoothly.

Database: MySQL

MySQL manages the Automated Resource Management Systems data MySQL retrieves all the Automated Resource Management Systems data, such as resource records and user information.

Development Tool: Visual Studio Code

The Automated Resource Management System was developed using Visual Studio Code, Visual Studio Code is a tool for writing code and it has some features.
The Automated Resource Management Systems code was written using Visual Studio Codes; these features helped us write the code for the Automated Resource Management System. It works well because of these tools and technologies the Automated Resource Management System relies on these tools and technologies to do its job.
It is a tool and these technologies made it possible for the Automated Resource Management System to work the way it does. We are happy with the Automated Resource Management System and the tools and technologies we used to build it. The Automated Resource Management System would not have been possible, without these tools and technologies we are really glad we chose them for the Automated Resource Management System.

3.5 Data Gathering Techniques

To get the information they needed for the Automated Resource Management System the researchers used a few ways to gather data.

Observation: This is where we watched how the company handles its resources now. They paid attention to how resources are written down, followed, given out and taken care of. This helped them see what is good and what is bad about the way things are done by hand. They saw where things are slow, where mistakes can happen and where things can go wrong when handling resources. By watching how things are done the researchers got an idea of how work really gets done and they used this to design a better automated system.

Interview: We talked to the people who handle resources like the people in charge, staff members and others who are involved. The researchers wanted to know what these people think about the system and what problems they have now. By talking to them the researchers got a good understanding of what the users need and want. This helped them figure out what the system needs to do and make sure it will solve the problems that exist now and handle resources.

To make sure the Automated Resource Management System works well we did a lot of testing while it was being developed. We wanted to find mistakes, check if the system does what it is supposed to do and make sure it meets the needs of the people who will use it.

3.6 Testing Procedure

Unit Testing: We tested each part of the Automated Resource Management System separately. This means we looked at things like registering resources, tracking inventory, managing users and generating reports. We tested each of these features on its own to make sure it works correctly. This helped us find and fix problems which made the whole system better and more stable.

System Testing: After we put all the parts of the Automated Resource Management System together we tested the thing. We wanted to see if all the parts work well together and if the system does what it is supposed to do when it is being used normally. We tested a lot of things like how the different parts interact with each other, if the data is accurate, if the system is secure and how well the whole system performs.

User Acceptance Testing (UAT): We also had the people who will actually use the Automated Resource Management System test it. This included administrators and staff members. They looked at how the system works, how easy it is to use and if it really helps manage resources. We used their feedback to find things that need to be improved and to make sure the system meets their needs. When this testing was finished we knew the Automated Resource Management System was ready to be used.

3.7 Evaluation Criteria

To figure out if the Automated Resource Management System is any good we need to look at things. We want to know if the Automated Resource Management System actually does what people need it to do and if it does it right.

Functionality: The Automated Resource Management Systems functionality is about whether it can do everything it is supposed to do. We will check if the Automated Resource Management System can manage resources, track inventory, handle users and make reports correctly. We also want to make sure that what the Automated Resource Management System produces is accurate and complete.

Usability: Using the Automated Resource Management System should be easy. We need to know if people can navigate the Automated Resource Management System without getting confused, understand what it can do and get their work done without any trouble. If the Automated Resource Management System is easy to use, people can work efficiently with training and they will be happy.

Reliability: The Automated Resource Management System needs to work all the time. We will see if the Automated Resource Management System can work without any errors, crashes or losing information. The Automated Resource Management System should work well all the time. Give us accurate results.

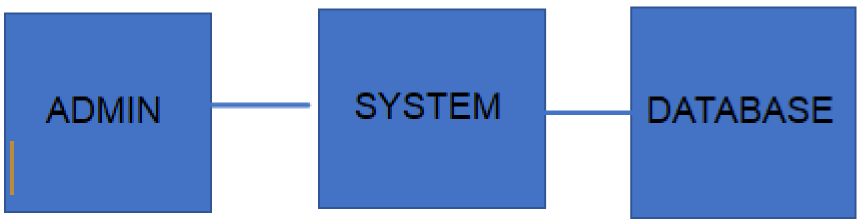
Security: The Automated Resource Management System needs to keep our data and resources safe from people who should not have access to them. We will look at how well the Automated Resource Management System checks who is using it, controls access, keeps data private and handles security measures. Security is very important because it keeps our information safe.

Efficiency: The Automated Resource Management System needs to use resources in a way that gets tasks done quickly. We will look at how fast the Automated Resource Management System responds, how quickly it processes information and how well it performs overall. If the Automated Resource Management System is efficient it gets things done quickly makes people more productive and helps manage resources

The results of these evaluation criteria will help us understand how well the Automated Resource Management System works, how good it is and if it is acceptable. The evaluation will also give us feedback to make the Automated Resource Management System better, in the future.

SUPPORTING DIAGRAMS (Included within Chapter 3)

Data Flow Diagram (DFD – Level 0)



Based on the figure above, it shows how data moves between the administrator, the system and the database. This diagram gives us a view of the Automated Resource Management System.

The DFD Level 0 is a way to see how the administrator, the system and the database work together in the Automated Resource Management System. The administrator uses the Automated Resource Management System to update and manage resource information. They do this by putting in updates to the Automated Resource Management System.

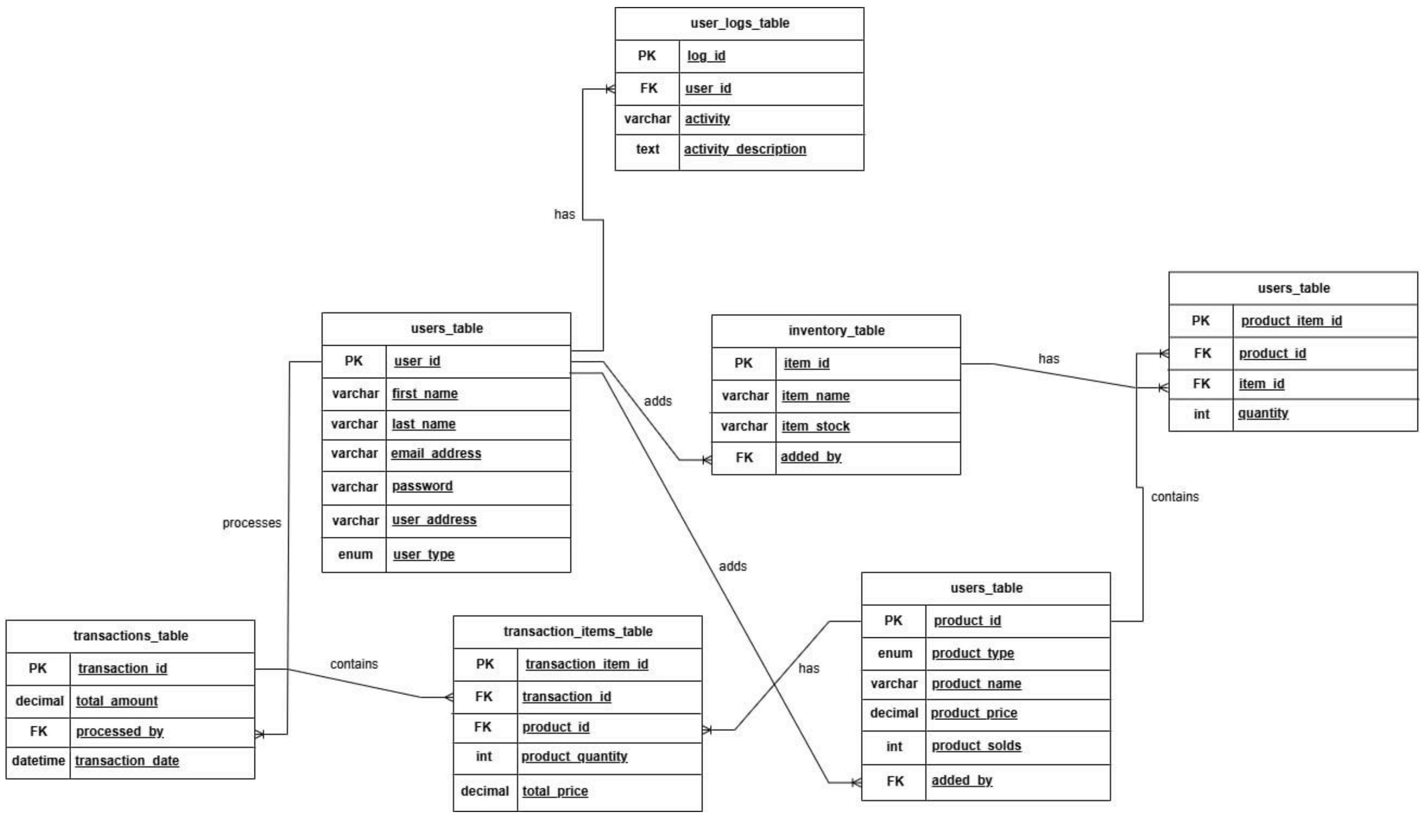
The system takes the data from the Automated Resource Management System, processes it and then stores it. The system also retrieves the data when it is needed. It updates the data. Keep an eye on it. Then the system saves the information in the database. The database stores all the resource data.

When the administrator wants to see some information the system gets the records from the database. Show them. This diagram shows how data moves inside the Automated Resource Management System. It shows how everything works together to manage resources. The Automated Resource Management System makes things easier. It reduces mistakes because it does things automatically.

The administrator manages the Automated Resource Management System. They put in information and update records. They watch resource availability. Manage system data. When the administrator gives the system some information the system works on it. The system checks the data stores it and retrieves it when needed. The system makes reports. Saves the information in the database. The database is a place to store all the information.

When the administrator needs to see records the system looks at the database. Gives the administrator the information. This way the Automated Resource Management System makes sure the data is correct and easy to get.

Entity Relationship Diagram (ERD)



Entity Attributes

- user_id (Primary Key)
- first_name
- last_name
- email_address
- password
- user_address
- user_type (Admin / Staff)
- log_id (Primary Key)
- user_id (Foreign Key → Users.user_id)
- activity
- activity_description
- item_id (Primary Key)
- item_name
- item_stock
- added_by (Foreign Key → Users.user_id)
- product_id (Primary Key)
- product_type
- product_name

- product_price I'm
- product_solds
- added_by (Foreign Key → Users.user_id)
- product_id (Foreign Key → Products.product_id)
- item_id (Foreign Key → Inventory.item_id)
- quantity
- transaction_id (Primary Key)
- total_amount
- processed_by (Foreign Key → Users.user_id)
- transaction_date
- transaction_item_id (Primary Key)
- transaction_id (Foreign Key → Transactions.transaction_id)
- product_id (Foreign Key → Products.product_id)
- product_quantity
- total_price

Relationships:

- A User can register several Inventory Items, meaning one user may handle the input or management of multiple stock records.
- A User can create multiple Products, showing that products are maintained and managed by system users such as administrators or staff.
- A User can handle multiple Transactions, where each transaction represents a completed process or sale.
- A Product can be included in many Transaction Items, meaning it can be sold repeatedly in different transactions.
- A Product can be included in many Transaction Items, meaning it can be sold repeatedly in different transactions.
- A Transaction can include multiple Transaction Items, where each item represents a specific product involved in that transaction.
- Each Transaction is managed by one User, identifying the person responsible for processing it.

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CHED BSIT ALIGNMENT STATEMENT

This capstone project complies with CHED BSIT standards by demonstrating competencies in application development, database management, systems analysis and design, and emerging technologies such as machine learning. The project follows outcomes-based education principles and integrates ethical considerations, user safety, and data privacy.